

Unit 06 (A): Normal Distribution

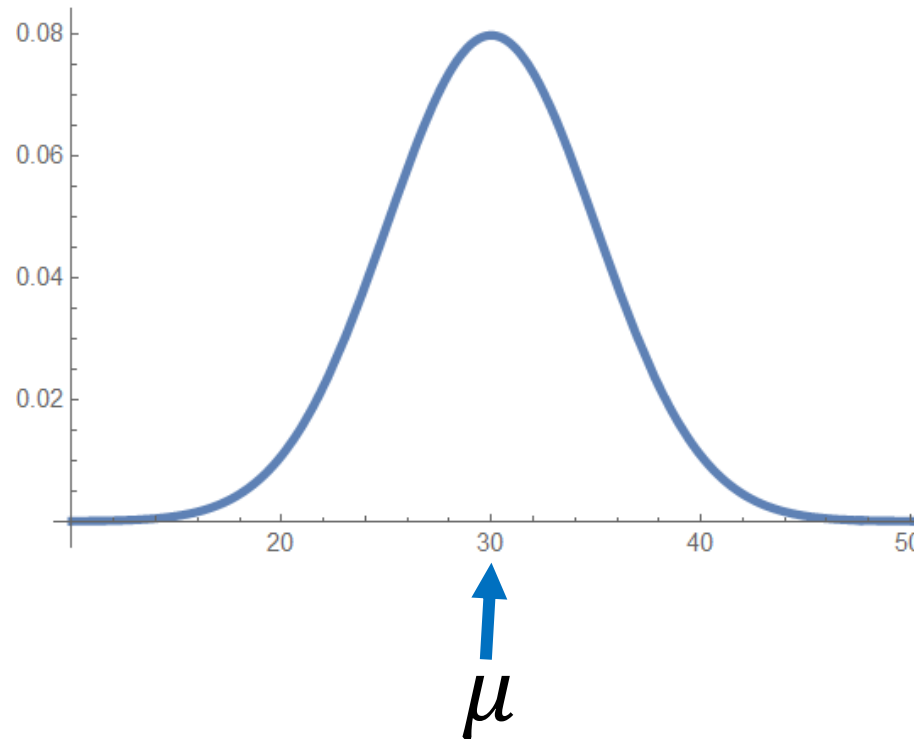
Dr. Md Mutasim Billah

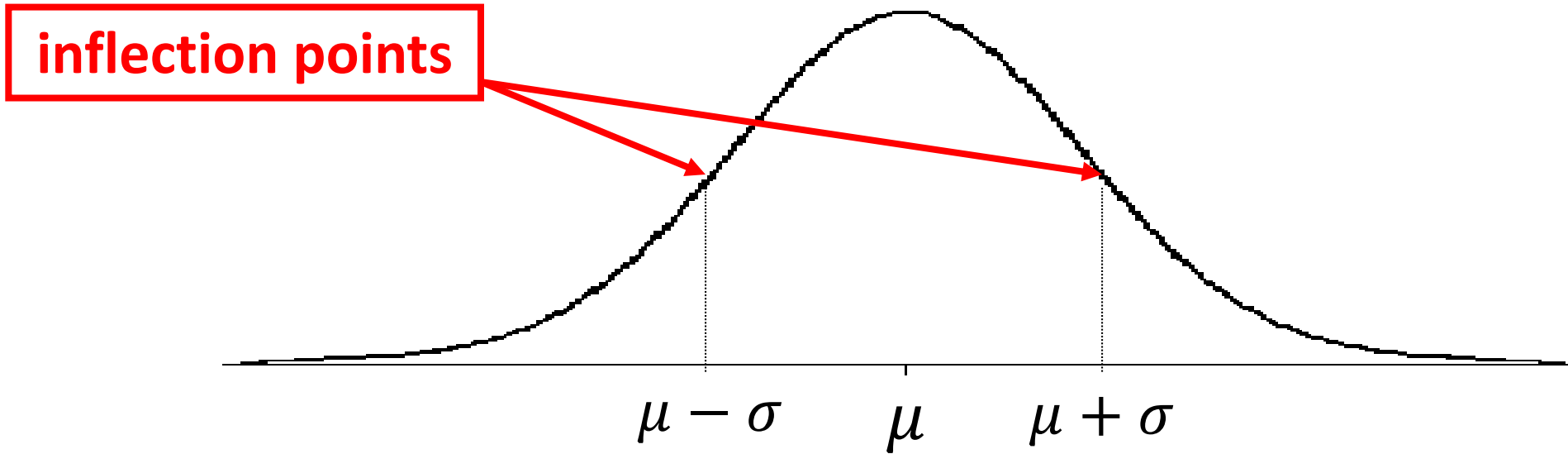
NORMAL DISTRIBUTION

The Normal Distribution (“bell curve”)

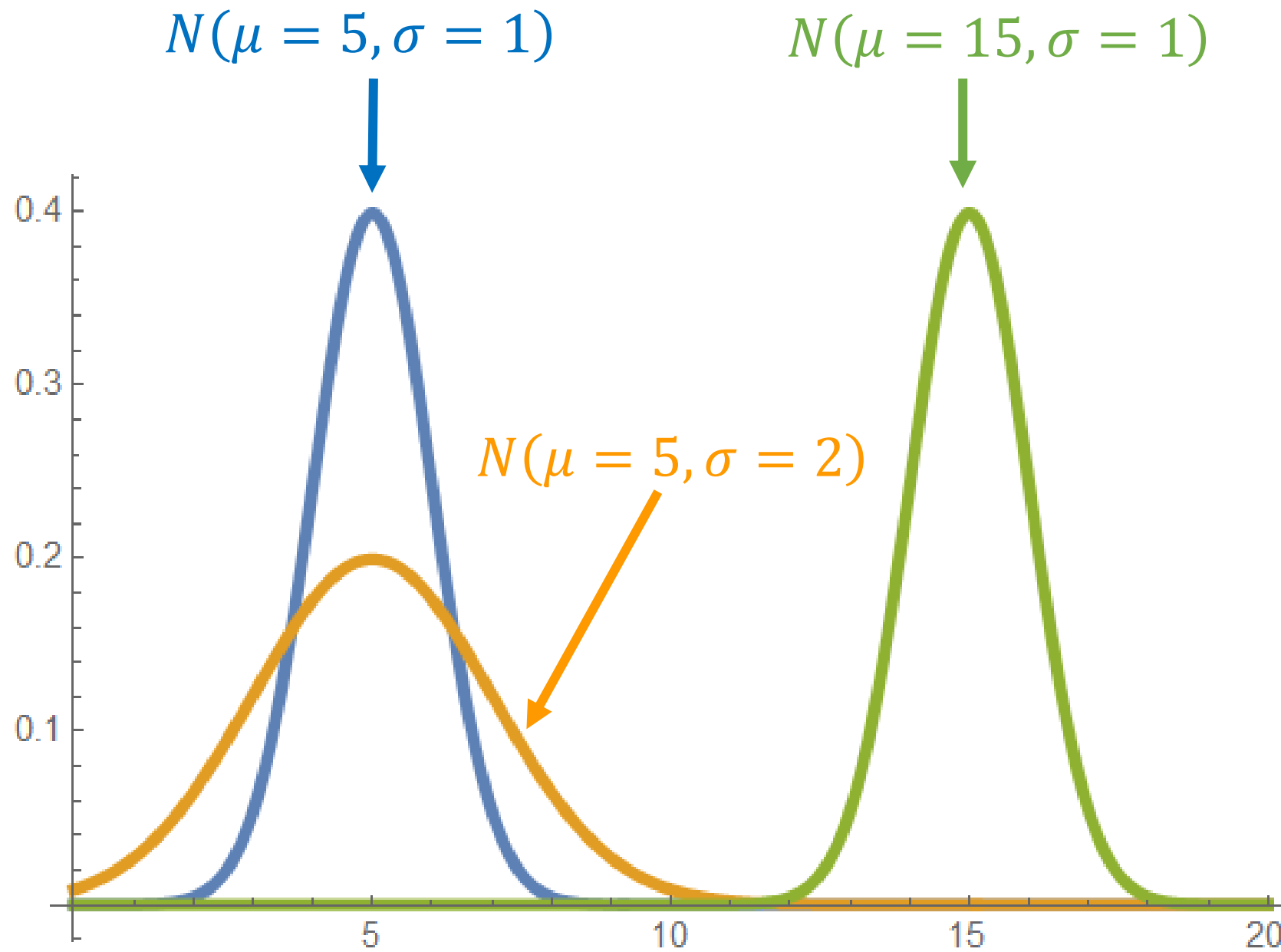
is the most famous and important continuous random variable

If X is normally distributed, its support is $(-\infty, \infty)$. The curve is centered around its mean μ and its width depends on its standard deviation σ





$$f(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad -\infty < x < \infty$$



One reason the normal distribution is famous and important is that many random variables are normal or approximately normal.

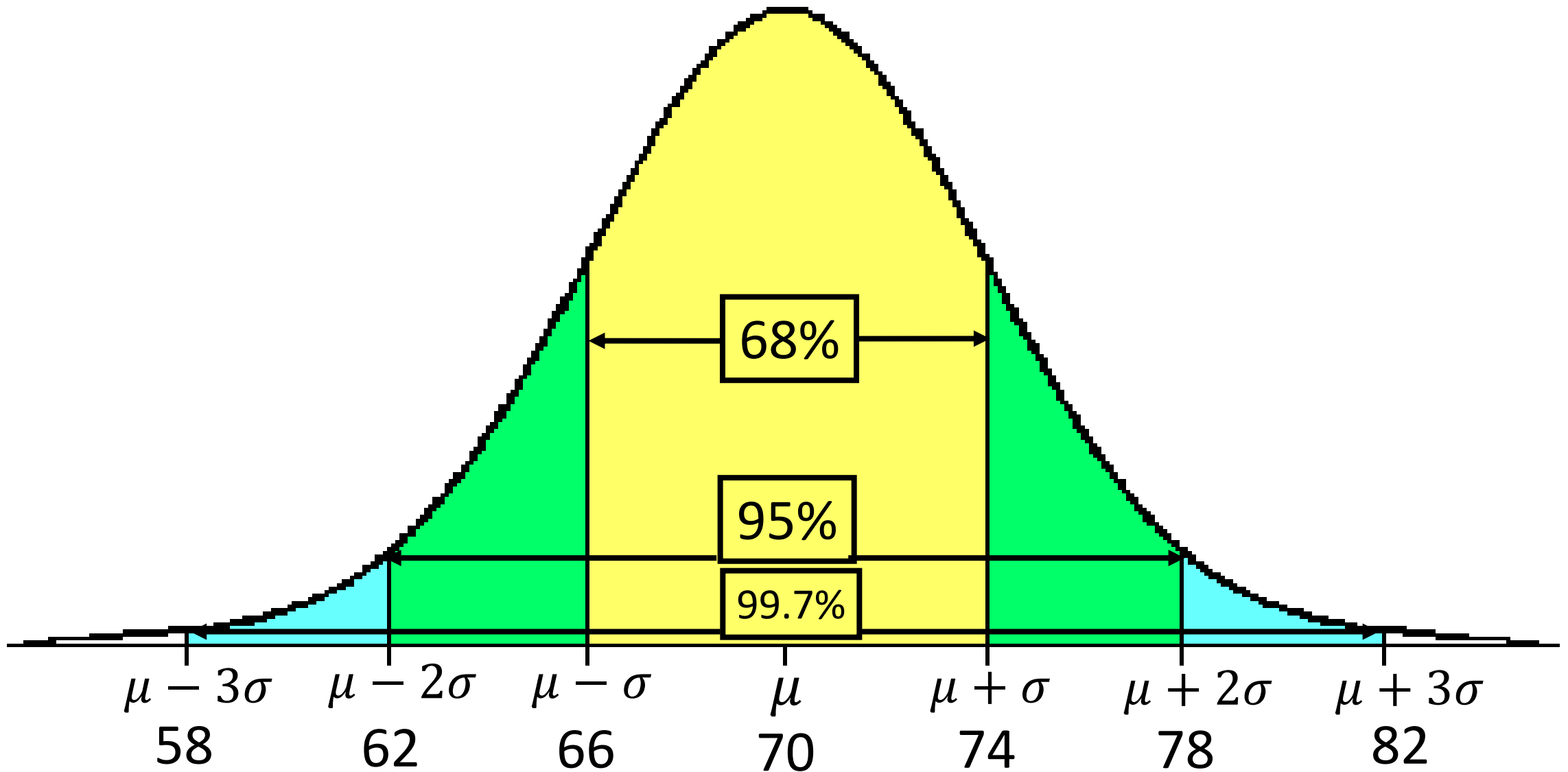
Suppose heights of adult American men are approximately normally distributed and you are about to meet a man whose height is 10σ (10 standard deviations) above the mean. How tall of a person would you be expecting? Someone who is “pretty tall” (maybe 6 and a half feet tall), or would you expect a giant (the tallest person who has ever lived)?

Heights of adult American men are approximately normal with a mean of $\mu = 70$ inches and a standard deviation of $\sigma = 4$ inches.

$$70 + 10(4) = 110 \text{ inches} \rightarrow 9'2''$$

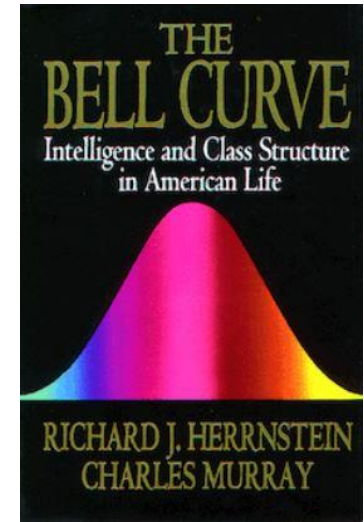
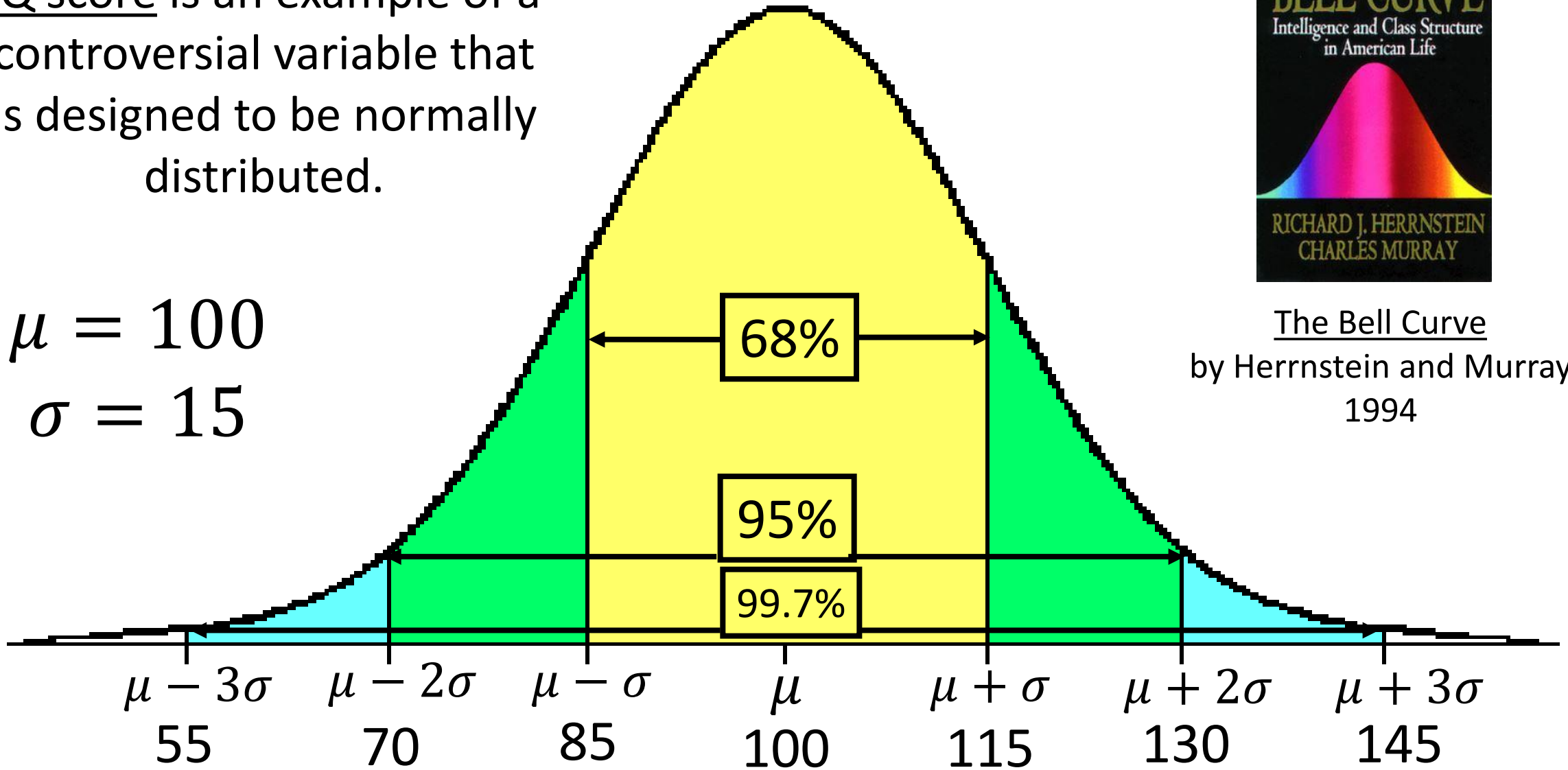
The tallest man who ever lived was 8'11"

Empirical Rule for normal distributions



IQ score is an example of a controversial variable that is designed to be normally distributed.

$$\mu = 100$$
$$\sigma = 15$$



The Bell Curve
by Herrnstein and Murray
1994

STANDARD NORMAL DISTRIBUTION

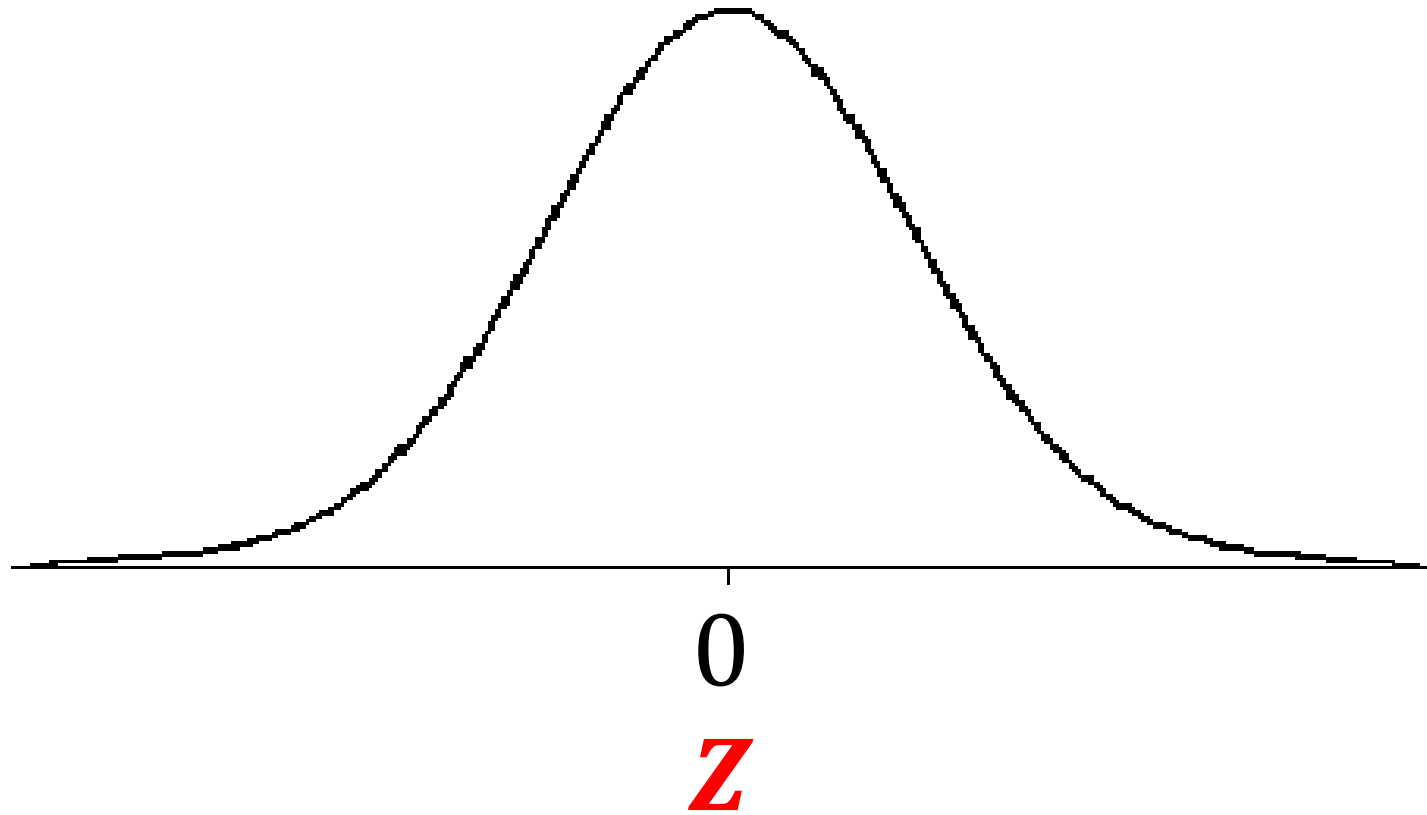
There are an endless number of normal curves since any real number is a possible choice for μ and any positive real number is a choice for σ . But there is a very special normal curve where $\mu = 0$ and $\sigma = 1$. We call it the **standard normal curve** and we use Z instead of X .

$$Z \sim N(\mu = 0, \sigma = 1)$$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad -\infty < z < \infty$$

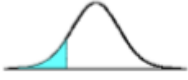
The pdf simplifies a bit when $\mu = 0$ and $\sigma = 1$, but it is still a function that is impossible to integrate with pencil and paper.

So how can we find areas under the Z curve?



$$P(Z < 1.55) = ?$$

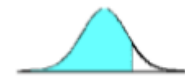
Z-Table
negative Z values



	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-1.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-1.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-1.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-1.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-1.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-1.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-1.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-1.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-1.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-1.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-0.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-0.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-0.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-0.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-0.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-0.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-0.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-0.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-0.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
0.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.0	.5000	.4980	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

negative Z values

Z-Table
positive Z values



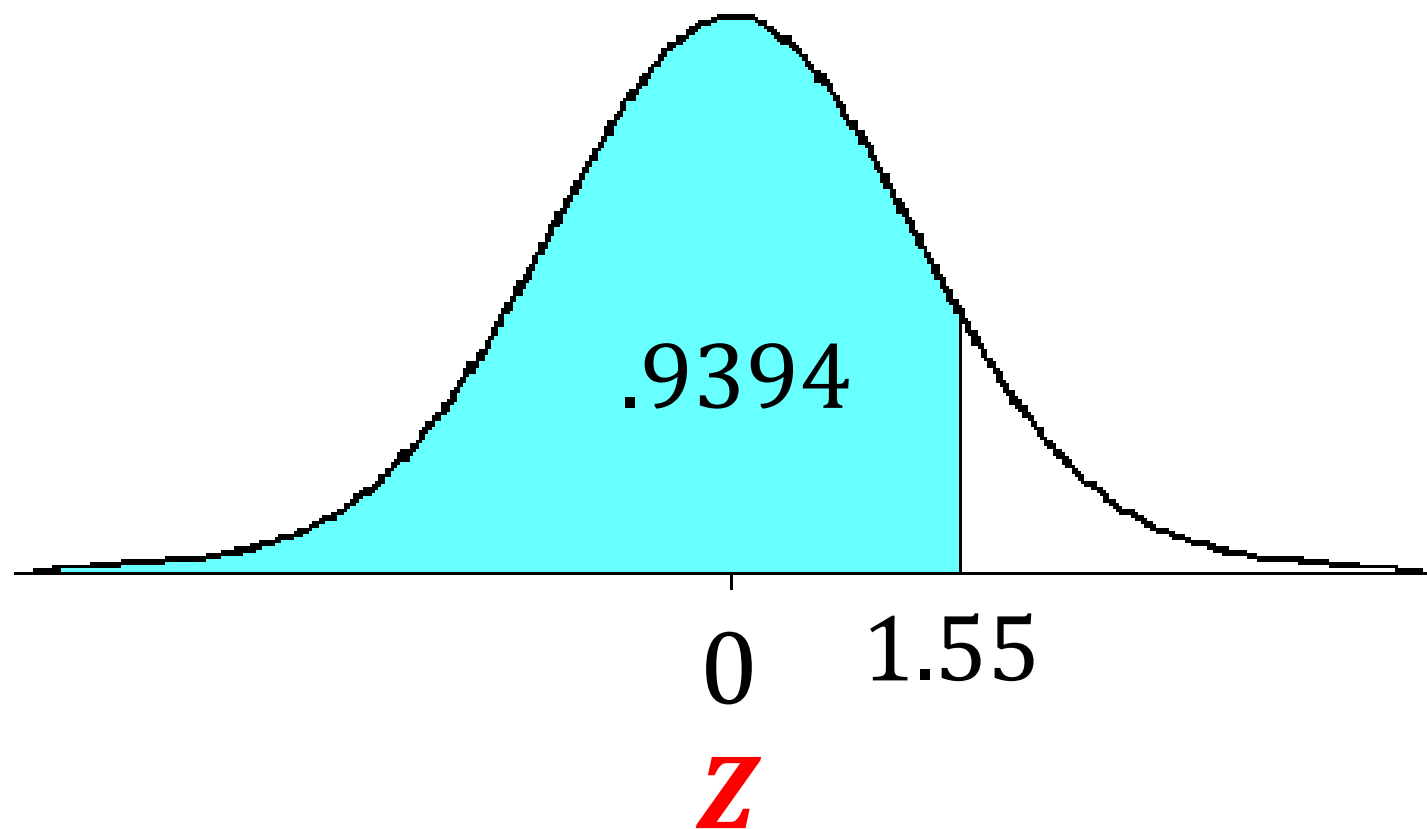
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

positive Z values

positive Z values



	0.00	0.01	0.02	0.03	0.04	0.05	0.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406



$$P(Z < 1.55) = .9394$$

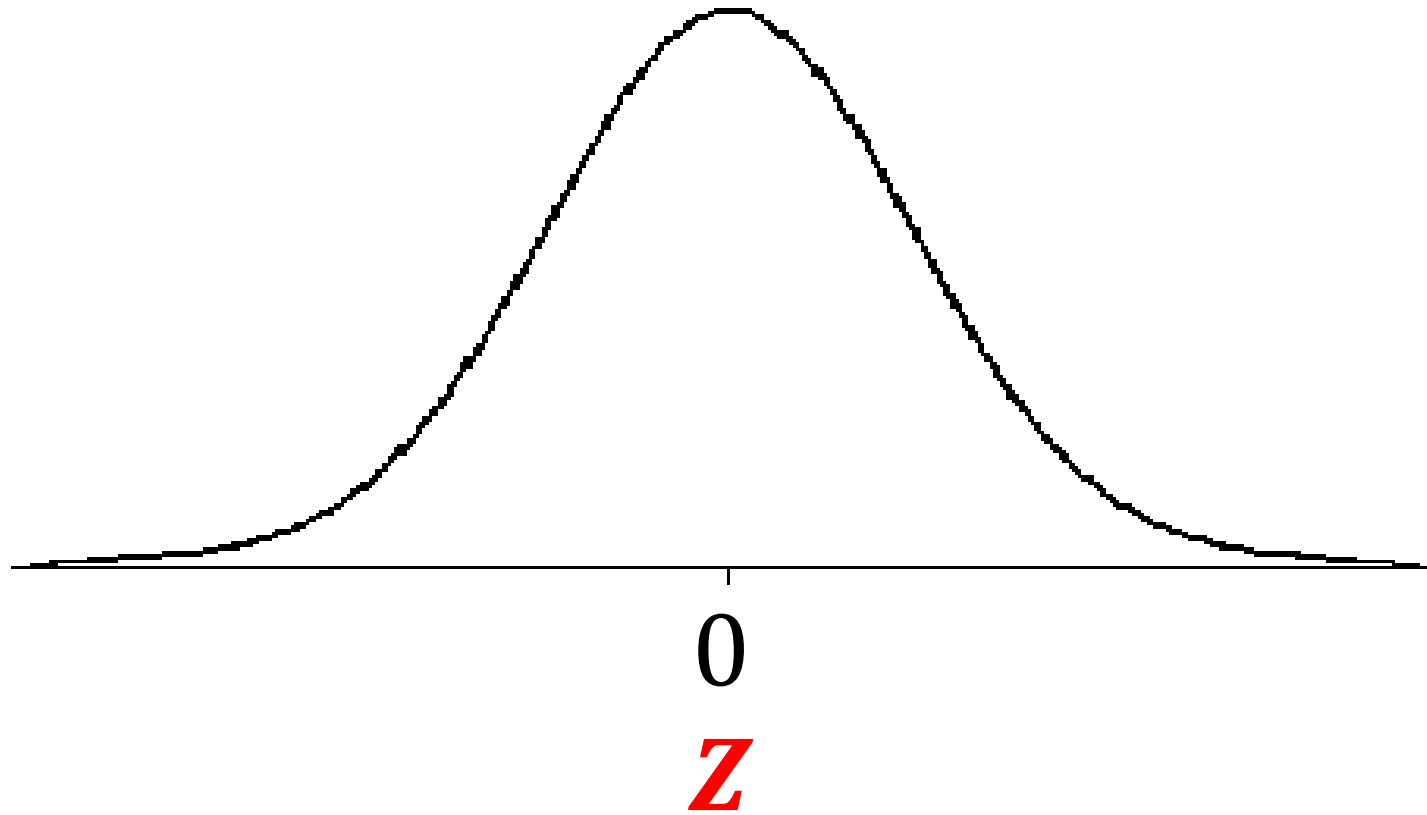
Standard Normal Table (Z-Table)

Remember:

The area under curve = 1

The curve is symmetric around zero.

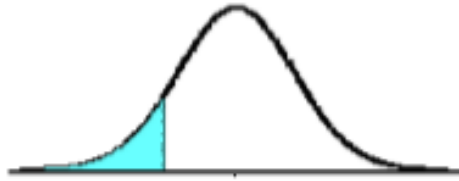
The table always gives you area to the LEFT of a z value. (Be aware: many different textbooks have Z tables that report areas in different ways.)



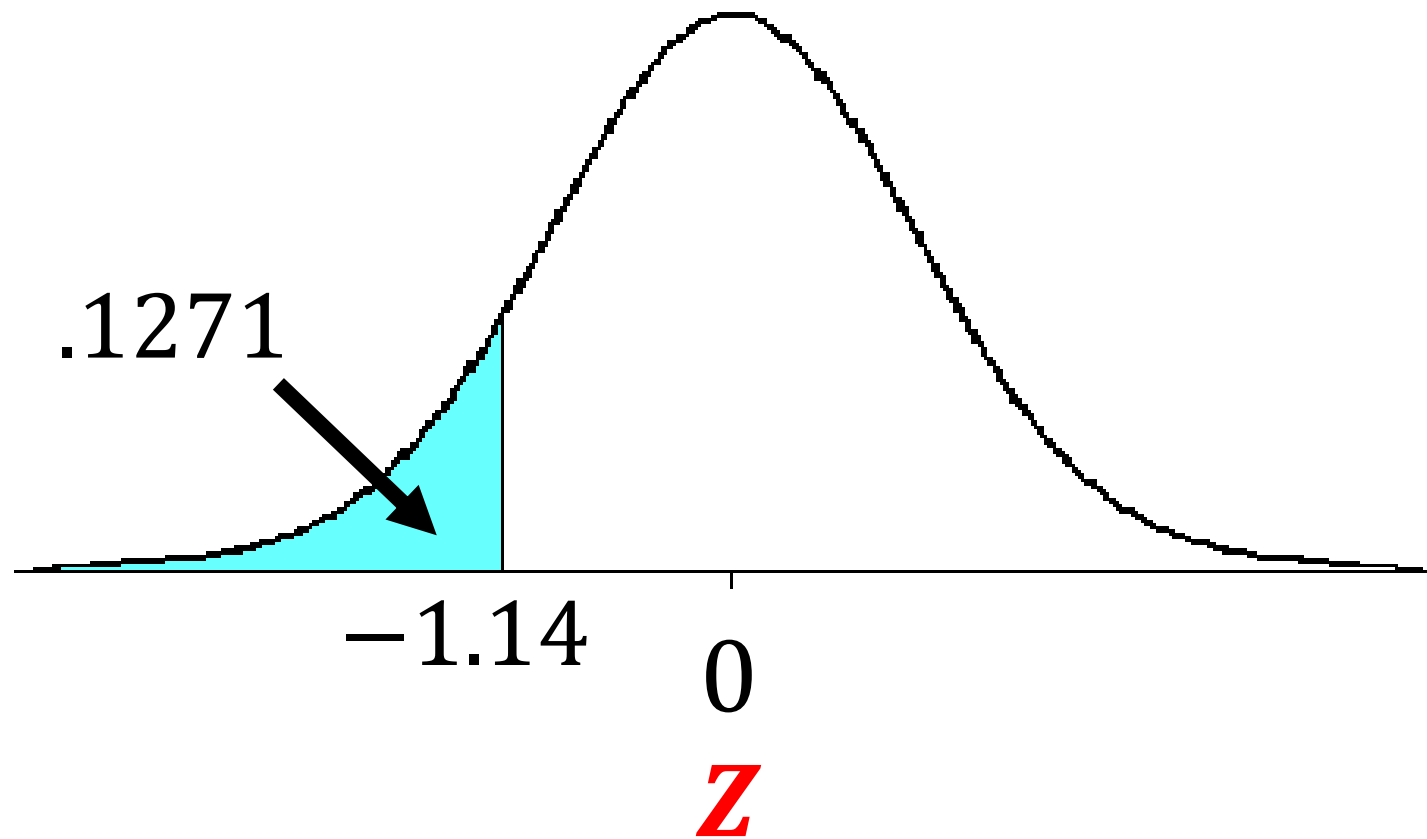
$$P(Z < -1.14) = ?$$

Z-Table

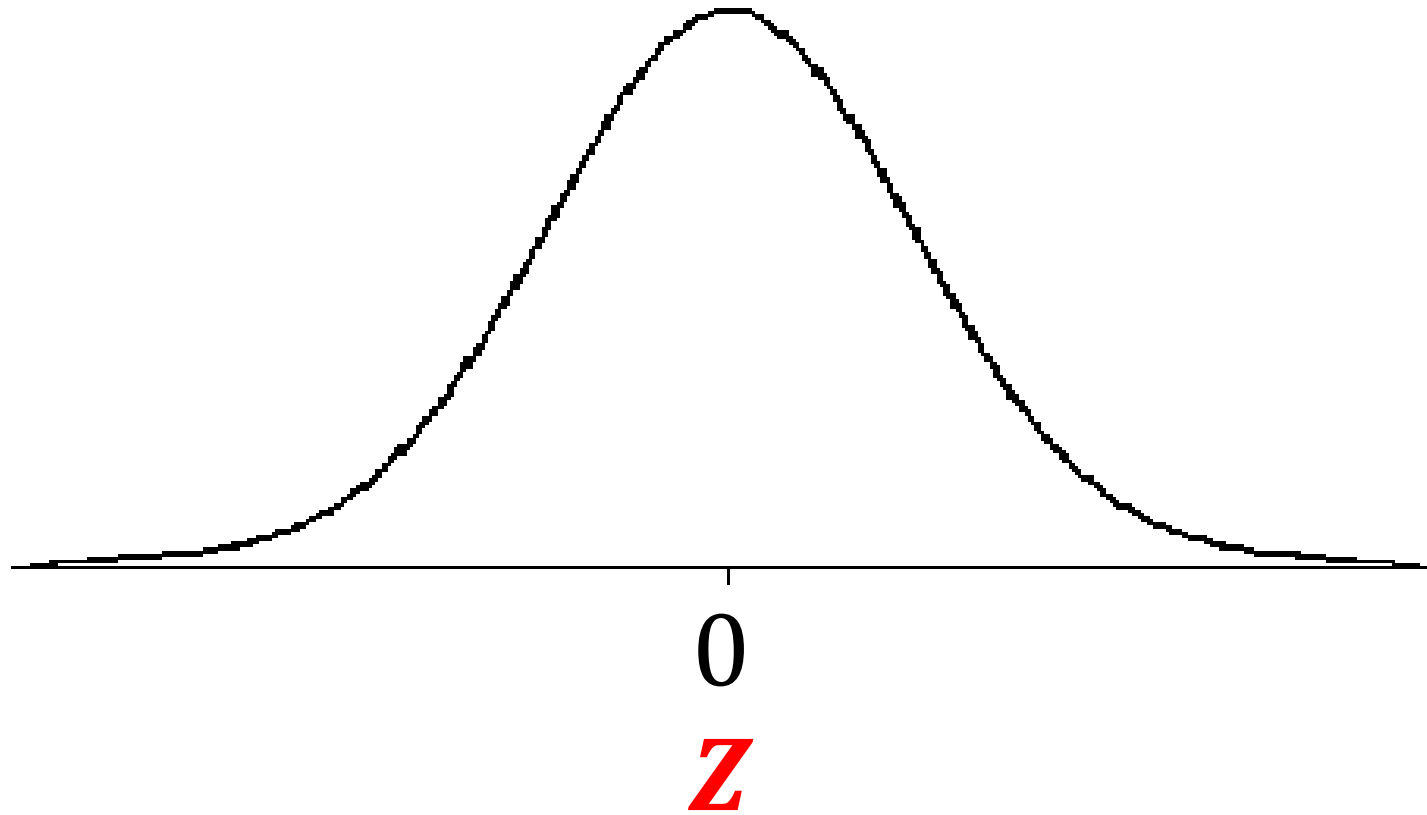
negative Z values



	0.00	0.01	0.02	0.03	0.04	0.05
-3.0	.0013	.0013	.0013	.0012	.0012	.0011
-2.9	.0019	.0018	.0018	.0017	.0016	.0016
-2.8	.0026	.0025	.0024	.0023	.0023	.0022
-2.7	.0035	.0034	.0033	.0032	.0031	.0030
-2.6	.0047	.0045	.0044	.0043	.0041	.0040
-2.5	.0062	.0060	.0059	.0057	.0055	.0054
-2.4	.0082	.0080	.0078	.0075	.0073	.0071
-2.3	.0107	.0104	.0102	.0099	.0096	.0094
-2.2	.0139	.0136	.0132	.0129	.0125	.0122
-2.1	.0179	.0174	.0170	.0166	.0162	.0158
-2.0	.0228	.0222	.0217	.0212	.0207	.0202
-1.9	.0287	.0281	.0274	.0268	.0262	.0256
-1.8	.0359	.0351	.0344	.0336	.0329	.0322
-1.7	.0446	.0436	.0427	.0418	.0409	.0401
-1.6	.0548	.0537	.0526	.0516	.0505	.0495
-1.5	.0668	.0655	.0643	.0630	.0618	.0606
-1.4	.0808	.0793	.0778	.0764	.0749	.0735
-1.3	.0968	.0951	.0934	.0918	.0901	.0885
-1.2	.1151	.1131	.1112	.1093	.1075	.1056
-1.1	.1357	.1335	.1314	.1292	.1271	.1251
-1.0	.1587	.1562	.1539	.1515	.1492	.1469



$$P(Z < -1.14) = .1271$$

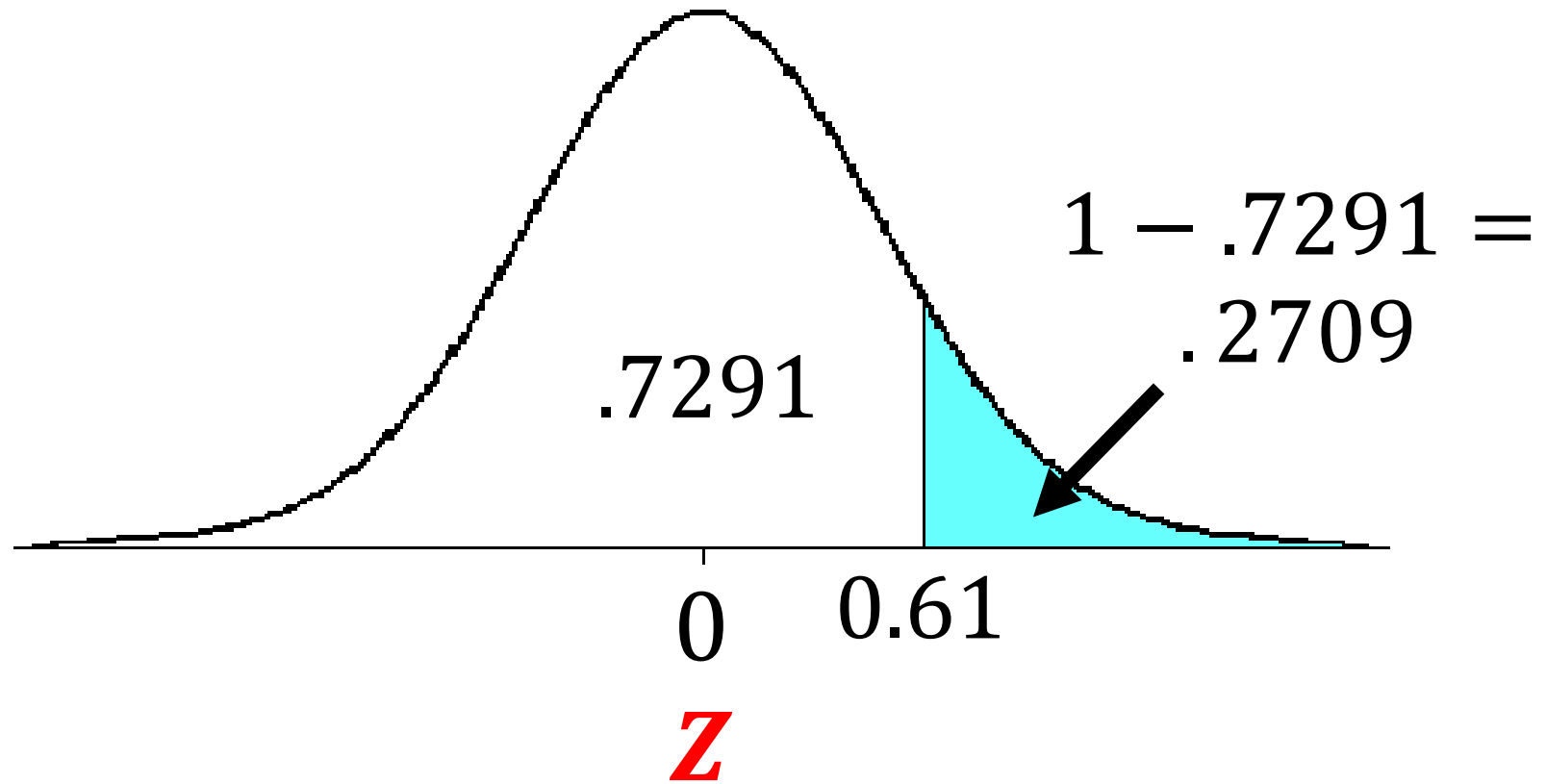


$$P(Z > 0.61) = ?$$

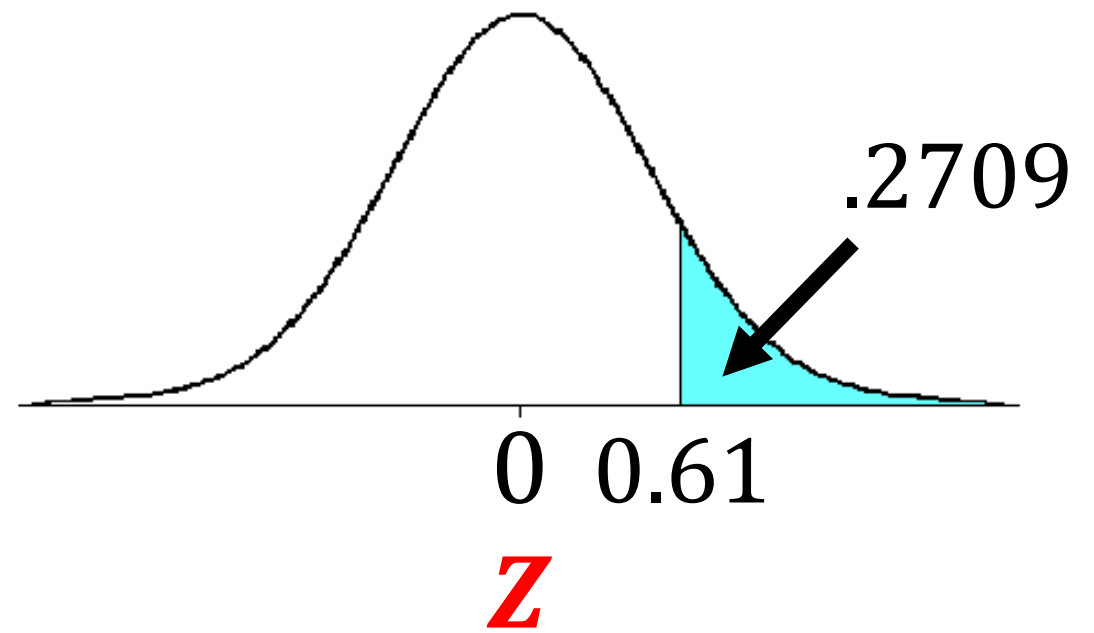
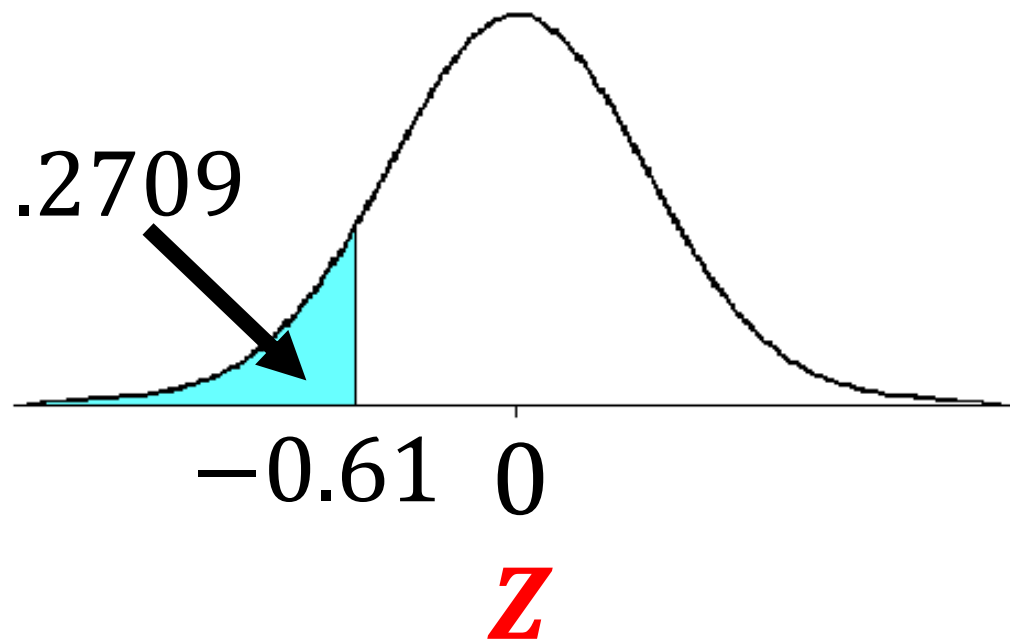
positive Z values



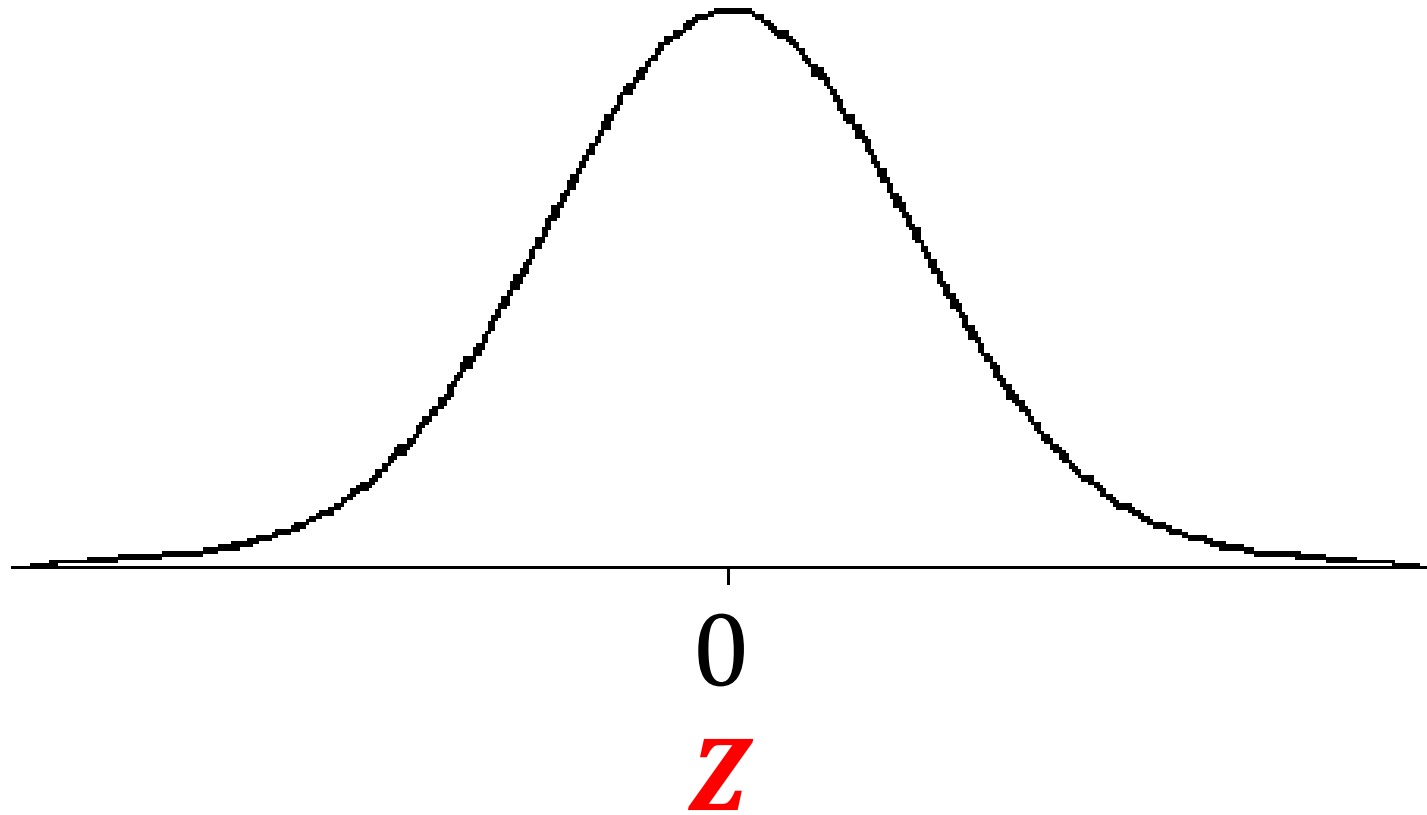
	0.00	0.01	0.02	0.03	0.04	0.05	0.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454
0.7	.7580	.7614	.7642	.7673	.7704	.7734	.7764
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406



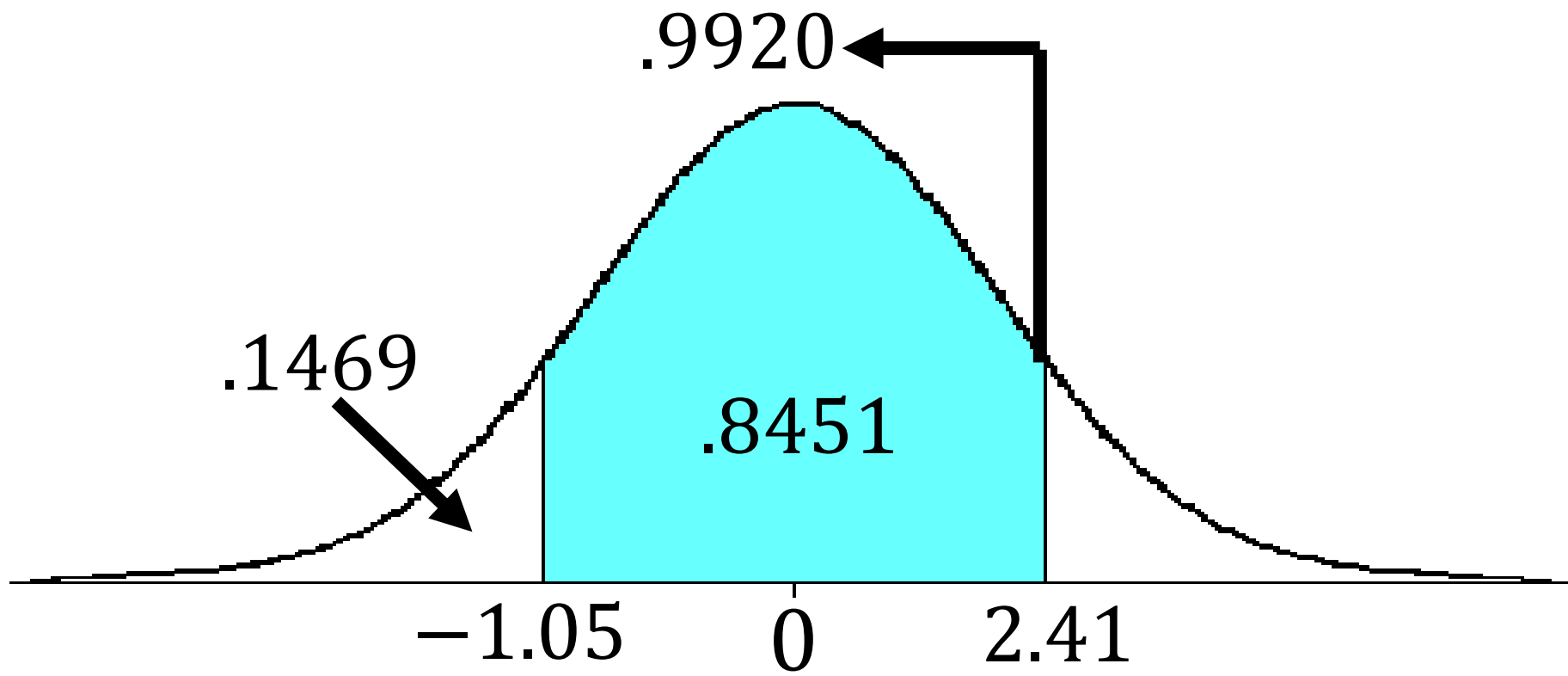
$$P(Z > 0.61) = .2709$$



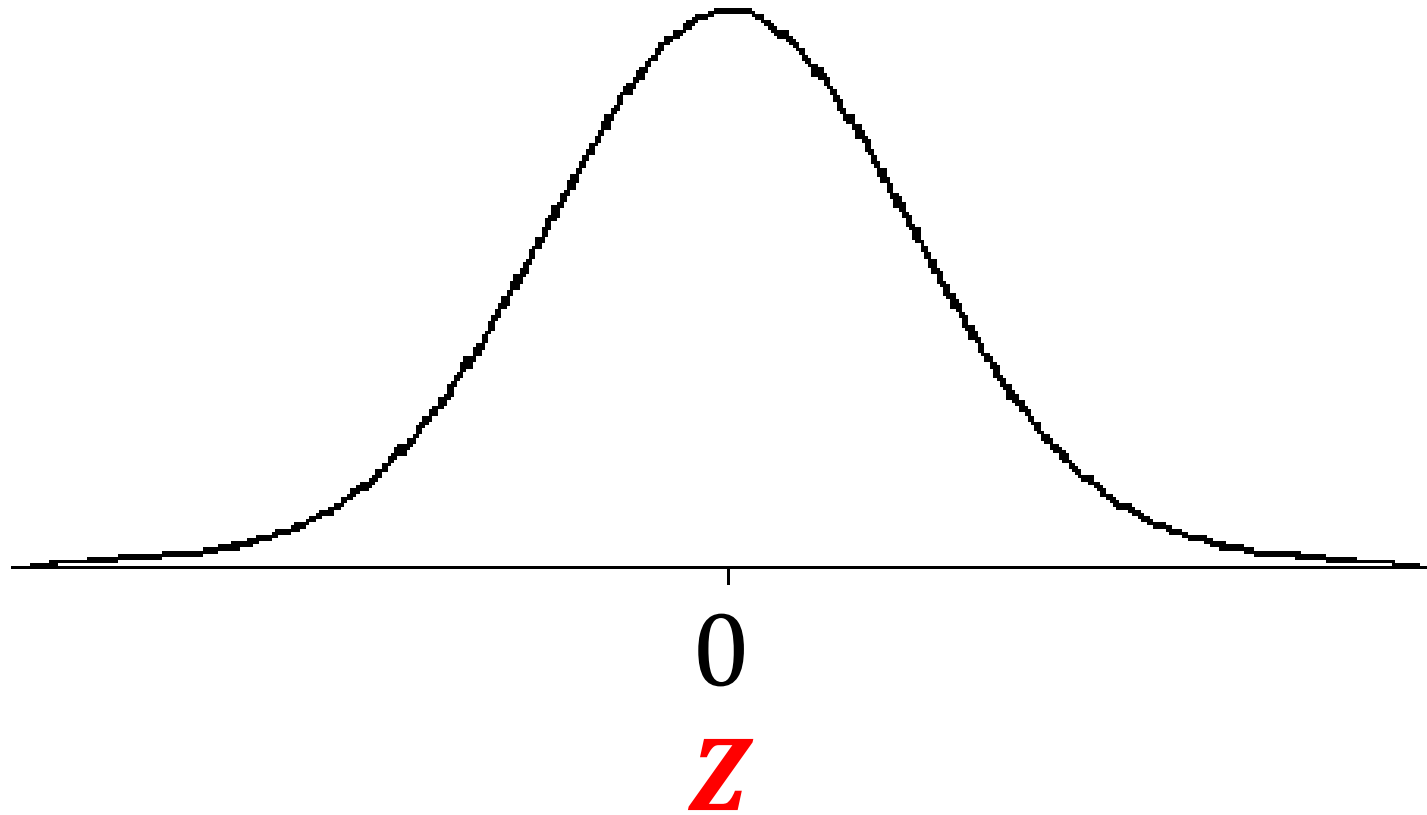
Due to symmetry, we know -0.61
must have an area to its left of $.2709$



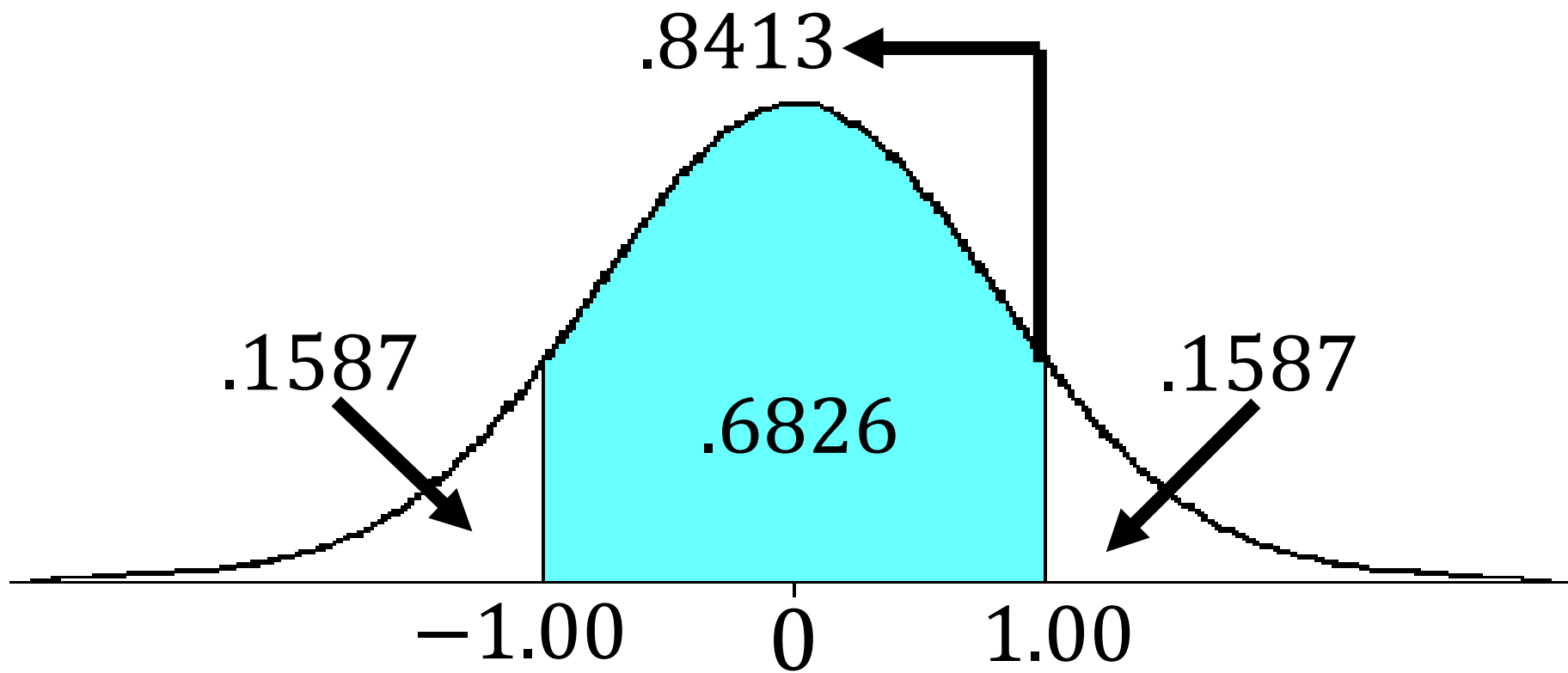
$$P(-1.05 < Z < 2.41) = ?$$



$$P(-1.05 < Z < 2.41) = .9920 - .1469 = .8451$$



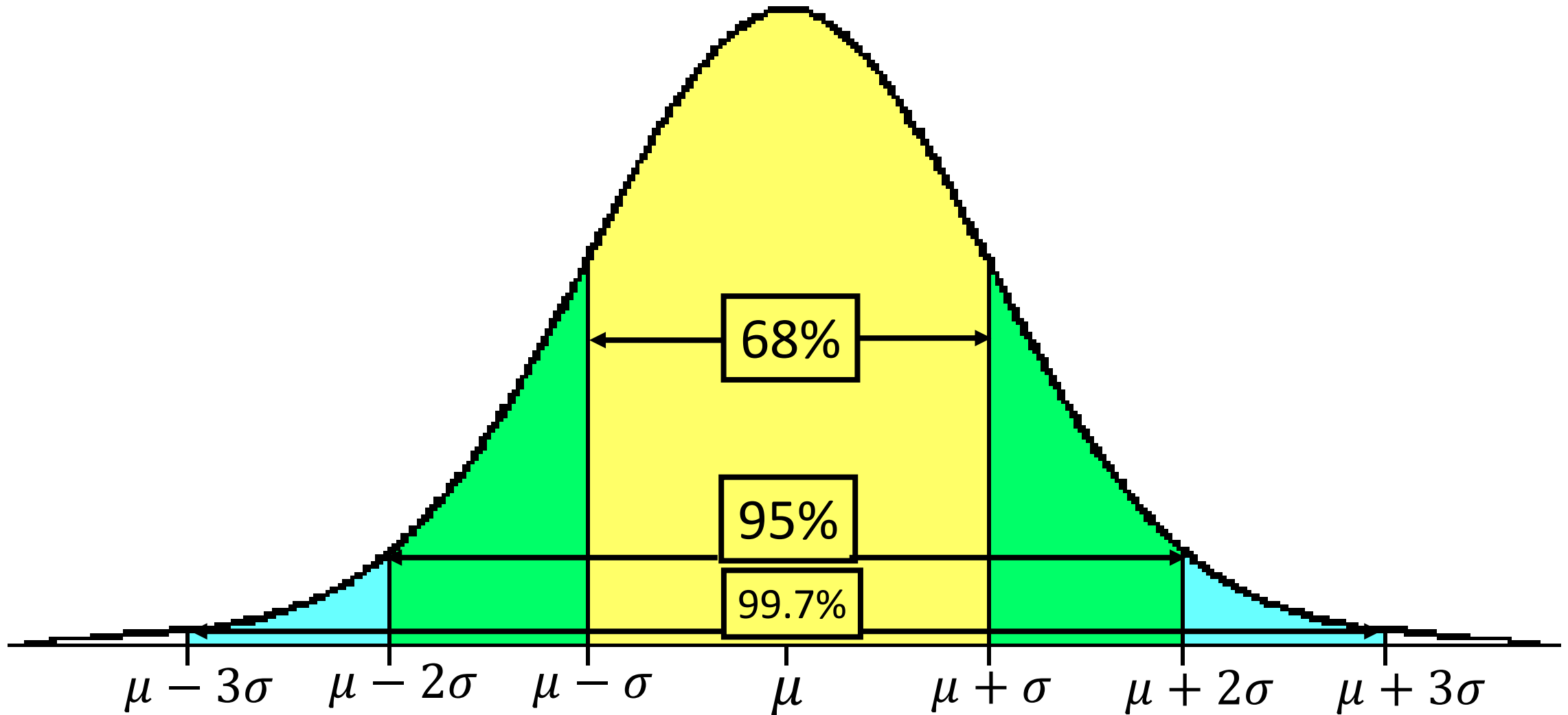
$$P(-1.00 < Z < 1.00) = ?$$



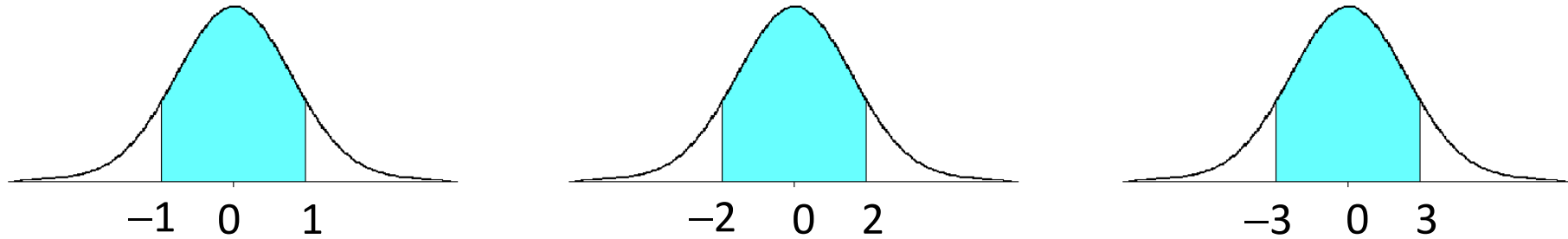
Z

$$\begin{aligned} P(-1.00 < Z < 1.00) &= \\ .8413 - .1587 &= .6826 = \\ 1 - 2(.1587) &= .6826 \end{aligned}$$

Empirical Rule for normal distributions



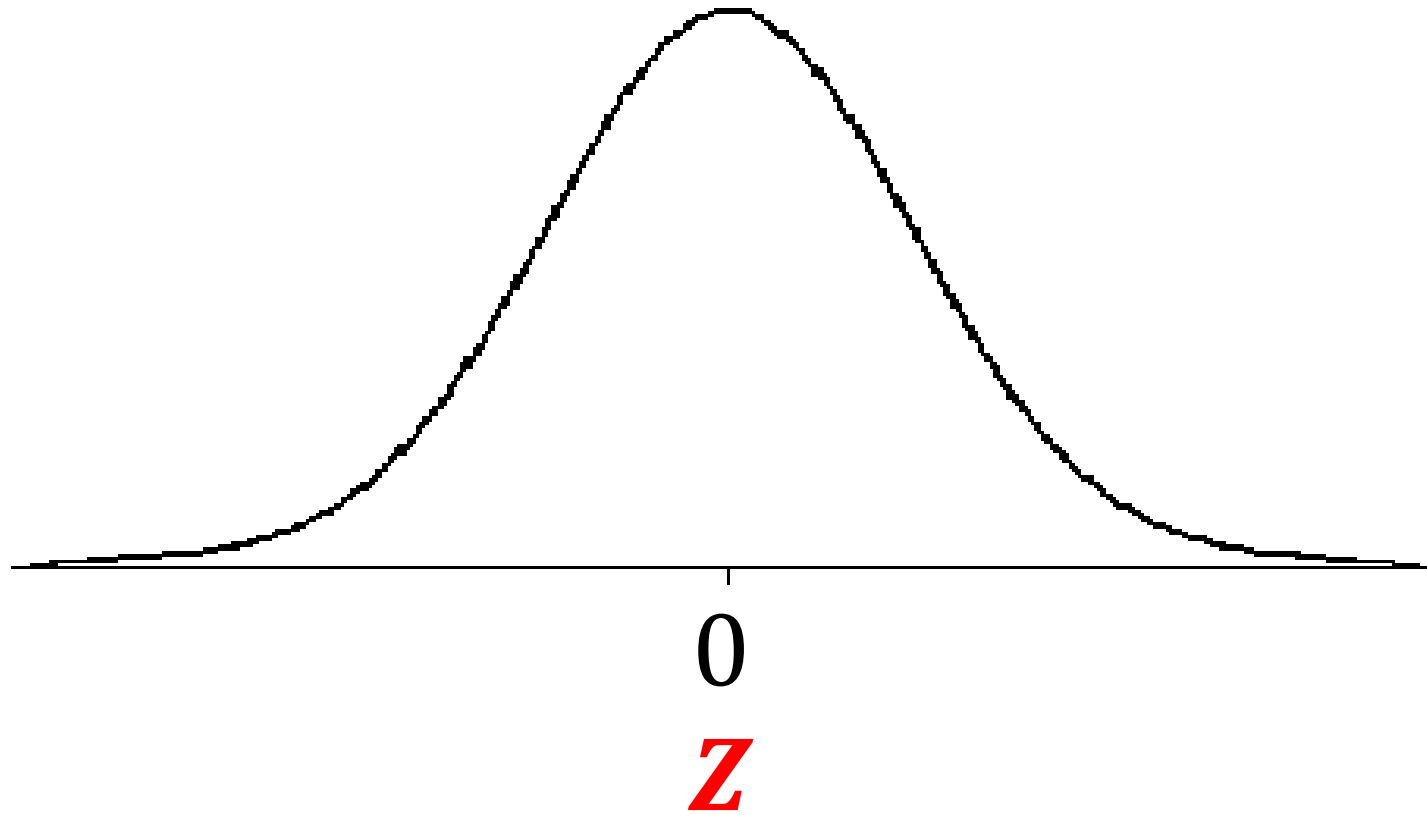
Empirical Rule for normal distributions



$$P(-1.00 < Z < 1.00) = .6826 \approx 68\%$$

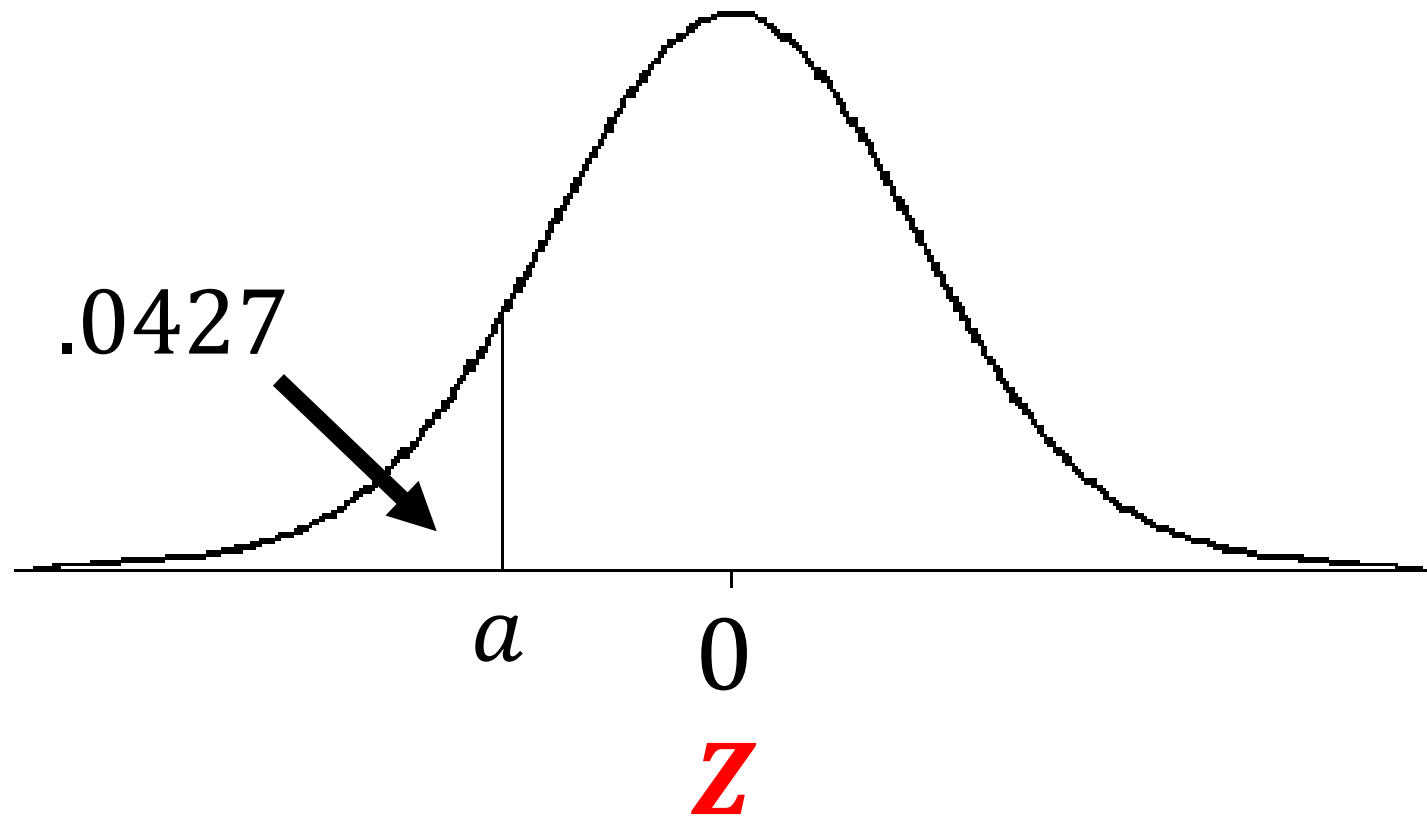
$$P(-2.00 < Z < 2.00) = .9544 \approx 95\%$$

$$P(-3.00 < Z < 3.00) = .9974 \approx 99.7\%$$



$$P(Z < a) = .0427$$

$$a = ?$$

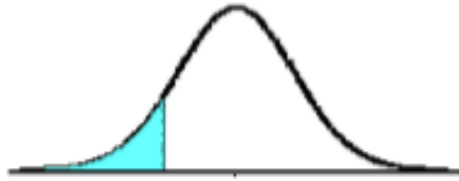


$$P(Z < a) = .0427$$

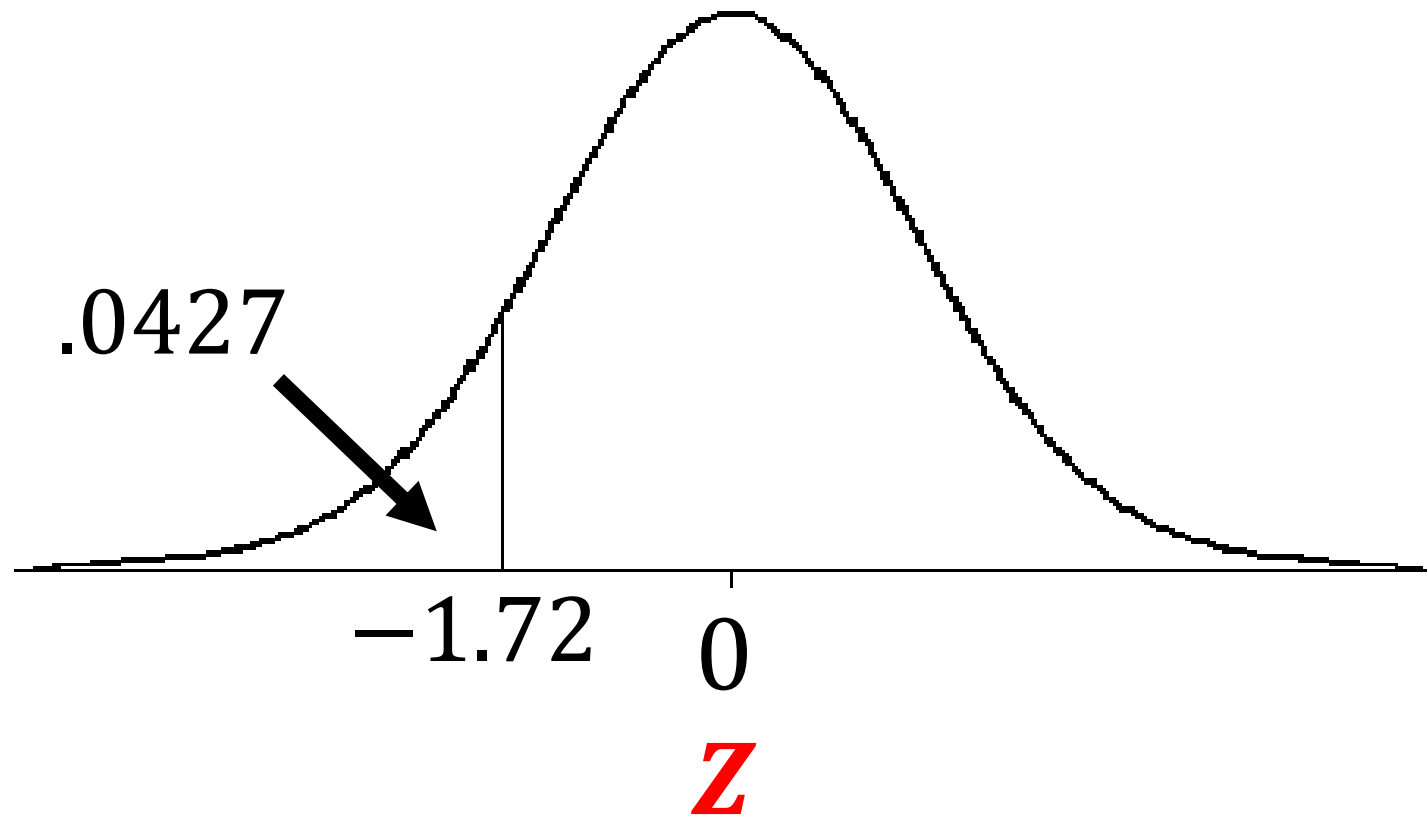
$$a = ?$$

Z-Table

negative Z values

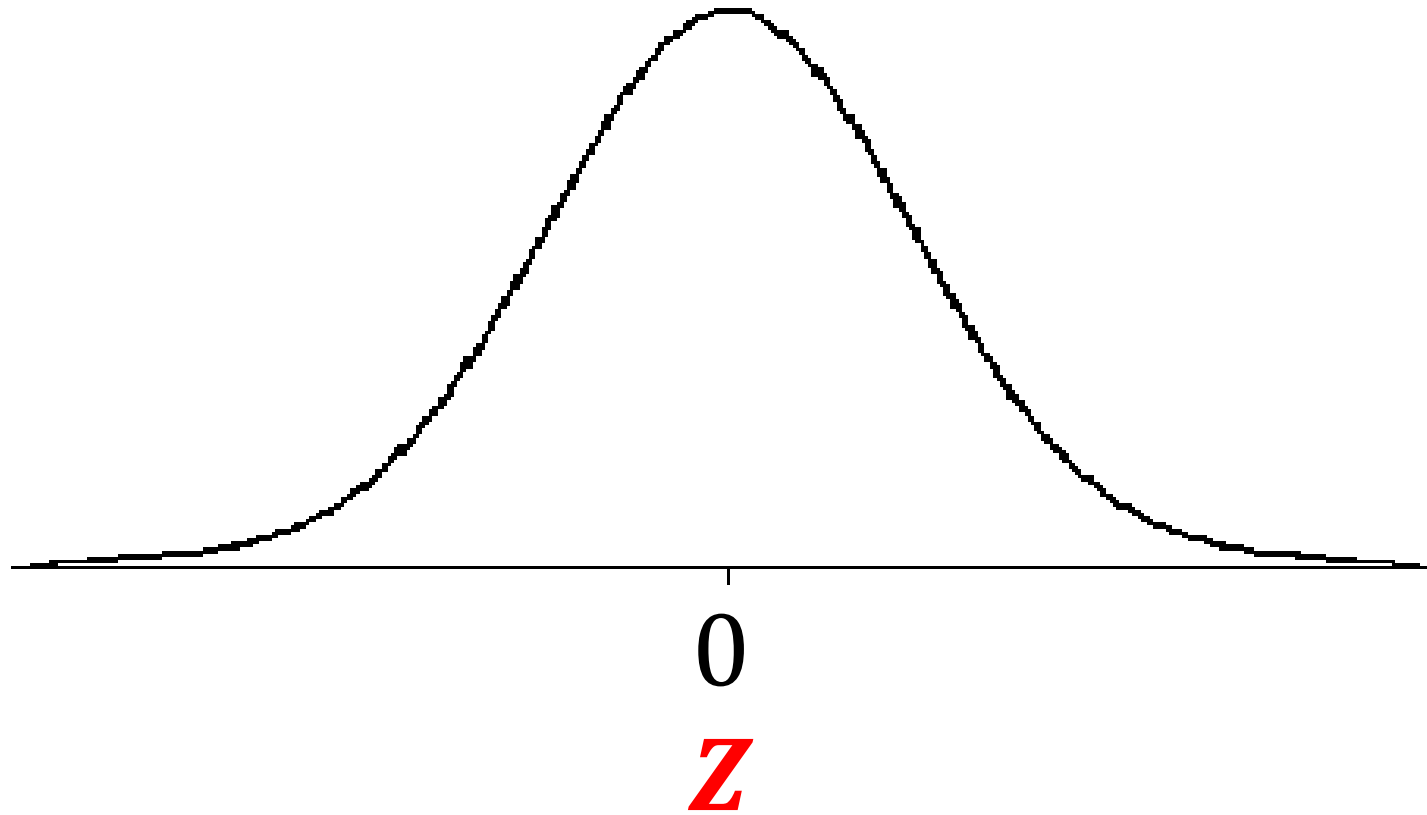


	0.00	0.01	0.02	0.03	0.04	0.05
-3.0	.0013	.0013	.0013	.0012	.0012	.0011
-2.9	.0019	.0018	.0018	.0017	.0016	.0016
-2.8	.0026	.0025	.0024	.0023	.0023	.0022
-2.7	.0035	.0034	.0033	.0032	.0031	.0030
-2.6	.0047	.0045	.0044	.0043	.0041	.0040
-2.5	.0062	.0060	.0059	.0057	.0055	.0054
-2.4	.0082	.0080	.0078	.0075	.0073	.0071
-2.3	.0107	.0104	.0102	.0099	.0096	.0094
-2.2	.0139	.0136	.0132	.0129	.0125	.0122
-2.1	.0179	.0174	.0170	.0166	.0162	.0158
-2.0	.0228	.0222	.0217	.0212	.0207	.0202
-1.9	.0287	.0281	.0274	.0268	.0262	.0256
-1.8	.0359	.0351	.0344	.0336	.0329	.0322
-1.7	.0446	.0438	.0427	.0418	.0409	.0401
-1.6	.0548	.0537	.0526	.0516	.0505	.0495
-1.5	.0668	.0655	.0643	.0630	.0618	.0606
-1.4	.0808	.0793	.0778	.0764	.0749	.0735
-1.3	.0968	.0951	.0934	.0918	.0901	.0885
-1.2	.1151	.1131	.1112	.1093	.1075	.1056
-1.1	.1357	.1335	.1314	.1292	.1271	.1251
-1.0	.1587	.1562	.1539	.1515	.1492	.1469

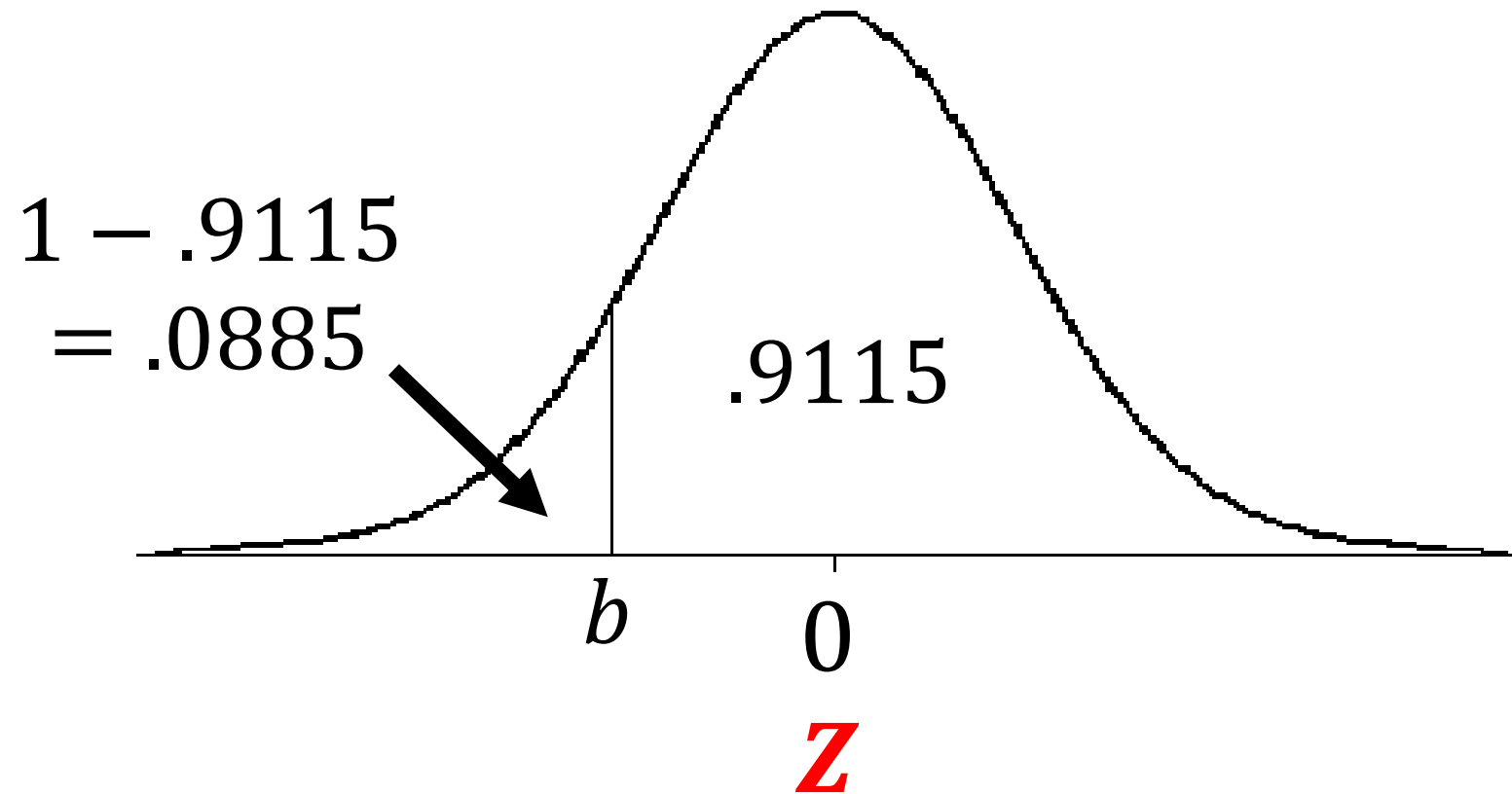


$$P(Z < a) = .0427$$

$$a = -1.72$$



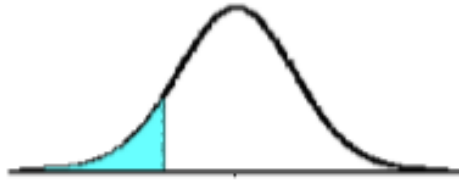
$$P(Z > b) = .9115$$
$$b = ?$$



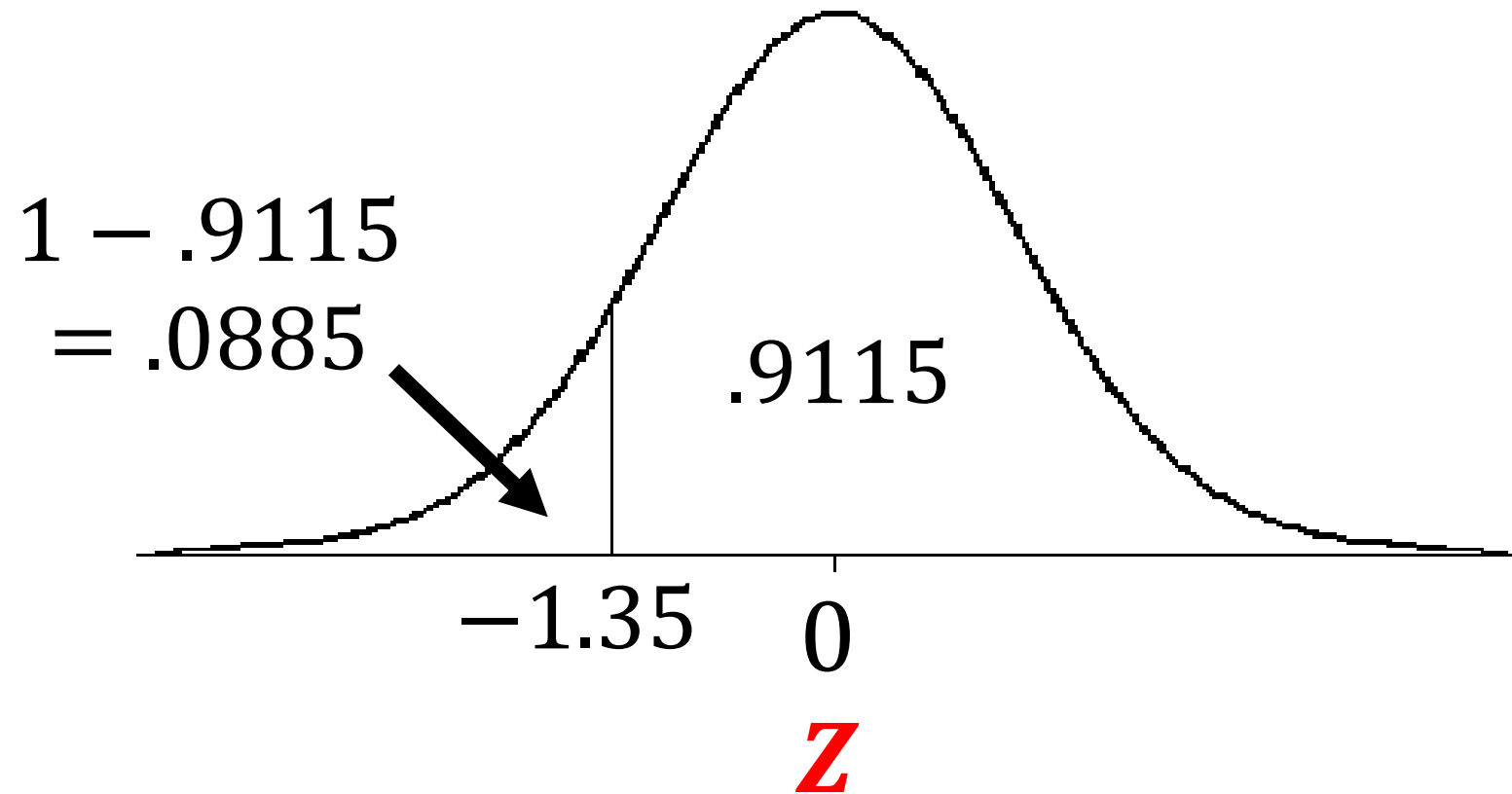
$$P(Z > b) = .9115$$
$$b = ?$$

Z-Table

negative Z values

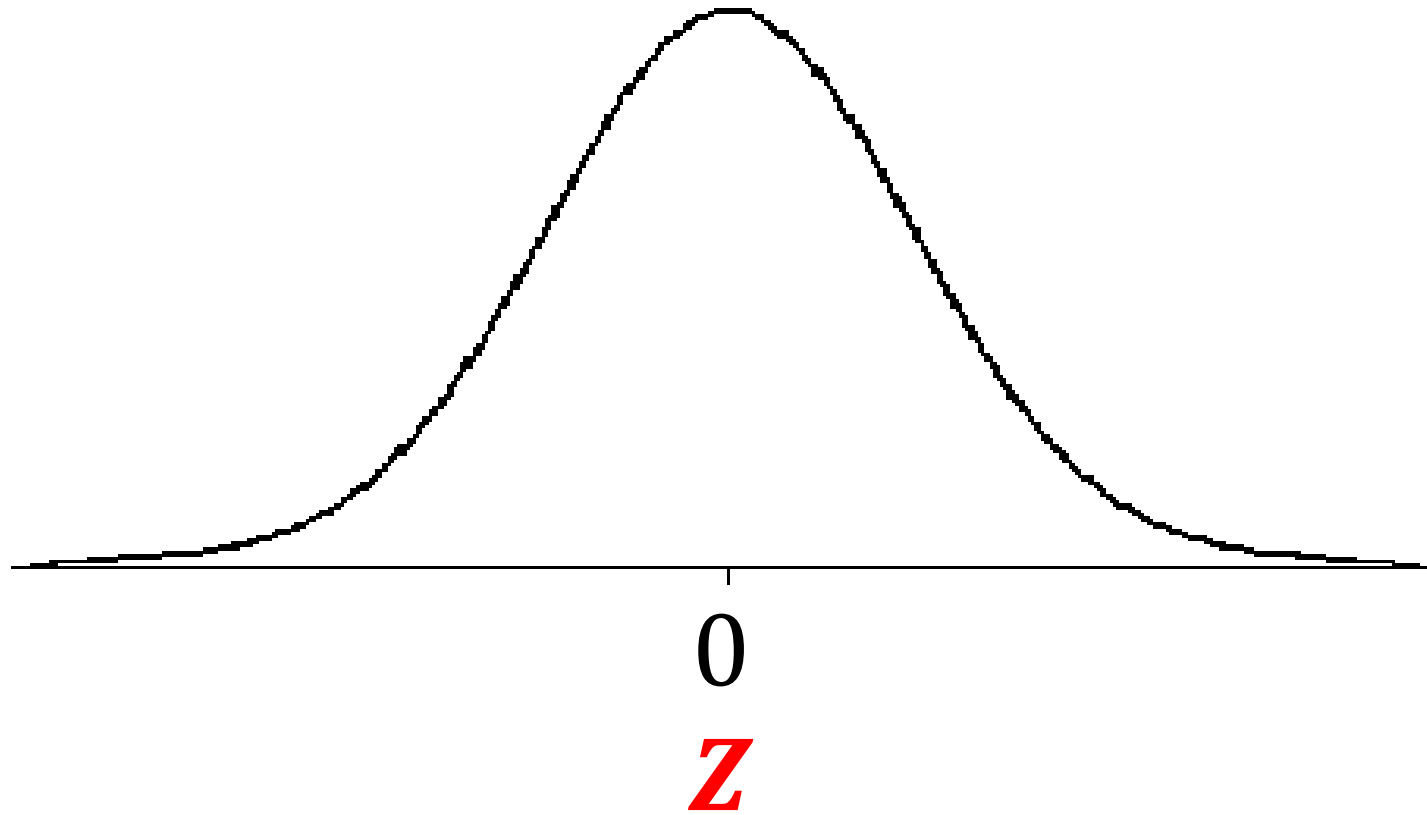


	0.00	0.01	0.02	0.03	0.04	0.05
-3.0	.0013	.0013	.0013	.0012	.0012	.0011
-2.9	.0019	.0018	.0018	.0017	.0016	.0016
-2.8	.0026	.0025	.0024	.0023	.0023	.0022
-2.7	.0035	.0034	.0033	.0032	.0031	.0030
-2.6	.0047	.0045	.0044	.0043	.0041	.0040
-2.5	.0062	.0060	.0059	.0057	.0055	.0054
-2.4	.0082	.0080	.0078	.0075	.0073	.0071
-2.3	.0107	.0104	.0102	.0099	.0096	.0094
-2.2	.0139	.0136	.0132	.0129	.0125	.0122
-2.1	.0179	.0174	.0170	.0166	.0162	.0158
-2.0	.0228	.0222	.0217	.0212	.0207	.0202
-1.9	.0287	.0281	.0274	.0268	.0262	.0256
-1.8	.0359	.0351	.0344	.0336	.0329	.0322
-1.7	.0446	.0436	.0427	.0418	.0409	.0401
-1.6	.0548	.0537	.0526	.0516	.0505	.0495
-1.5	.0668	.0655	.0643	.0630	.0618	.0606
-1.4	.0808	.0793	.0778	.0764	.0749	.0735
-1.3	.0968	.0951	.0934	.0918	.0901	.0885
-1.2	.1151	.1131	.1112	.1093	.1075	.1056
-1.1	.1357	.1335	.1314	.1292	.1271	.1251
-1.0	.1587	.1562	.1539	.1515	.1492	.1469



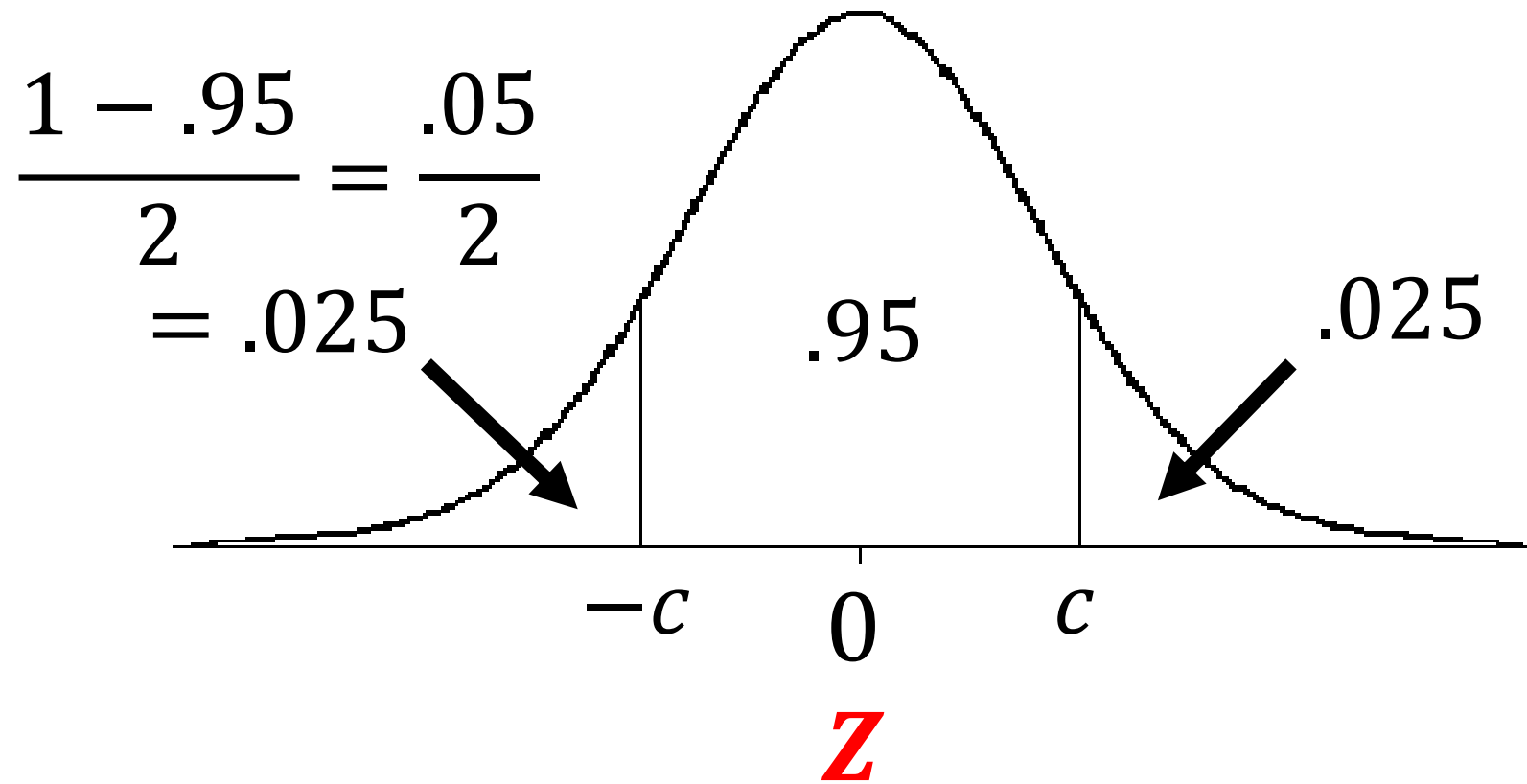
$$P(Z > b) = .9115$$

$$b = -1.35$$



$$P(-c < Z < c) = .95$$

$$c = ?$$

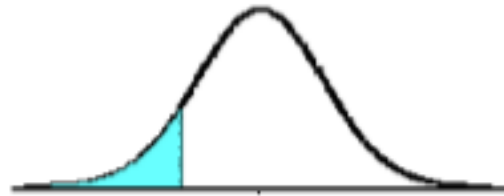


$$P(-c < Z < c) = .95$$

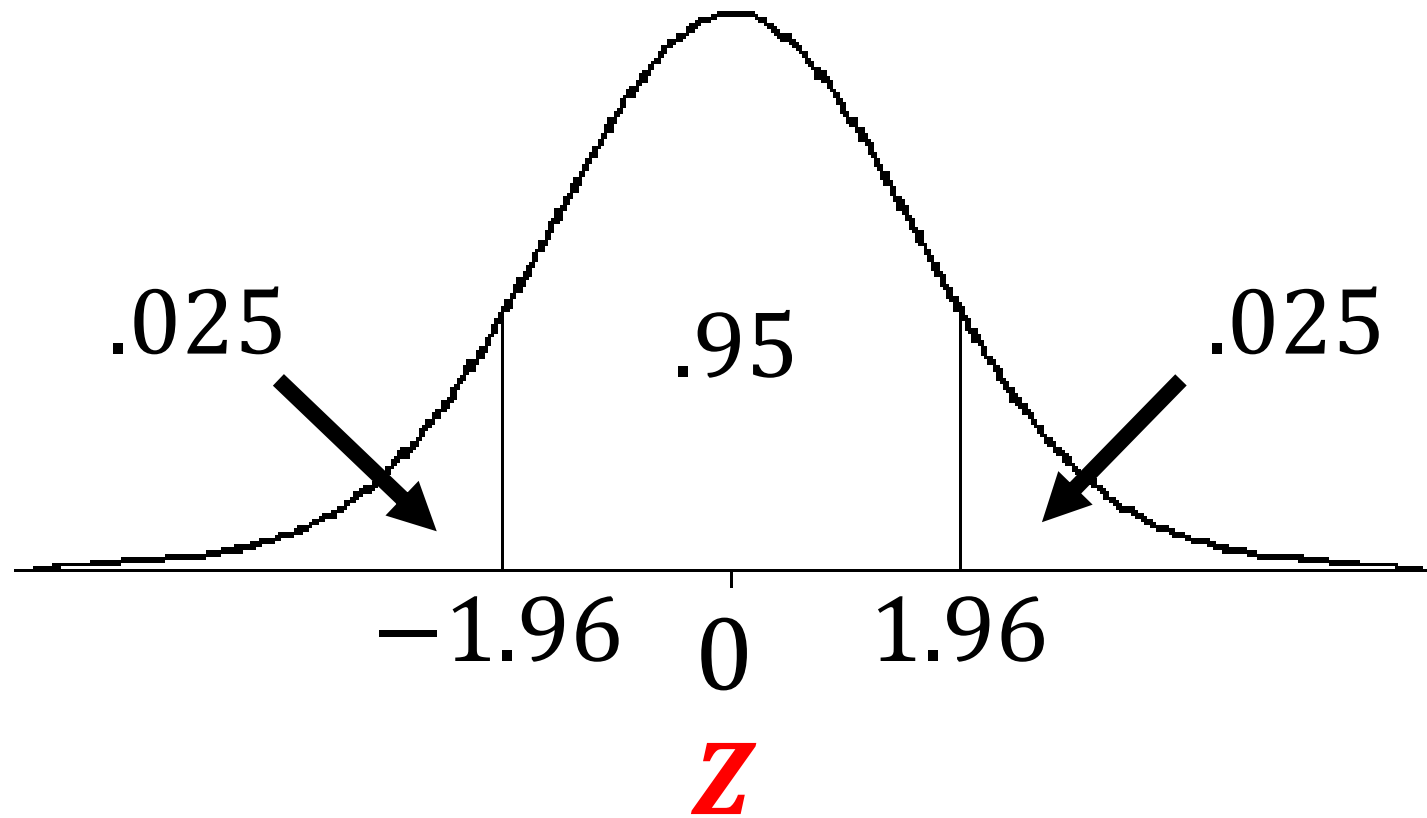
$$c = ?$$

Z-Table

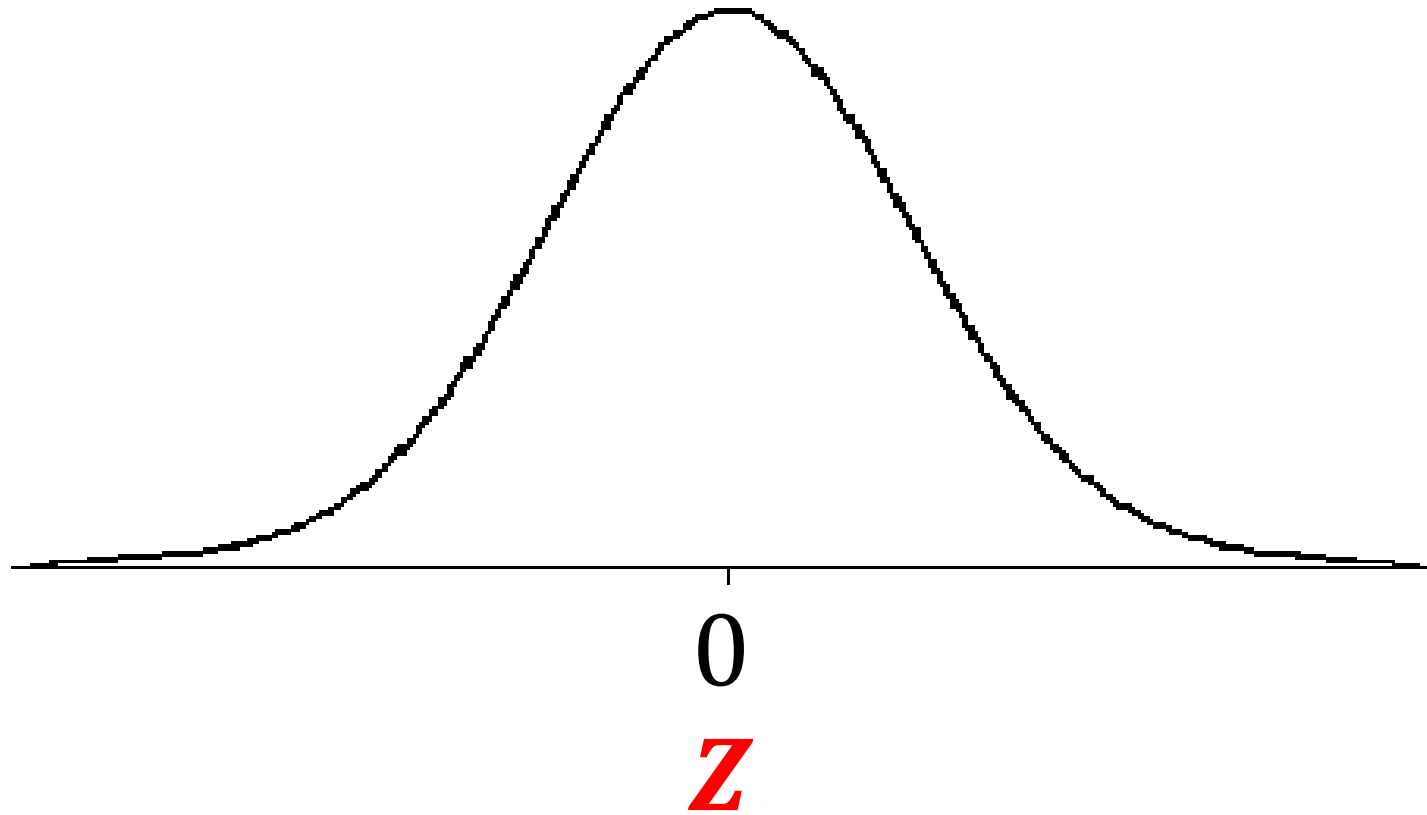
negative Z values



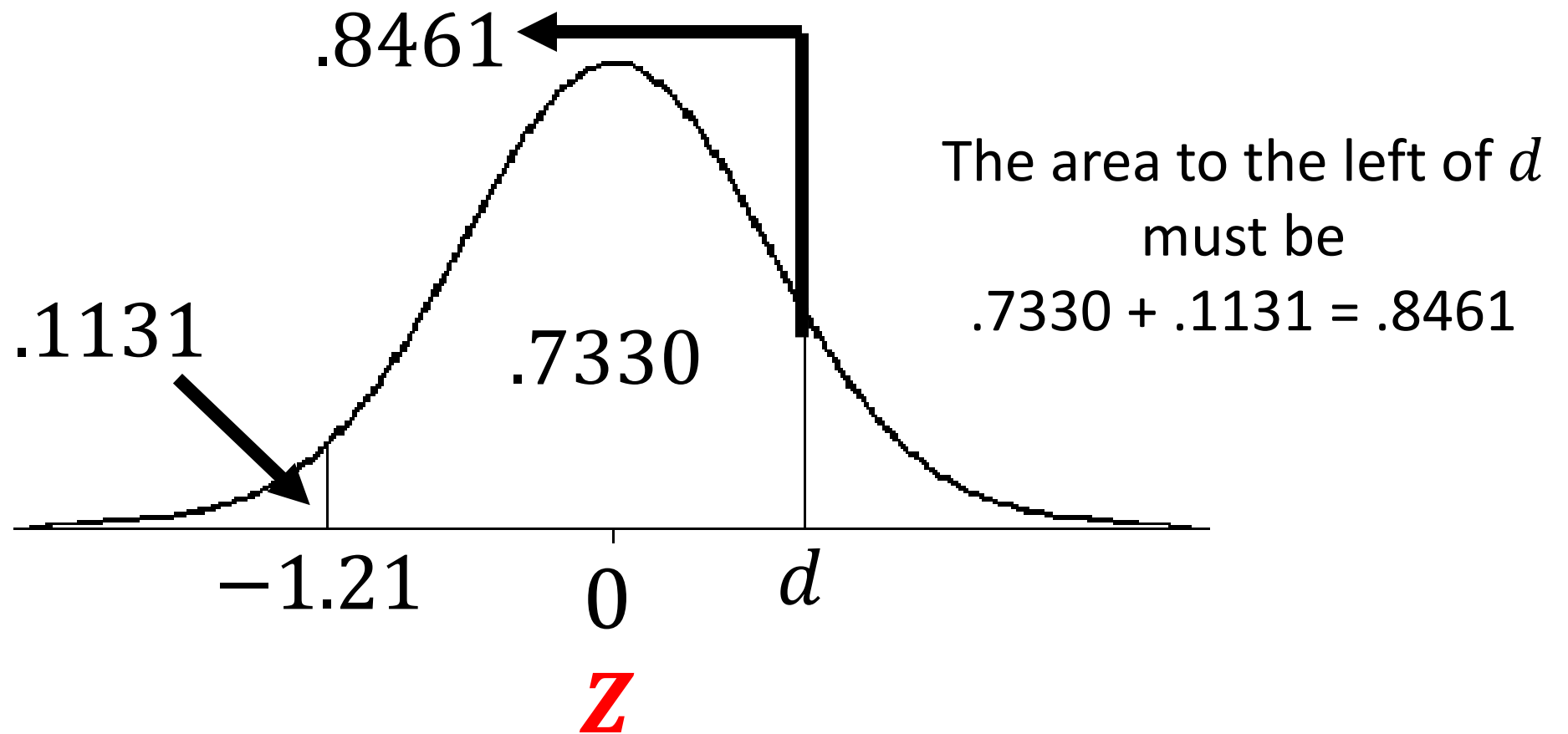
	0.00	0.01	0.02	0.03	0.04	0.05	0.06
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197
-1.9	.0287	.0284	.0274	.0269	.0262	.0256	.0250
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594



$$P(-c < Z < c) = .95$$
$$c = 1.96$$



$$P(-1.21 < Z < d) = .7330$$
$$d = ?$$



$$P(-1.21 < Z < d) = .7330$$

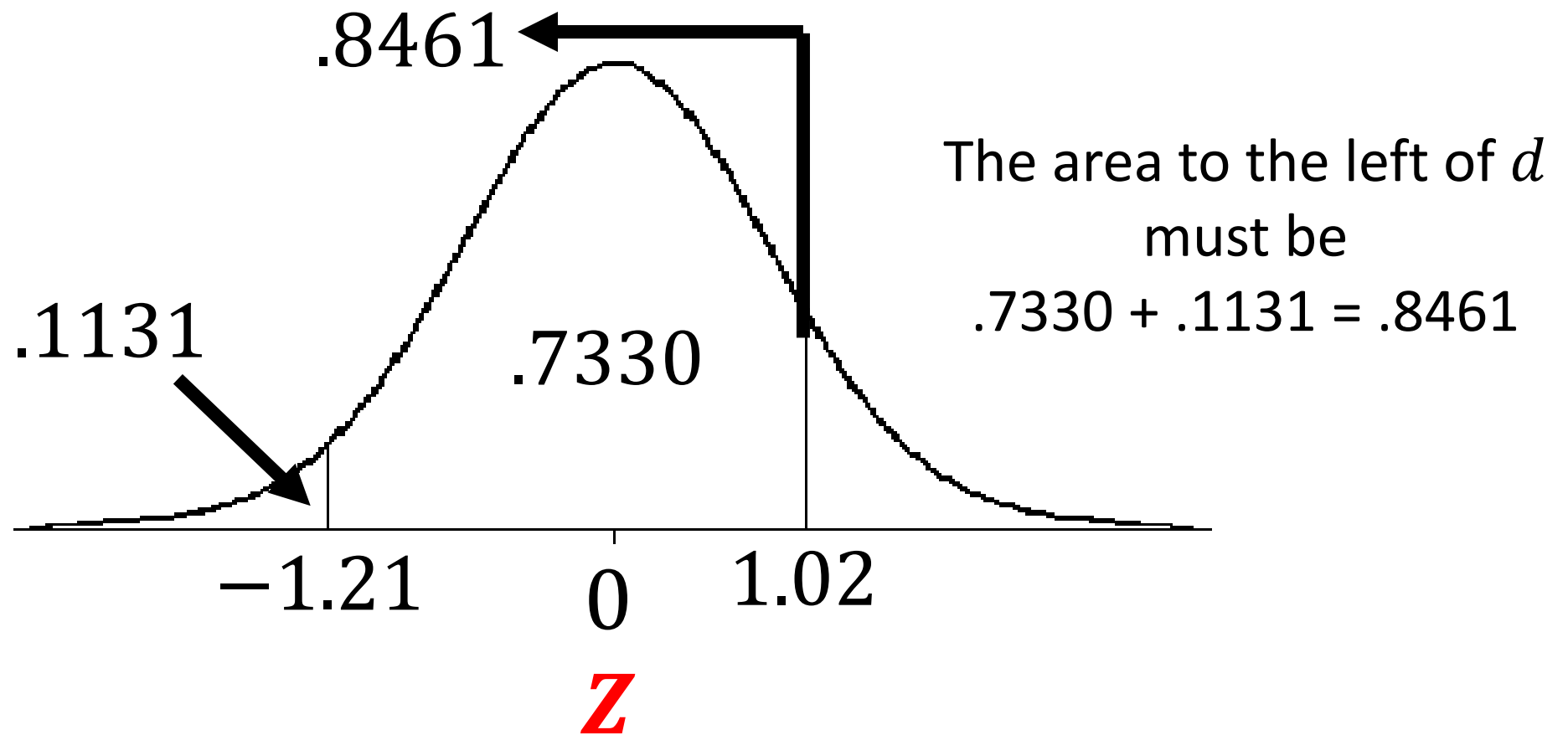
$$d = ?$$

Z-Table

positive Z values

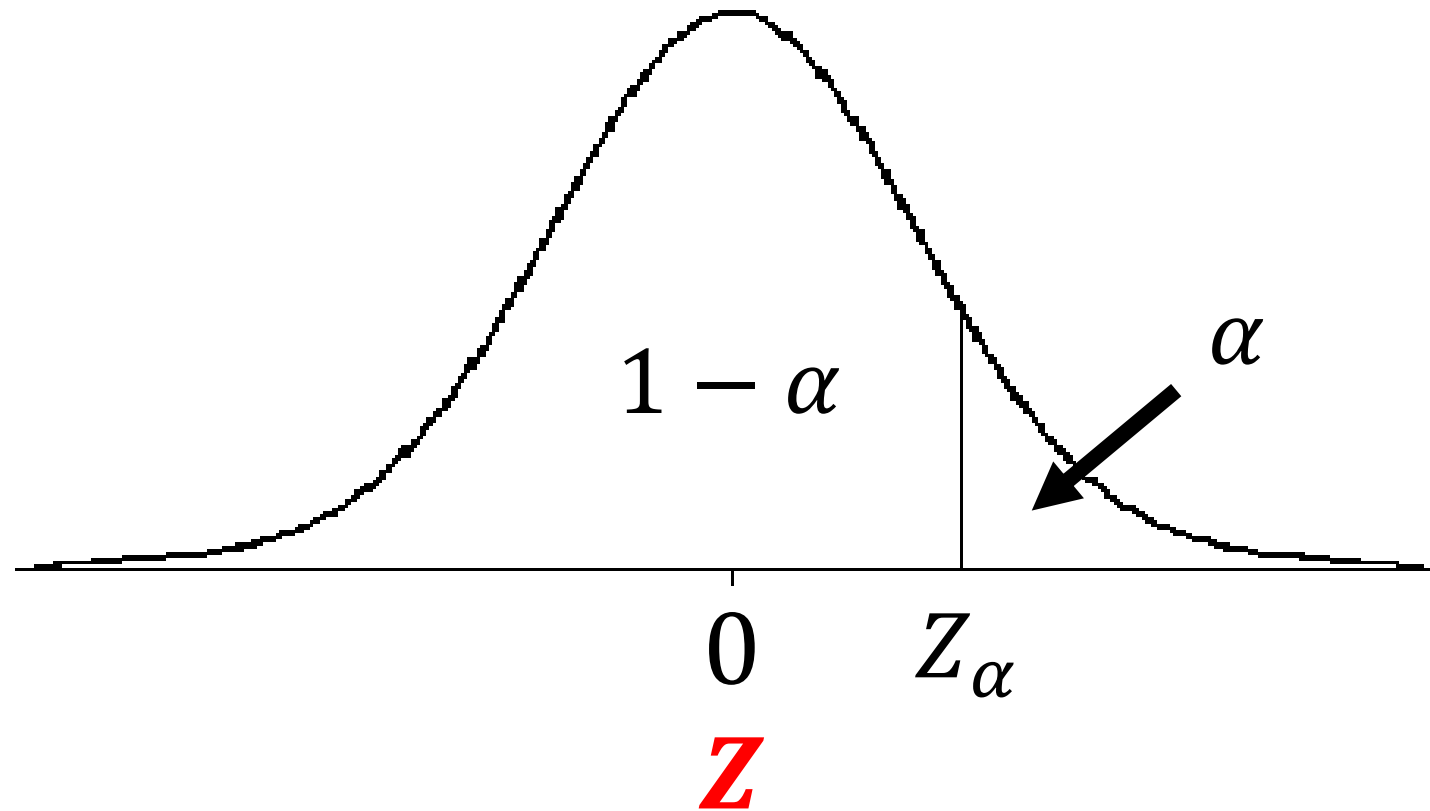


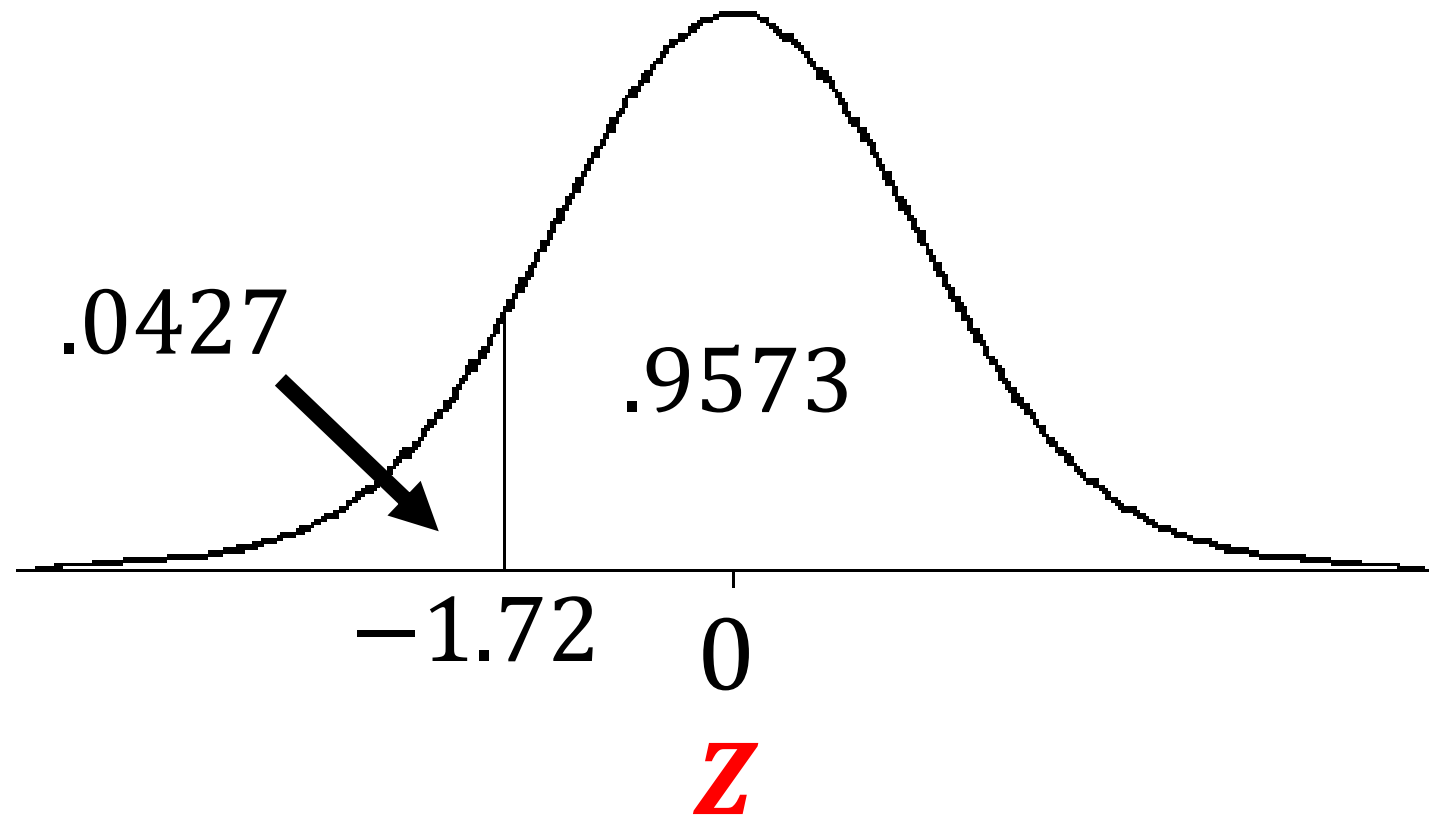
	0.00	0.01	0.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255
0.4	.6554	.6591	.6628
0.5	.6915	.6950	.6985
0.6	.7257	.7291	.7324
0.7	.7580	.7611	.7642
0.8	.7881	.7910	.7939
0.9	.8159	.8186	.8212
1.0	.8413	.8438	.8461



$$P(-1.21 < Z < d) = .7330$$
$$d = 1.02$$

Z_α = the z value with α area to its right

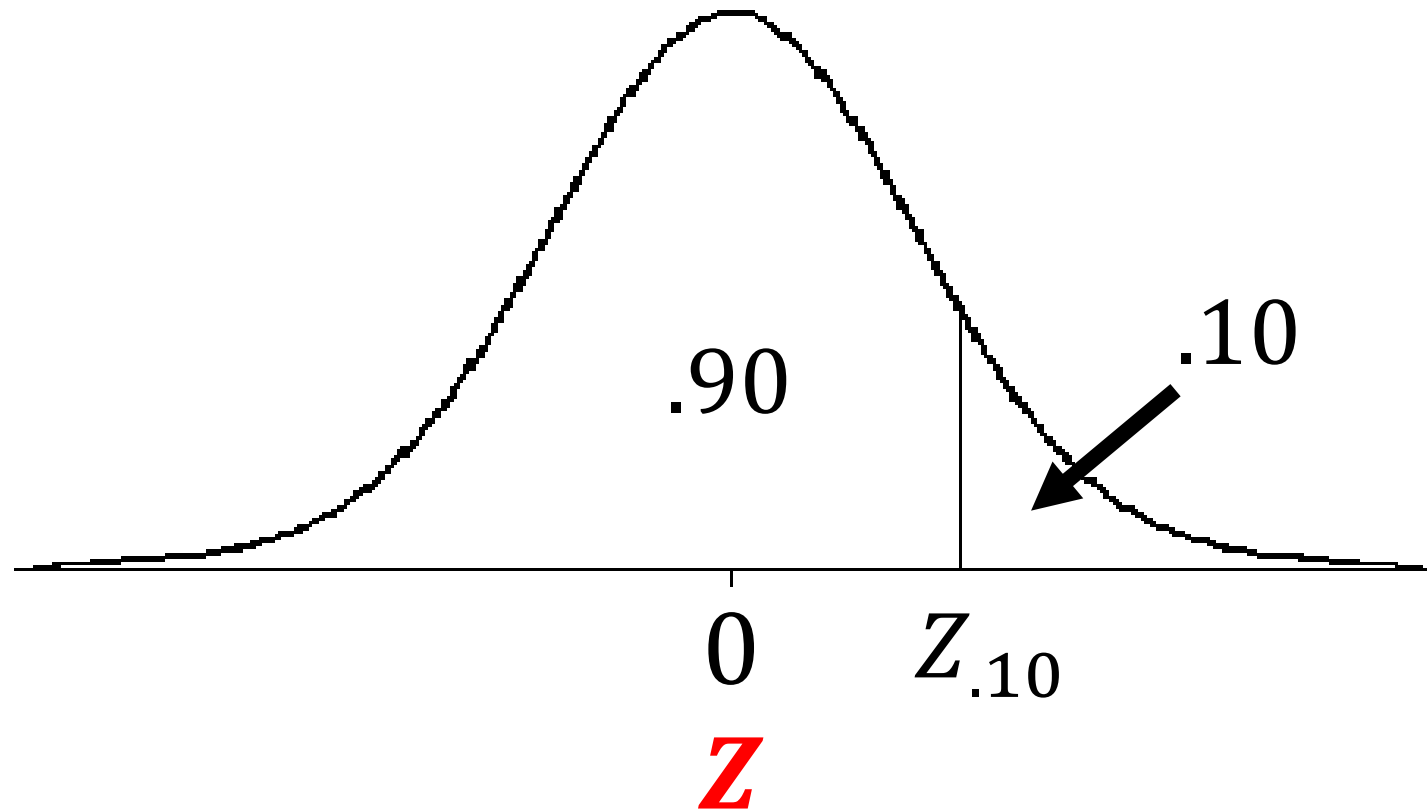




$$-1.72 = Z_{.9573}$$

What is the 90th percentile on the Z curve?

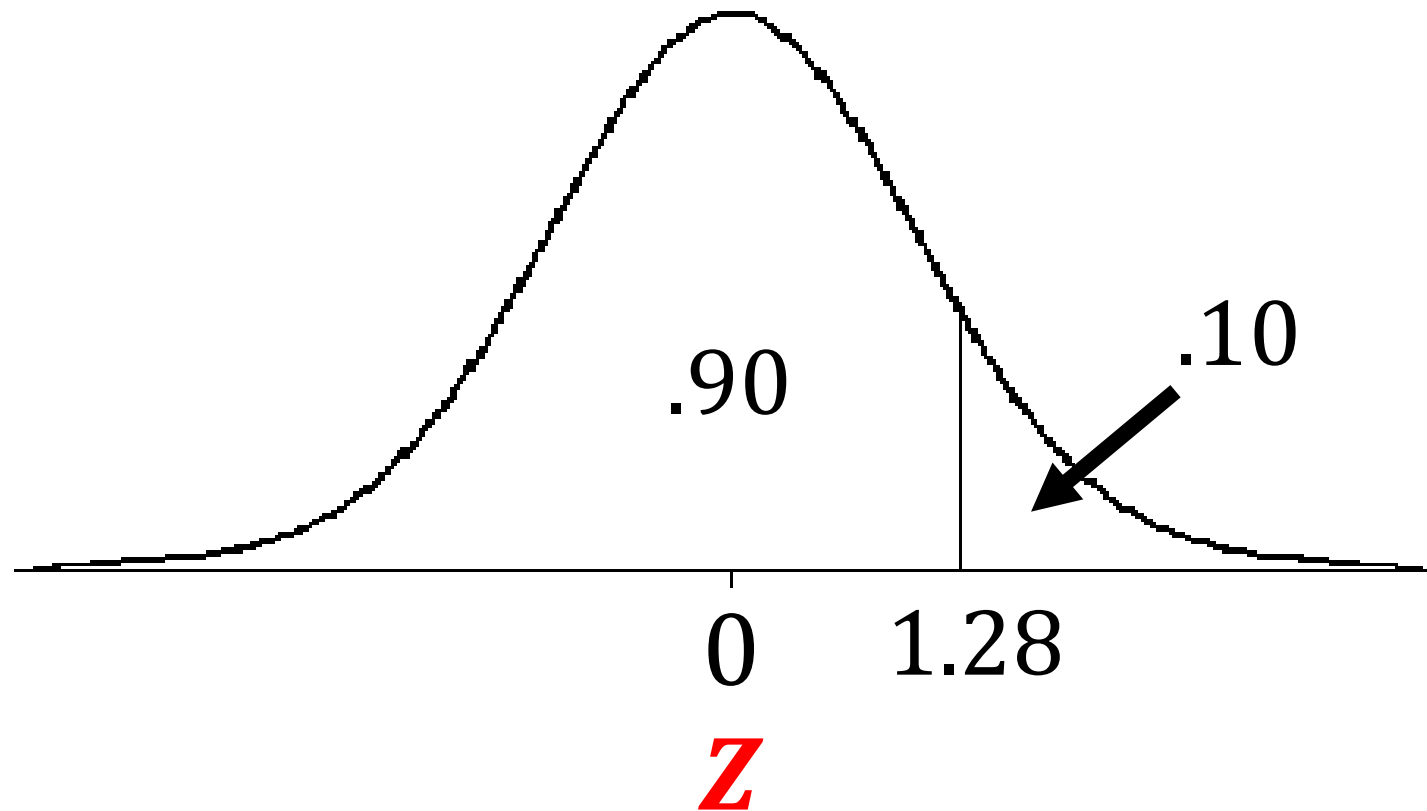
The 90th percentile would have an area to its left equal to .90

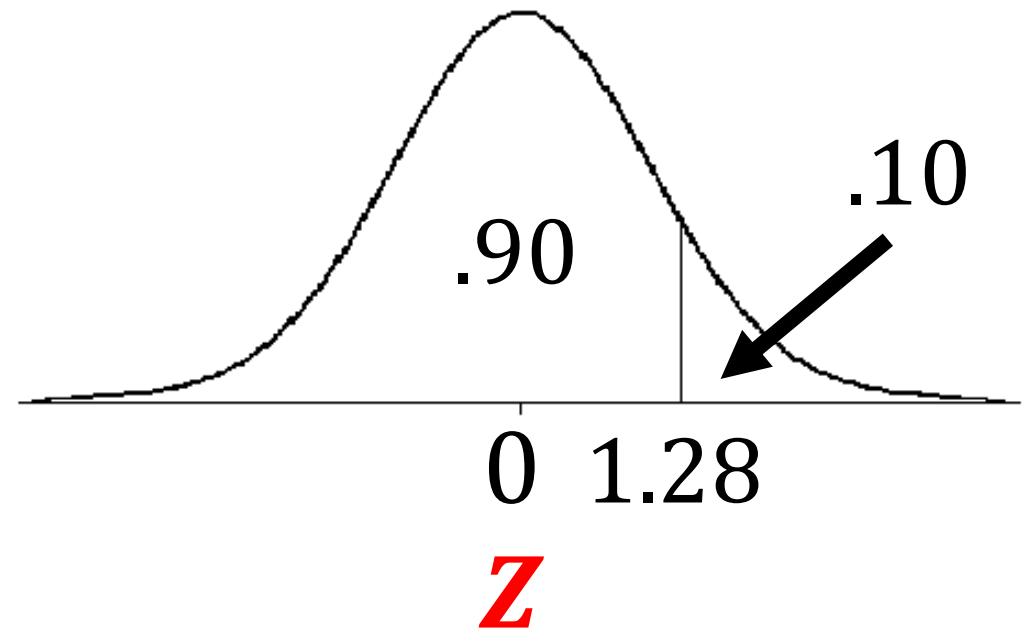
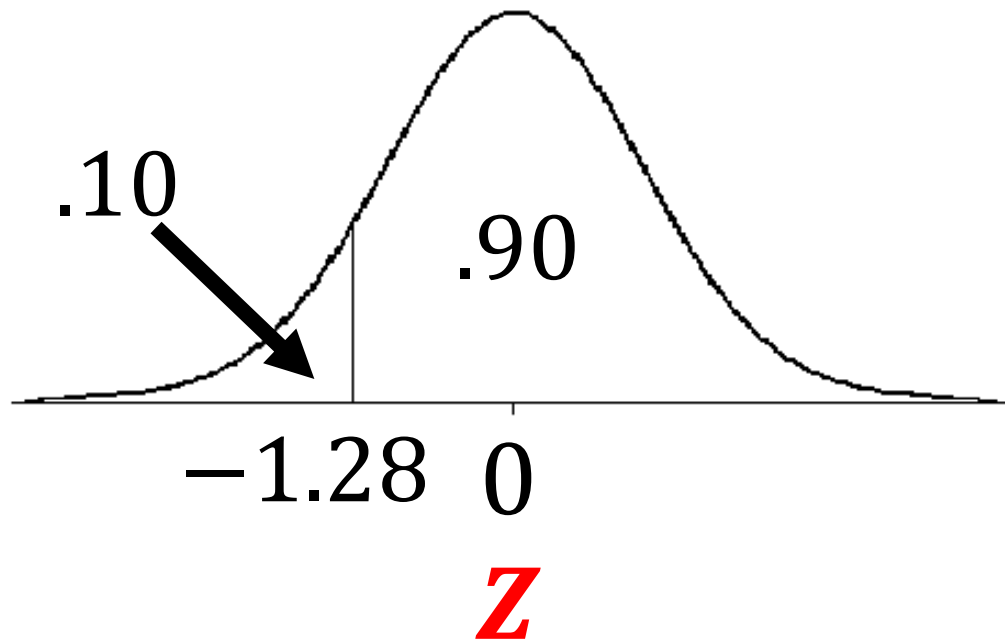


If you can't find the exact area you are looking for, it is okay to use the closest value.

	0.06	0.07	0.08	0.09
.5239	.5239	.5279	.5319	.5359
.5636	.5636	.5675	.5714	.5753
.6026	.6026	.6064	.6103	.6141
.6406	.6406	.6443	.6480	.6517
.6772	.6772	.6808	.6844	.6879
.7123	.7123	.7157	.7190	.7224
.7454	.7454	.7486	.7517	.7549
.7764	.7764	.7794	.7823	.7852
.8051	.8051	.8078	.8106	.8133
.8315	.8315	.8340	.8365	.8389
.8554	.8554	.8577	.8599	.8621
.8770	.8770	.8790	.8810	.8830
.8997	.8997	.8999	.8997	.9015
.9131	.9131	.9147	.9162	.9177
.9279	.9279	.9292	.9306	.9319
.9406	.9406	.9418	.9429	

90th percentile = 1.28

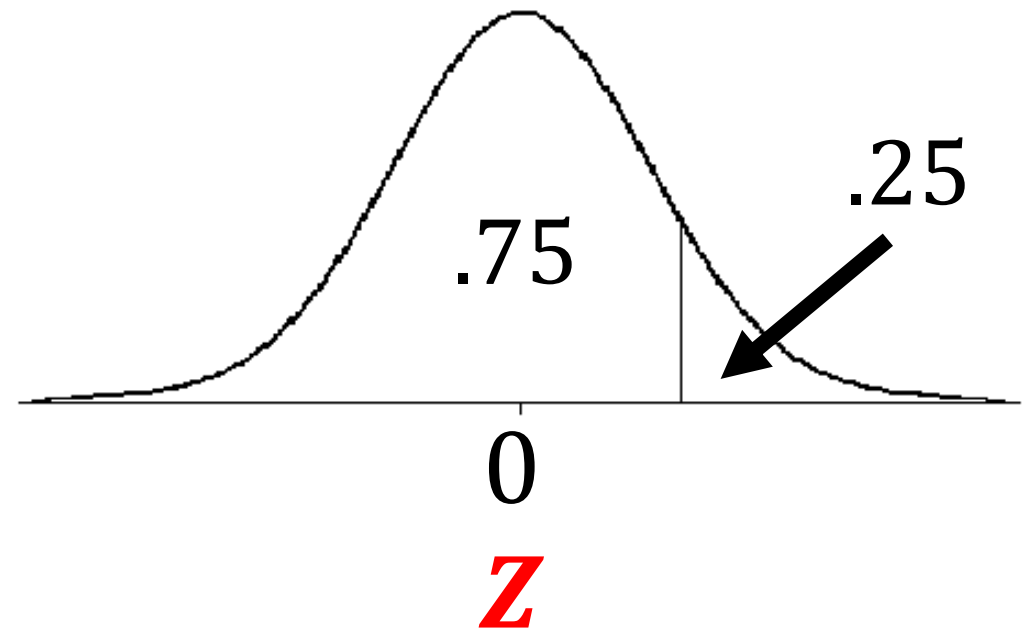
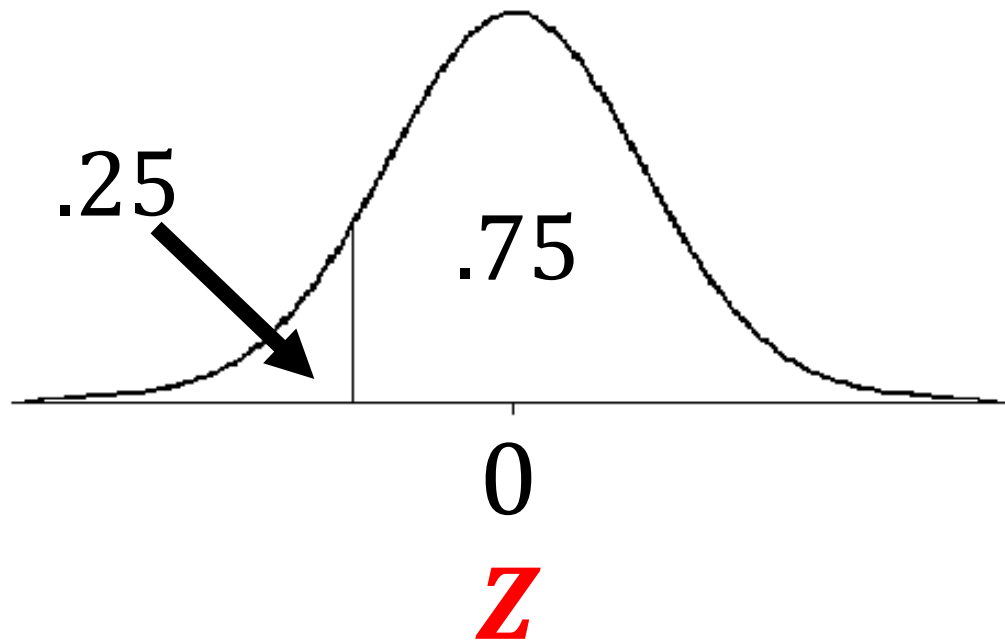




Due to symmetry, we know -1.28 is the 10th percentile.

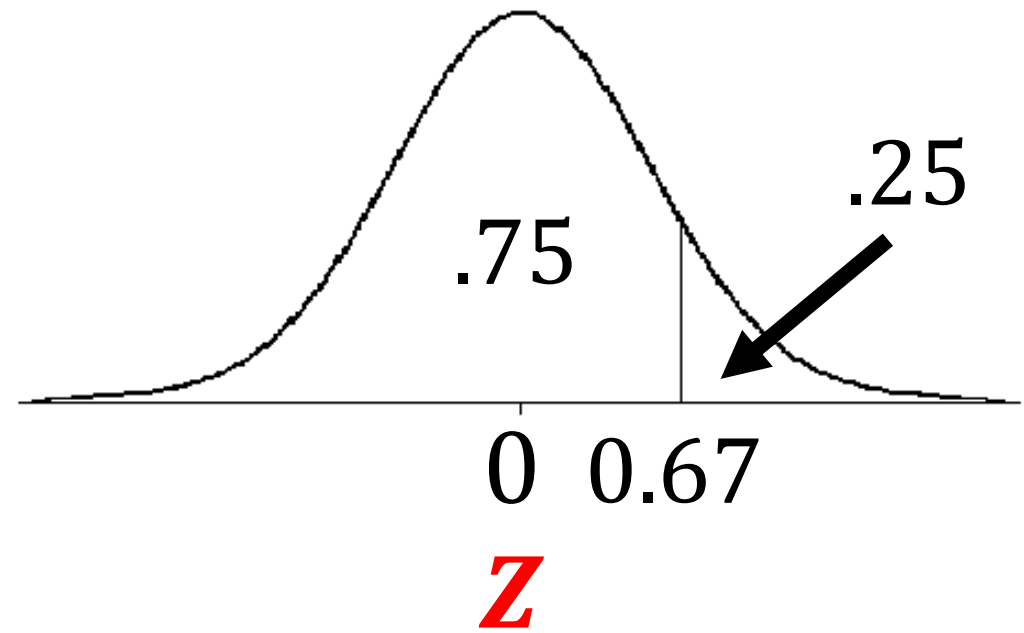
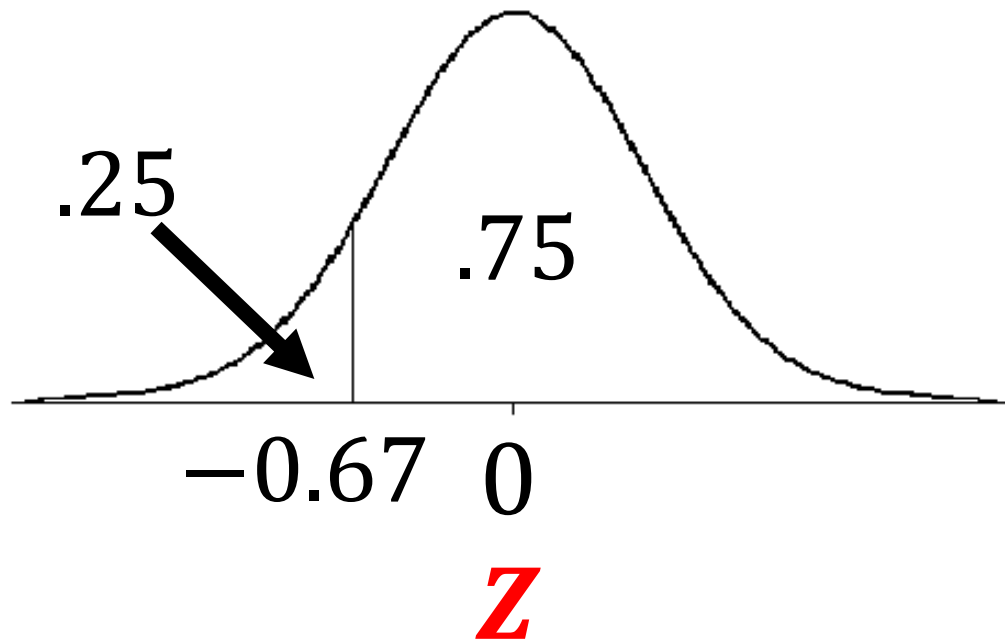
$$90^{\text{th}} \text{ percentile} = 1.28 = Z_{.10}$$

$$10^{\text{th}} \text{ percentile} = -1.28 = Z_{.90}$$



Find the 75th and 25th percentiles.

	0.06	0.07	0.08	0.09
	.5239	.5279	.5319	.5359
	.5636	.5675	.5714	.5753
	.6026	.6064	.6103	.6141
	.6406	.6443	.6480	.6517
	.6772	.6808	.6844	.6879
	.7123	.7157	.7190	.7224
0.6	.7454	.7486	.7517	.7549
	.7764	.7794	.7823	.7852
	.8051	.8078	.8106	.8133
	.8315	.8340	.8365	.8389
	.8554	.8577	.8599	.8621
	.8770	.8790	.8810	.8830
	.8962	.8980	.8997	.9015
	.9131	.9147	.9162	.9177
	.9279	.9292	.9306	.9319
	.9406	.9418	.9429	.9441



75^{th} percentile $= 0.67 = Z_{.25}$

25^{th} percentile $= -0.67 = Z_{.75}$

Z SCORES & OTHER NORMAL CURVES

	score	possible	μ	σ	Z
Buck	100	200	80	20	1
Darlene	75	100	80	10	-0.5

$$\frac{100 - 80}{20} = 1$$

$$\frac{75 - 80}{10} = -0.5$$

$$Z = \frac{X - \mu}{\sigma}$$

Calculating the Z score for a value is sometimes called “standardizing” the value, and we are measuring how many standard deviations above or below the mean the observation is.

$$Z = \frac{X - \mu}{\sigma}$$

observation > mean
positive Z score

observation < mean
negative Z score

standard normal = $Z \sim N(\mu = 0, \sigma = 1)$

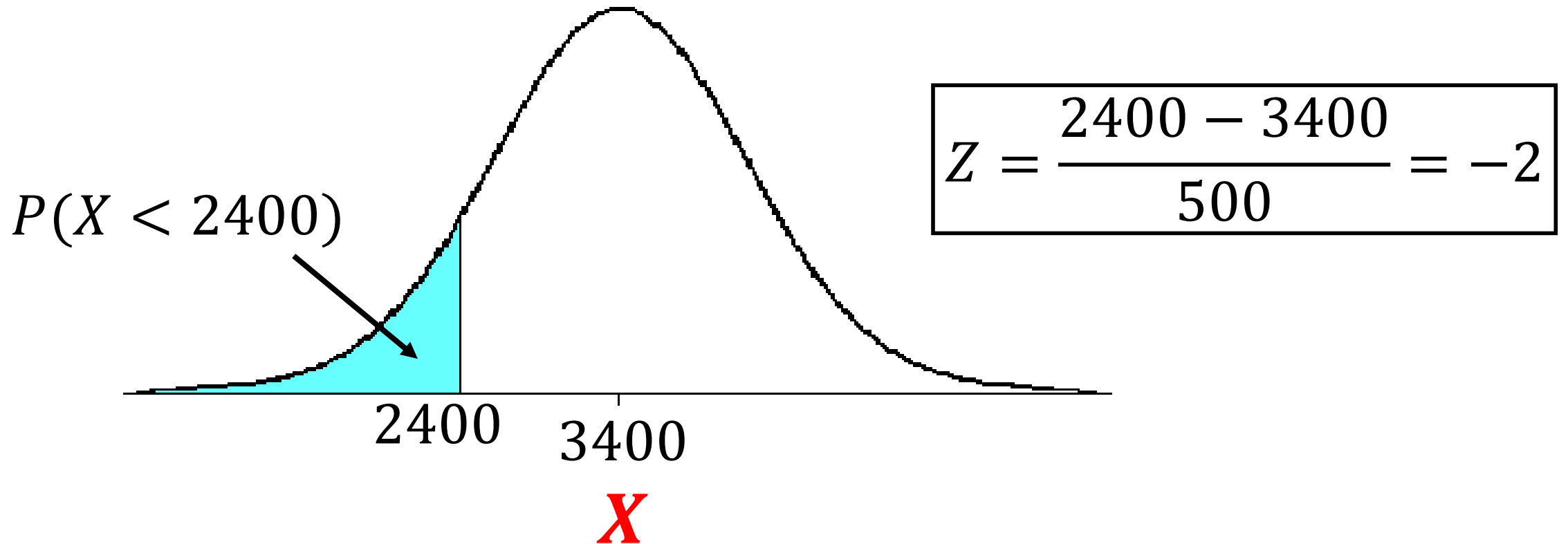
We have a table for this curve.

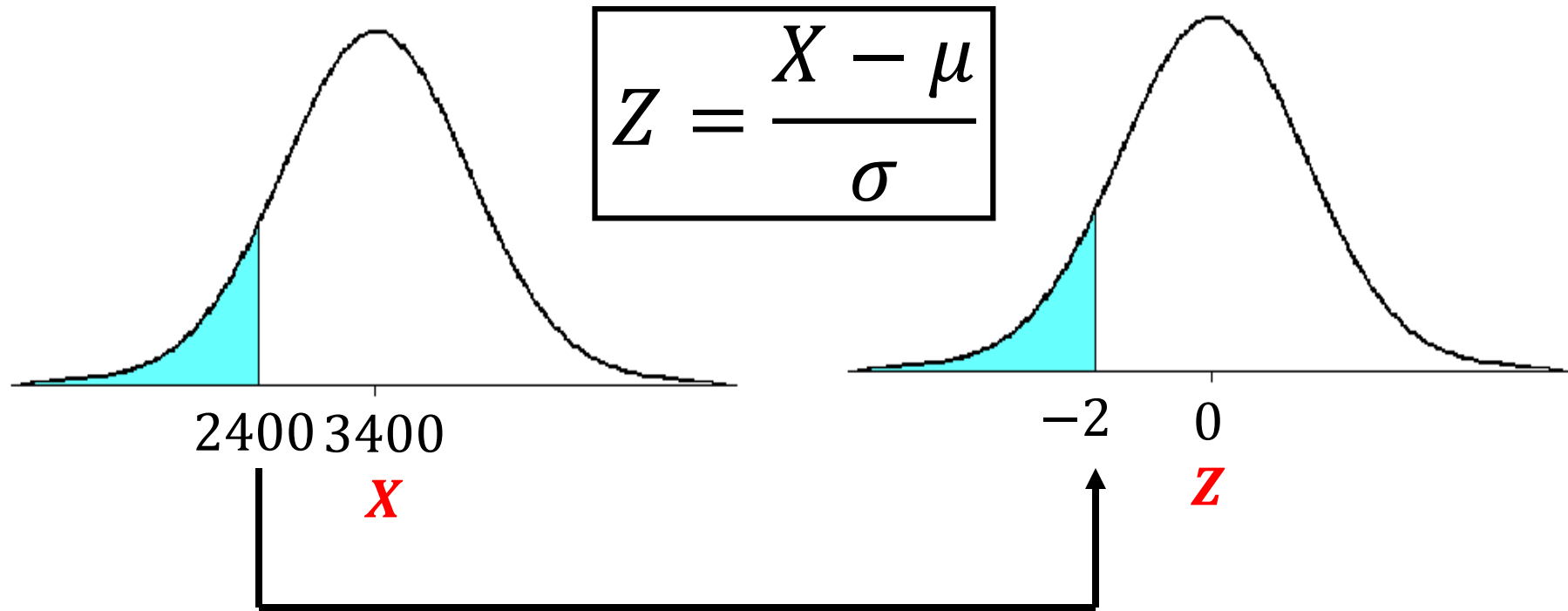
“regular” normal = $X \sim N(\mu, \sigma)$

No table!

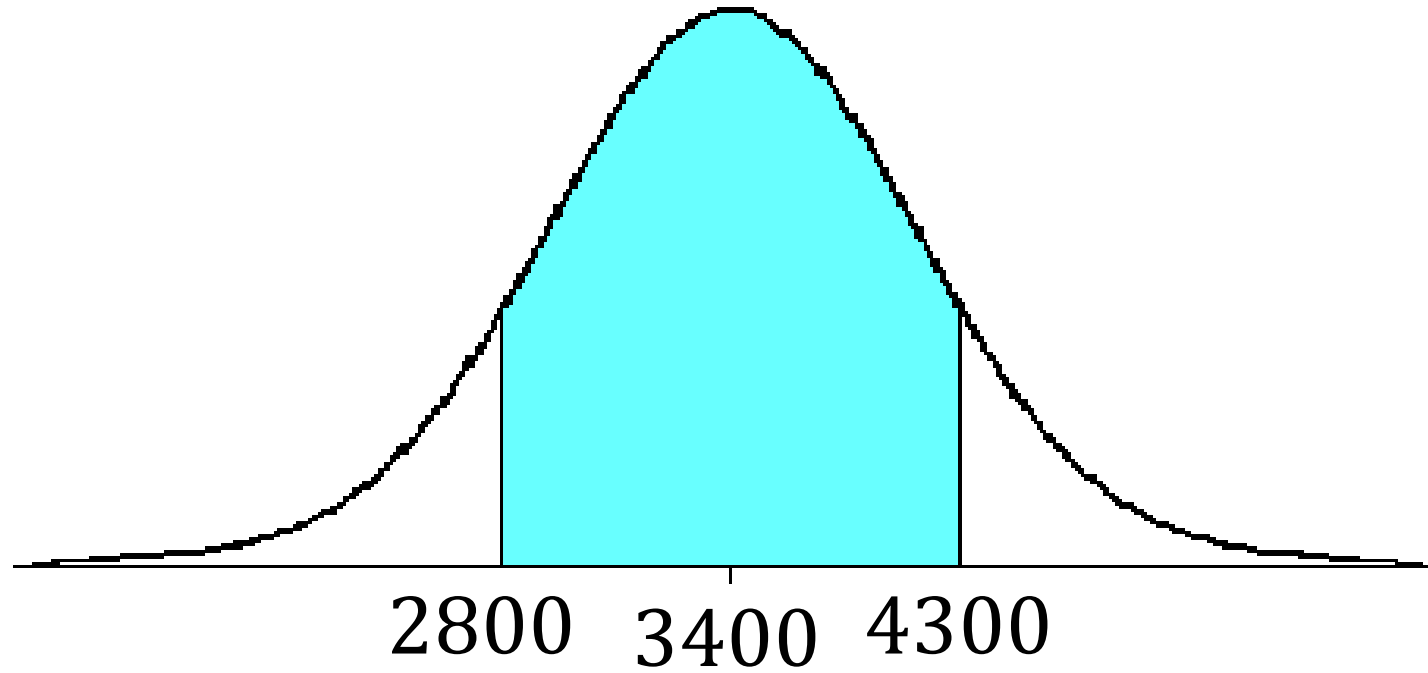
X = mass of a randomly selected newborn baby

$X \sim N(\mu = 3400 \text{ grams}, \sigma = 500 \text{ grams})$



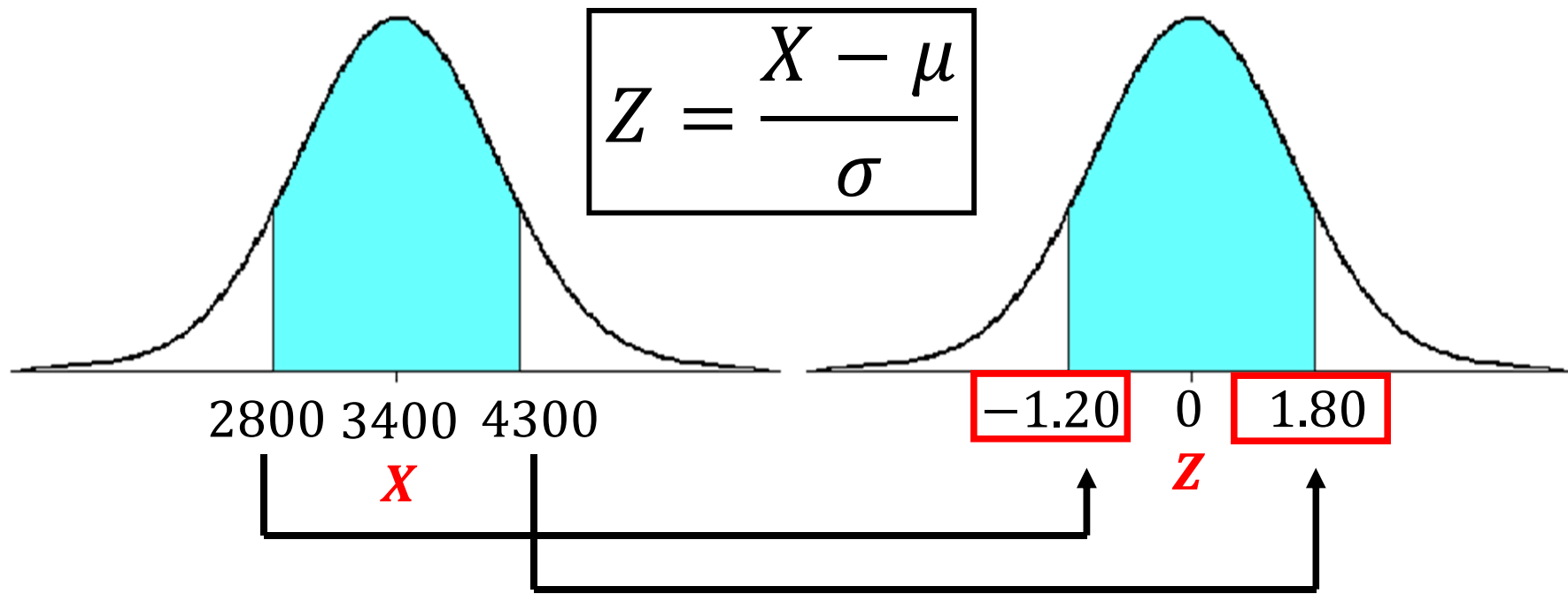


$$P(X < 2400) = P\left(Z < \frac{2400 - 3400}{500}\right) = P(Z < -2) = .0228$$



X

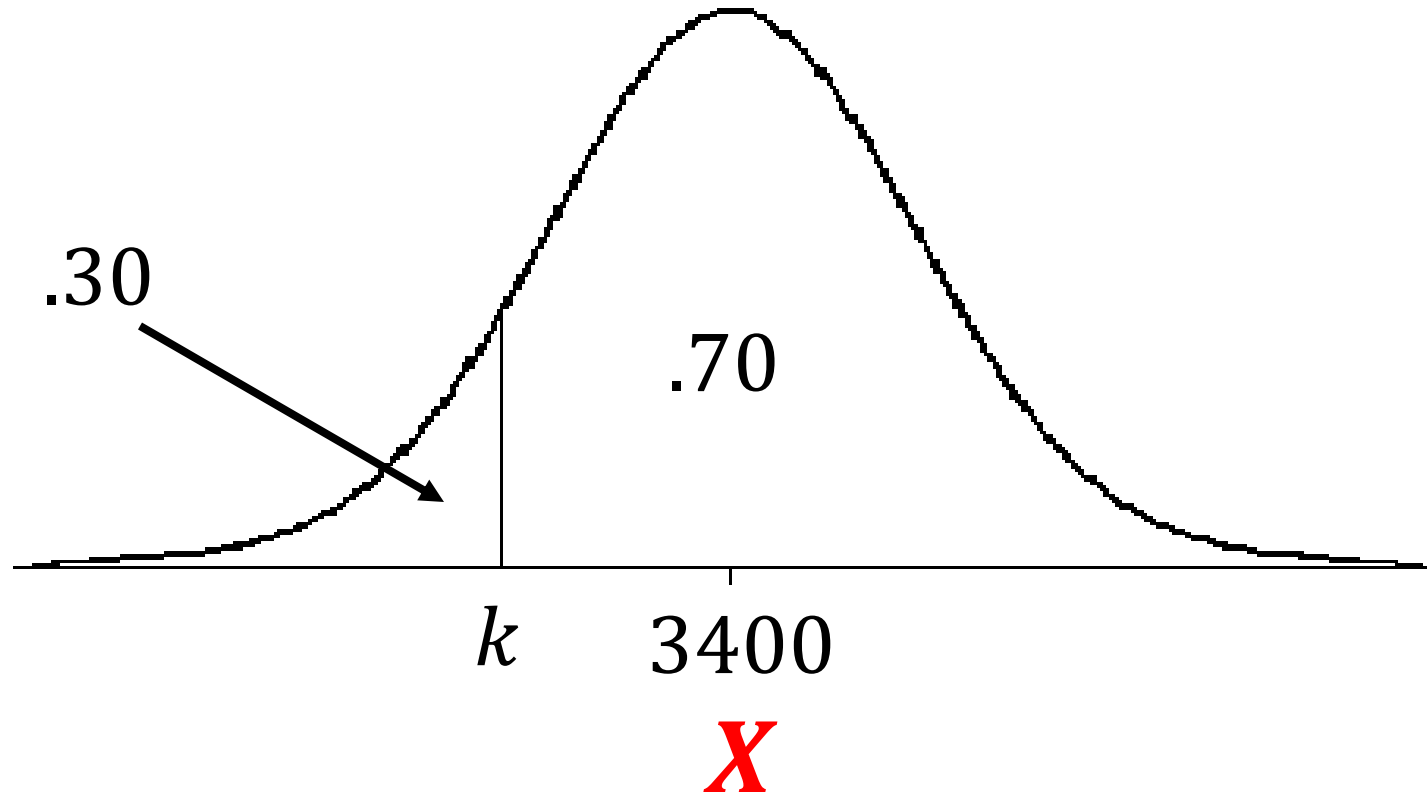
$$P(2800 < X < 4300) = ?$$

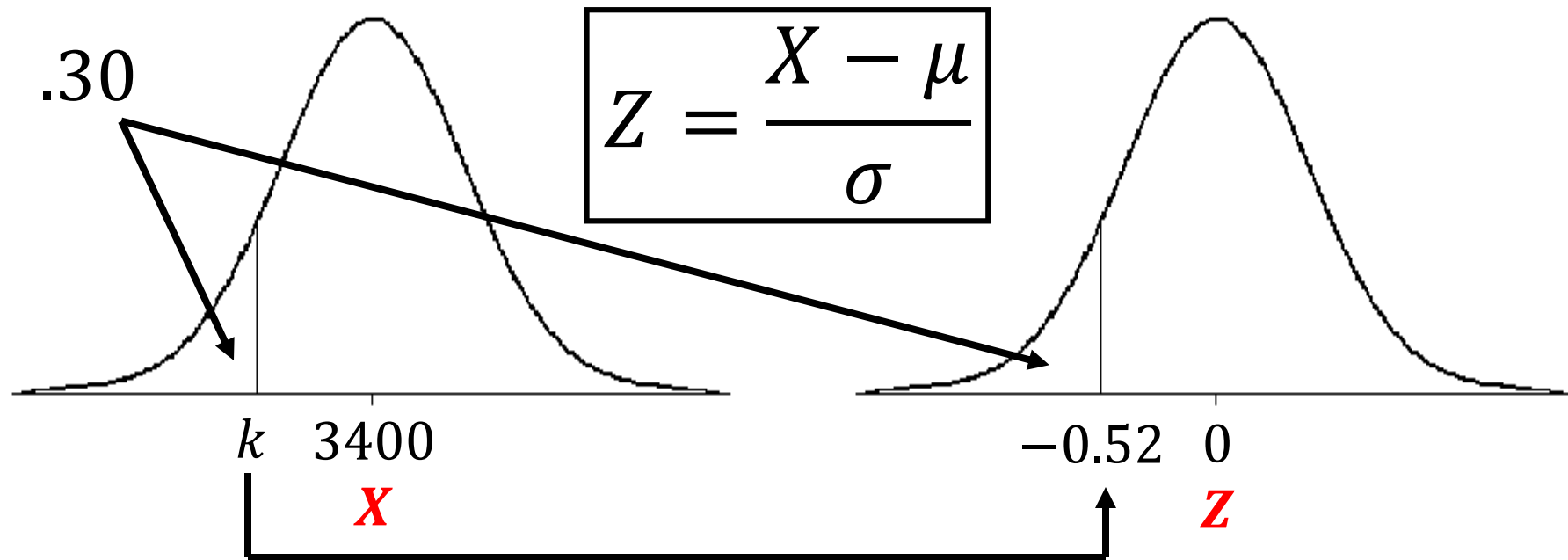


$$\begin{aligned} P(2800 < X < 4300) &= P\left(\frac{2800 - 3400}{500} < Z < \frac{4300 - 3400}{500}\right) \\ &= P(-1.20 < Z < 1.80) = .9641 - .1151 = .8490 \end{aligned}$$

A baby named Rowan has a mass of k grams.
70% of all the babies in the population are heavier than Rowan.

k is the 30th percentile. What is Rowan's mass in grams?





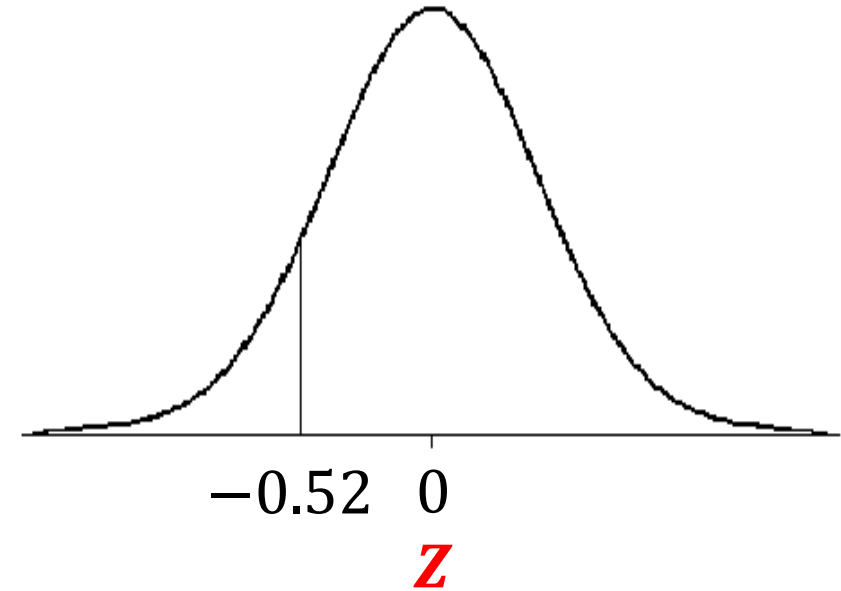
$$Z = \frac{k - 3400}{500} = -0.52$$

$$k = 3140 \text{ grams}$$

$$\frac{k - 3400}{500} = -0.52$$

$$k - 3400 = -0.52(500)$$

$$k = 3400 - 0.52(500) = 3140$$



percentile on any
normal curve

$$= \mu + Z\sigma$$

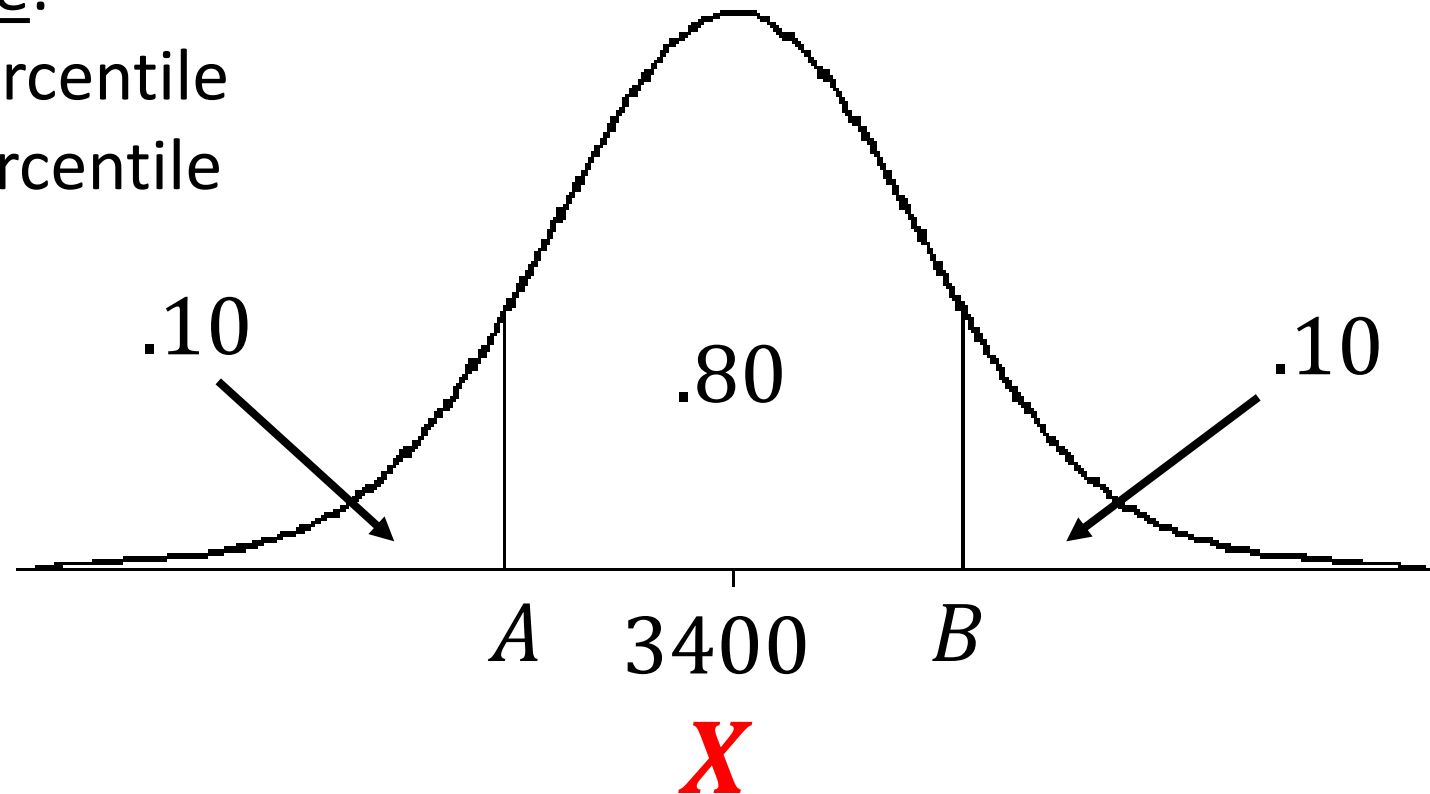
corresponding
percentile on Z-curve

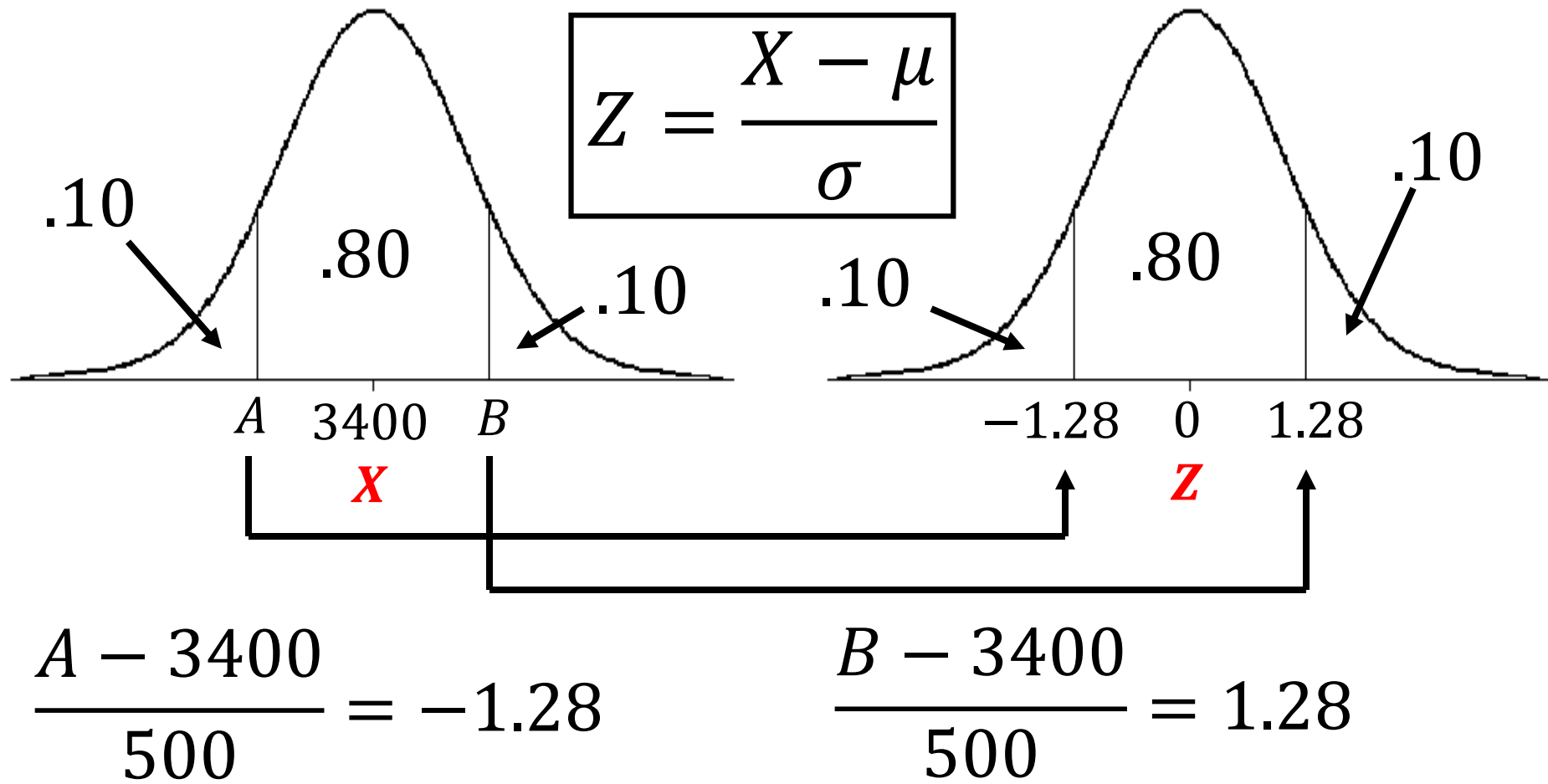
80% of all newborn babies have a mass in the interval (A, B) centered around the mean of $\mu = 3400$. Find A and B .

Notice:

$A = 10^{\text{th}}$ percentile

$B = 90^{\text{th}}$ percentile





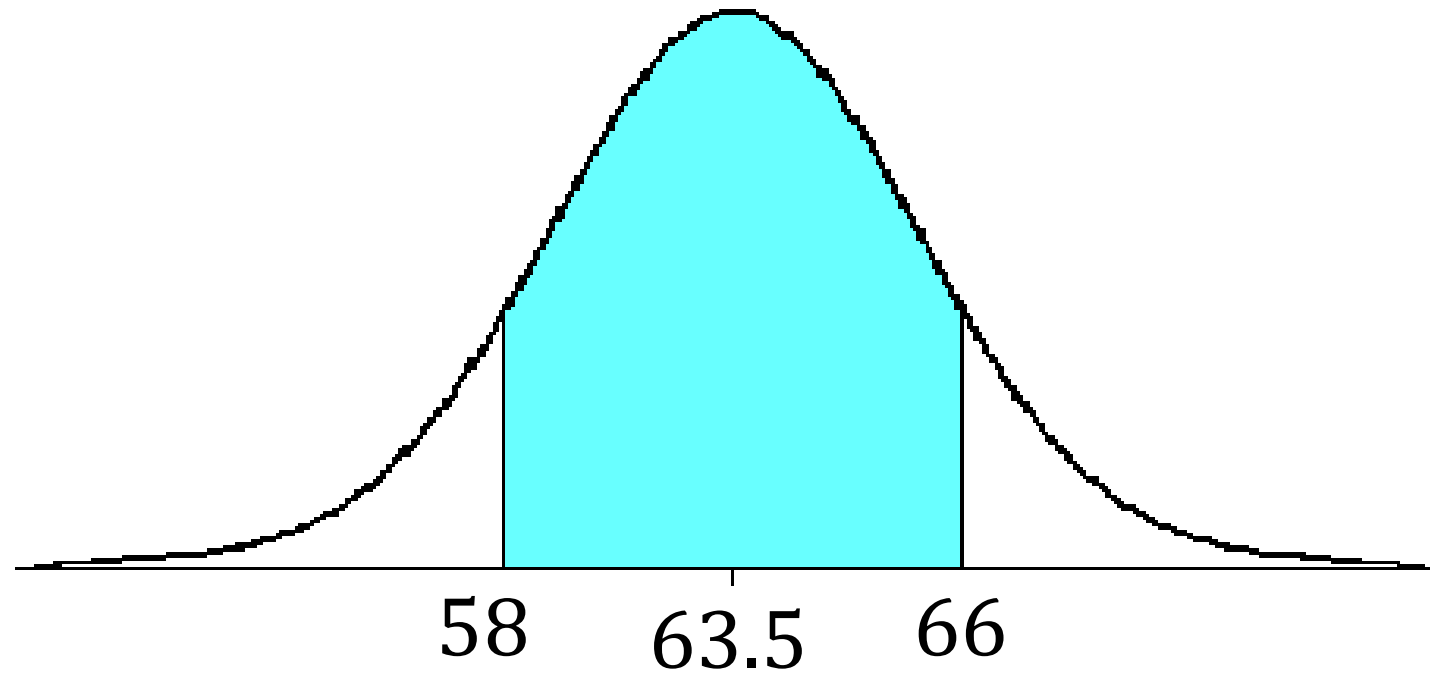
$A = 2760$ grams $B = 4040$ grams (2760, 4040)

X = height of a randomly selected woman
 $X \sim N(\mu = 63.5 \text{ inches}, \sigma = 2.5)$

Find:

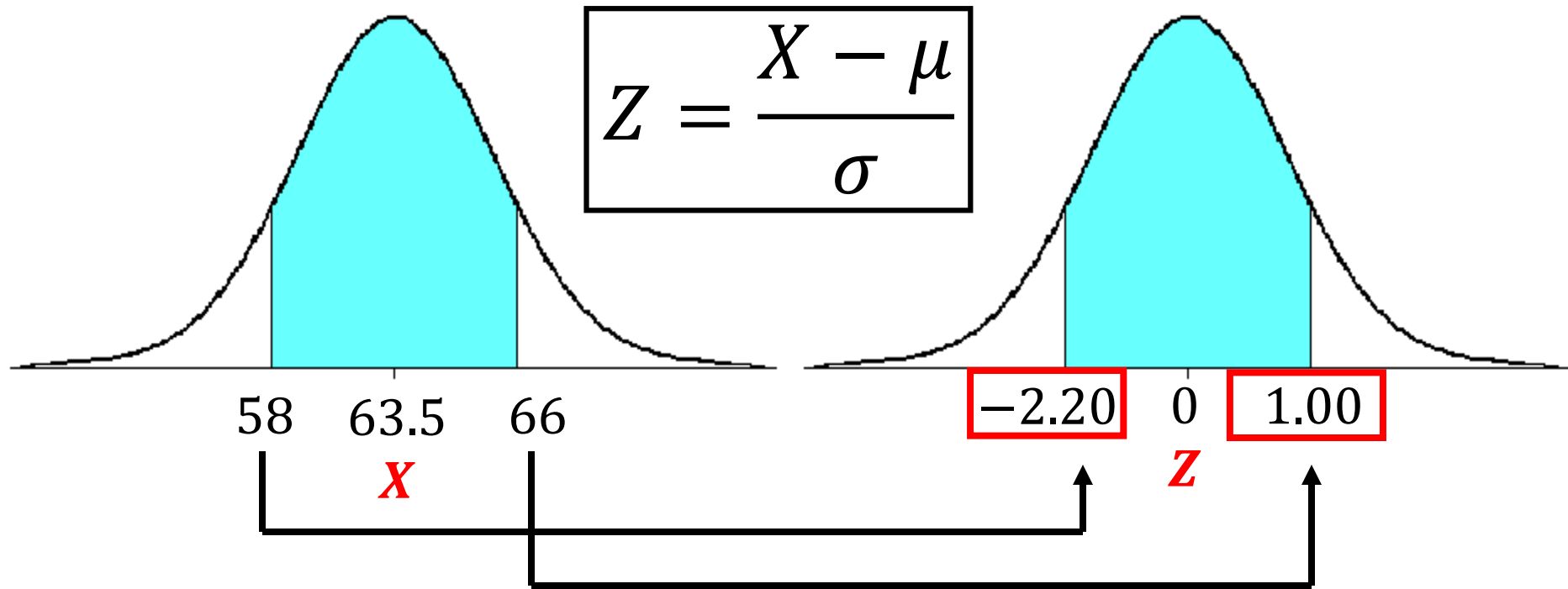
$$P(58 < X < 66)$$

68th percentile among the women's heights



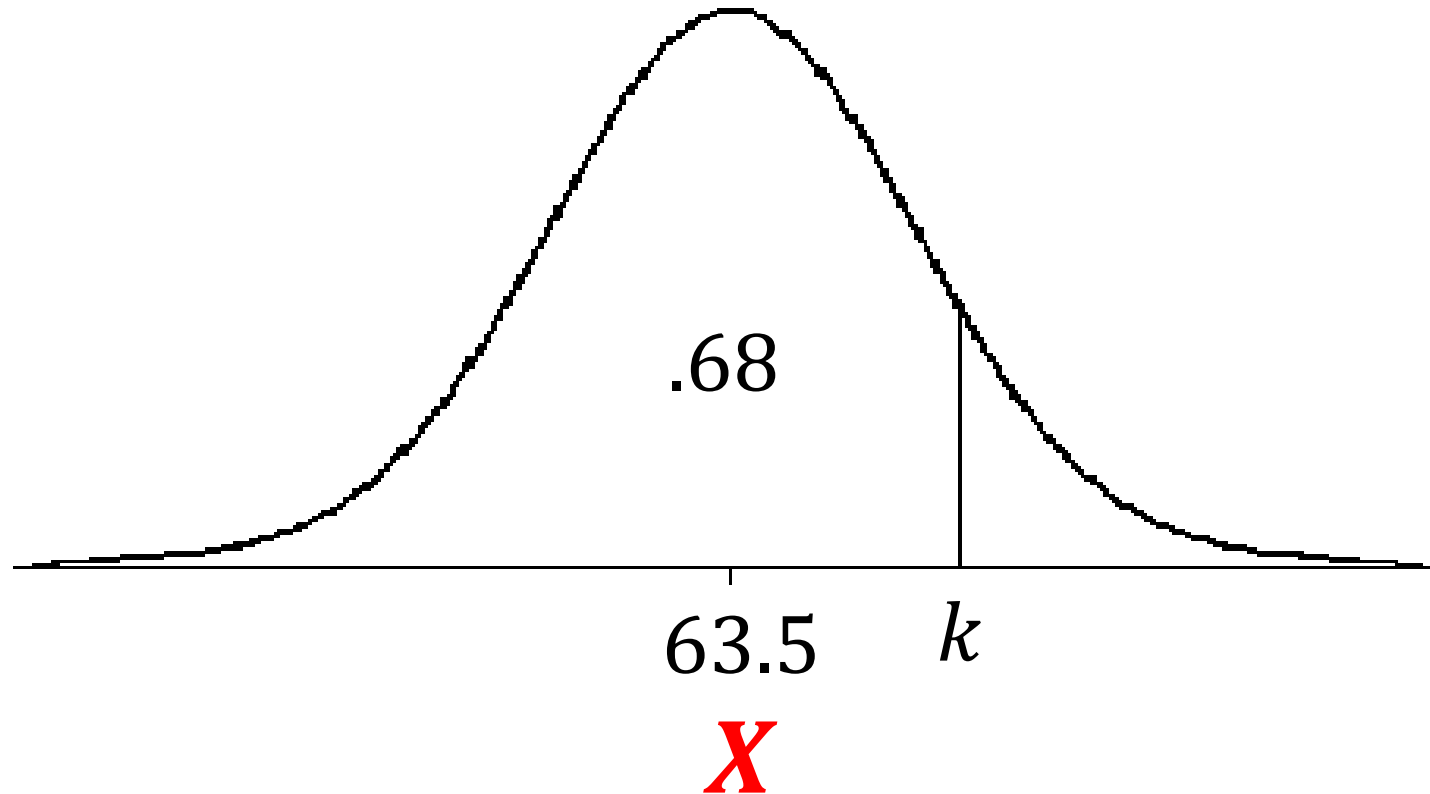
X

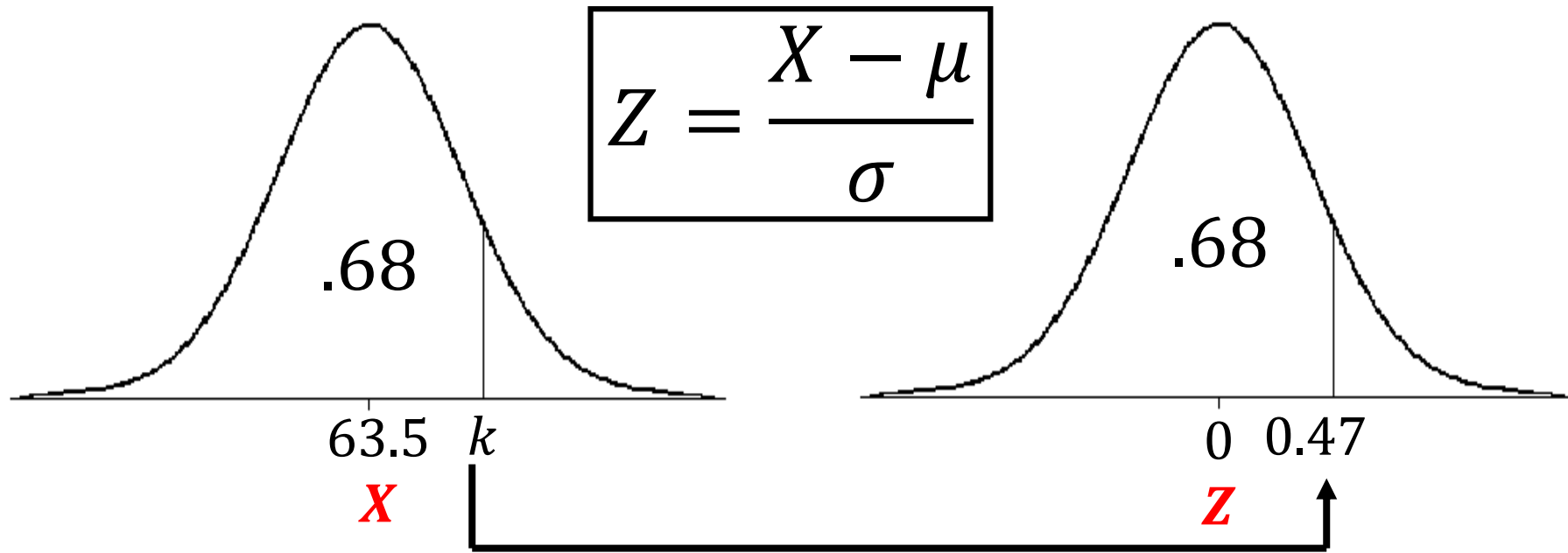
$$P(58 < X < 66) = ?$$



$$\begin{aligned}
 P(58 < X < 66) &= P\left(\frac{58 - 63.5}{2.5} < Z < \frac{66 - 63.5}{2.5}\right) \\
 &= P(-2.20 < Z < 1.00) = .8413 - .0139 = .8274
 \end{aligned}$$

68th percentile = ?





$$Z = \frac{k - 63.5}{2.5} = 0.47$$

$$k = 64.675 \text{ inches}$$

ASSESSING NORMALITY & QQ PLOTS

When using tools of statistical inference (confidence intervals and hypothesis tests) an important and very common question is:

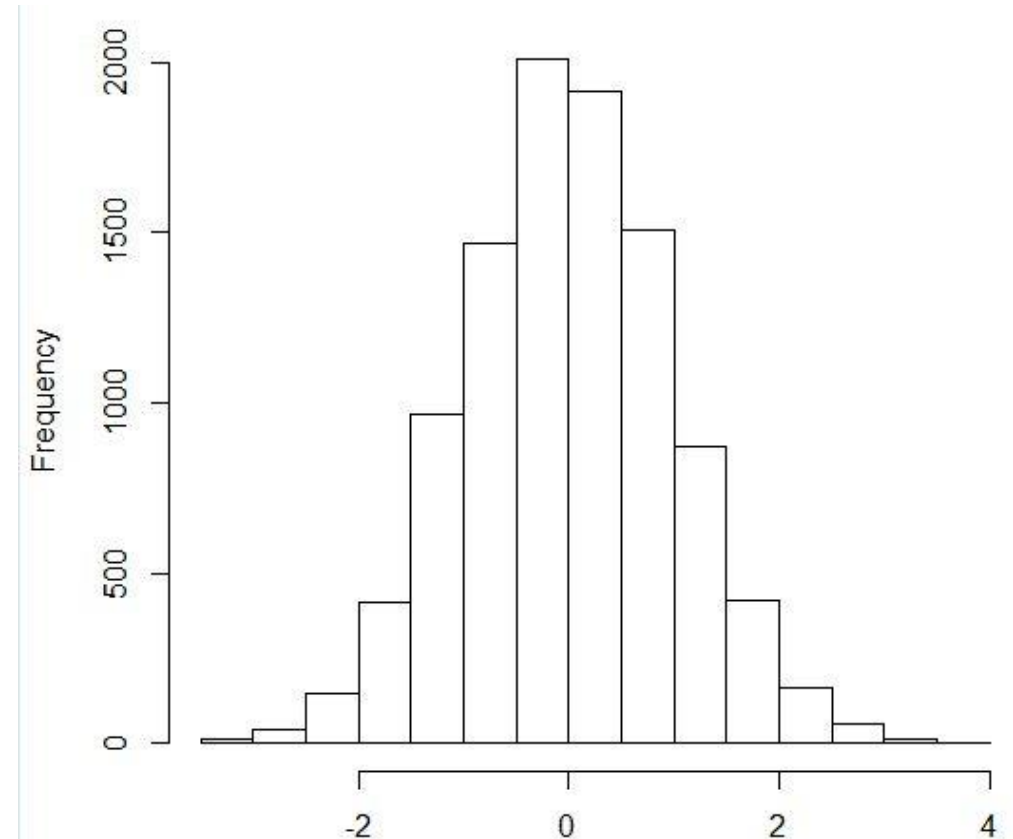
Is my data normally distributed?

How can we determine if a data set came from a normal distribution?

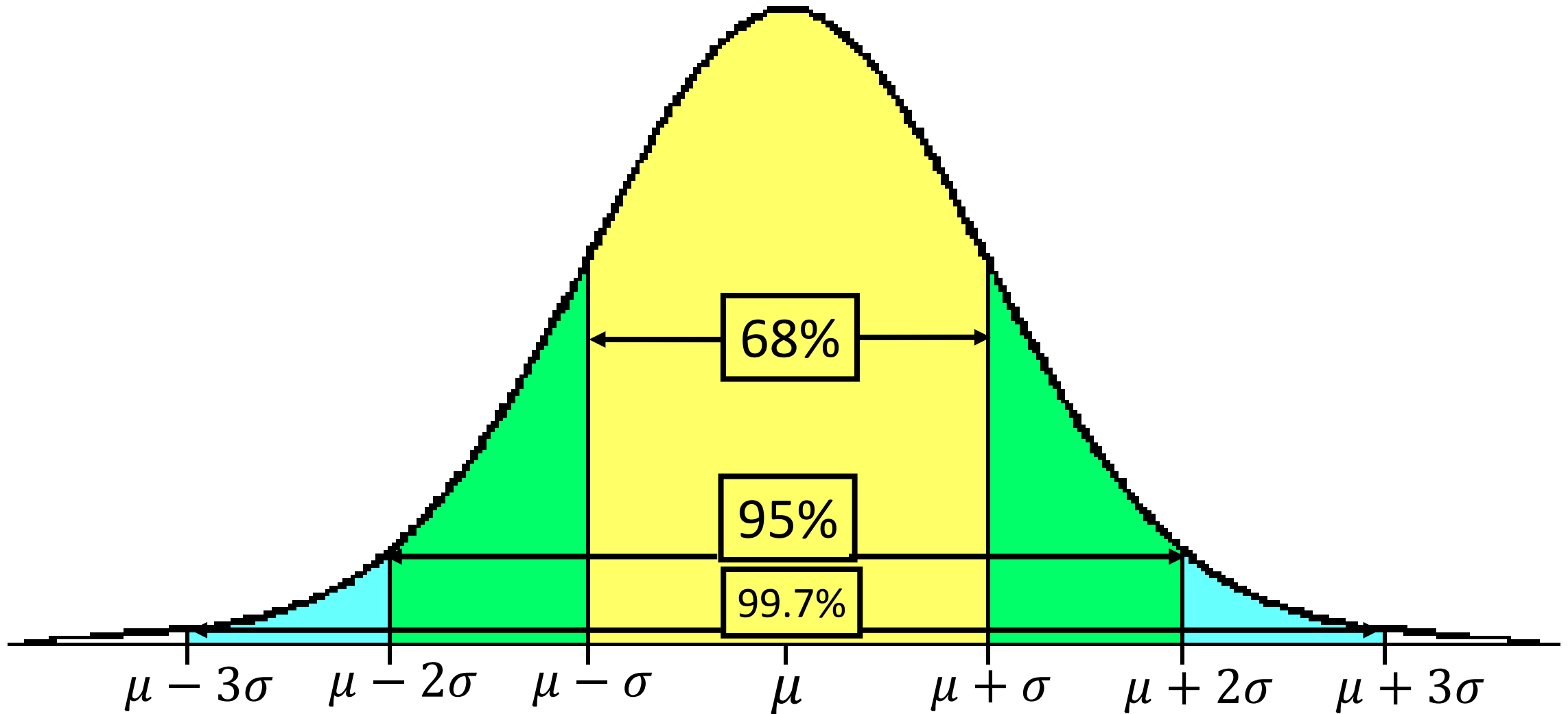
How can we determine if a data set came from a normal distribution?

1

Data from a normal distribution should yield a bell-shaped histogram.



Empirical Rule for normal distributions



How can we determine if a data set came from a normal distribution?

2

According to the Empirical Rule,

a data set of normally distributed data should have:

about 68% of its observations in the interval $(\bar{X} - s, \bar{X} + s)$

about 95% of its observations in the interval $(\bar{X} - 2s, \bar{X} + 2s)$

about 99.7% of its observations in the interval $(\bar{X} - 3s, \bar{X} + 3s)$

How can we determine if a data set came from a normal distribution?

3

Data from a normal distribution should yield a linear QQ Plot (also known as Normal Probability Plot).

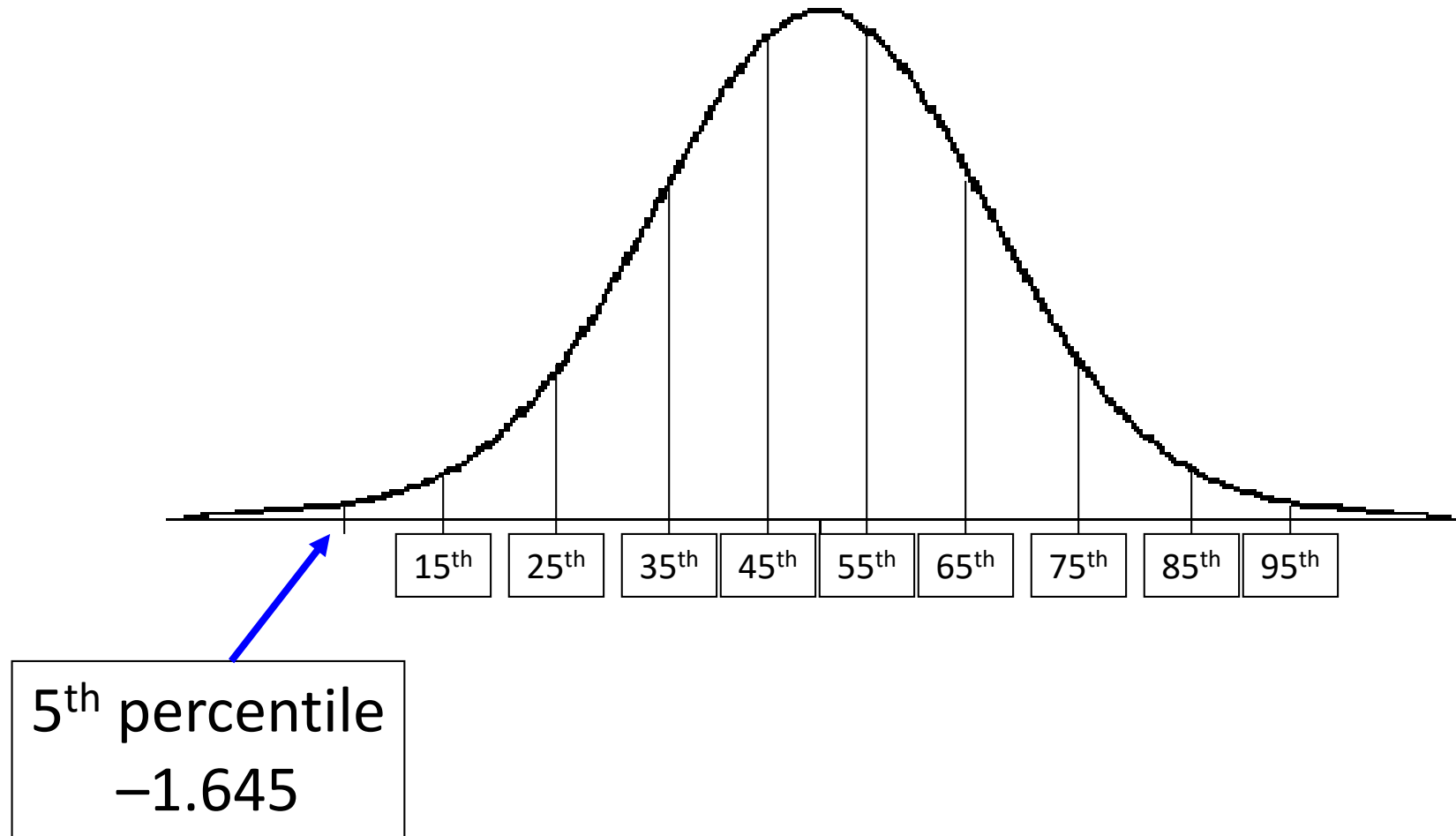
The QQ Plot is probably the most popular and reliable method to assess normality. How does a computer draw a QQ Plot?

Suppose you have a random sample of $n = 10$ football players. You would like to know if the population of football players weights is normally distributed.

Ordered data set:

{ 248, 253, 263, 272, 274, 286, 307, 310, 312, 323 }

If $n = 10$, the computer will draw 10 “slices” on the Z curve and find the corresponding percentiles.



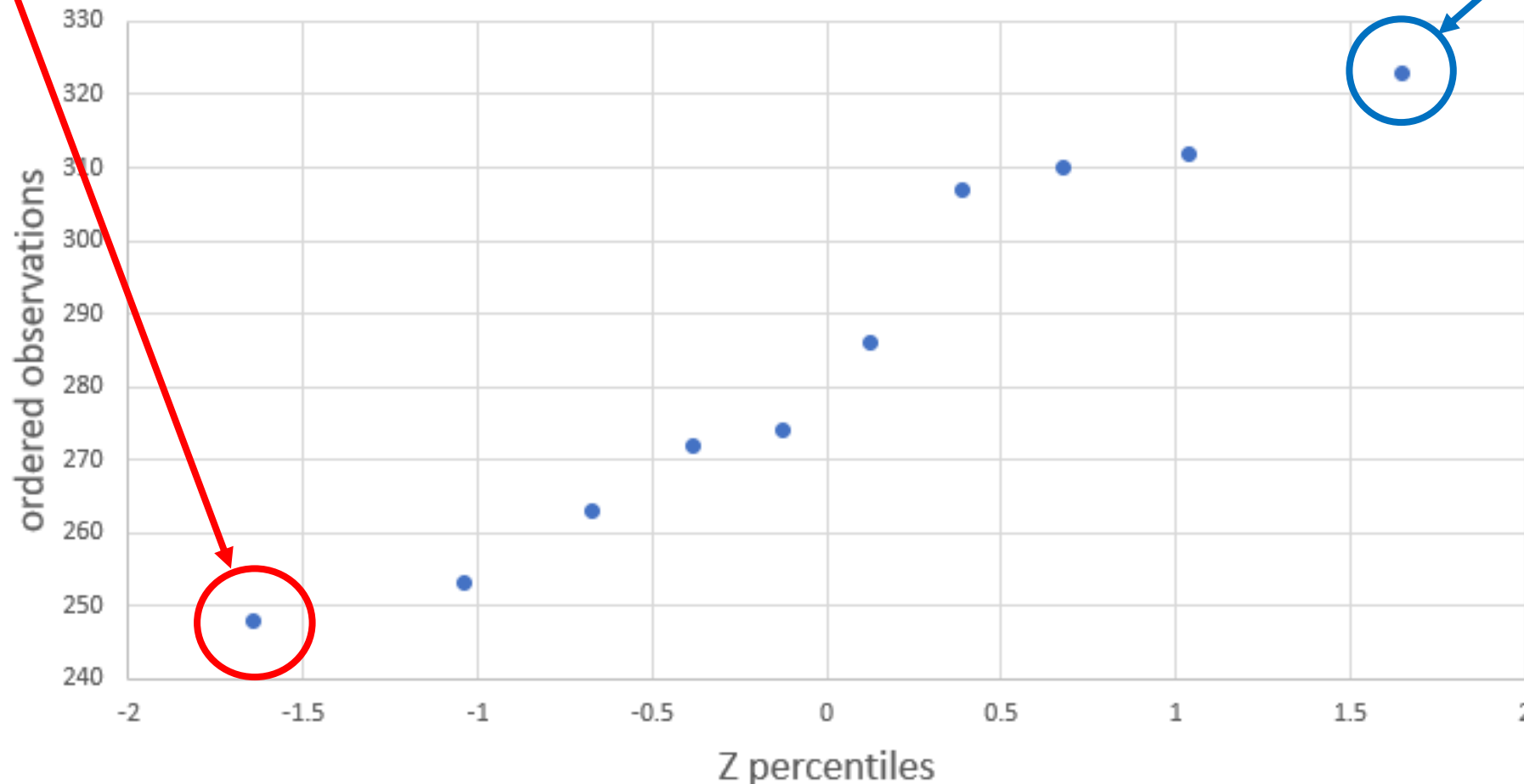
percentile	Z	weights (in order)
5th	-1.645	248
15th	-1.04	253
25th	-0.67	263
35th	-0.39	272
45th	-0.13	274
55th	0.13	286
65th	0.39	307
75th	0.67	310
85th	1.04	312
95th	1.645	323

The (QQ Plot) displays n points:

$X = Z$ percentile

$Y =$ ordered data value

QQ Plot for $n = 10$ football players weights

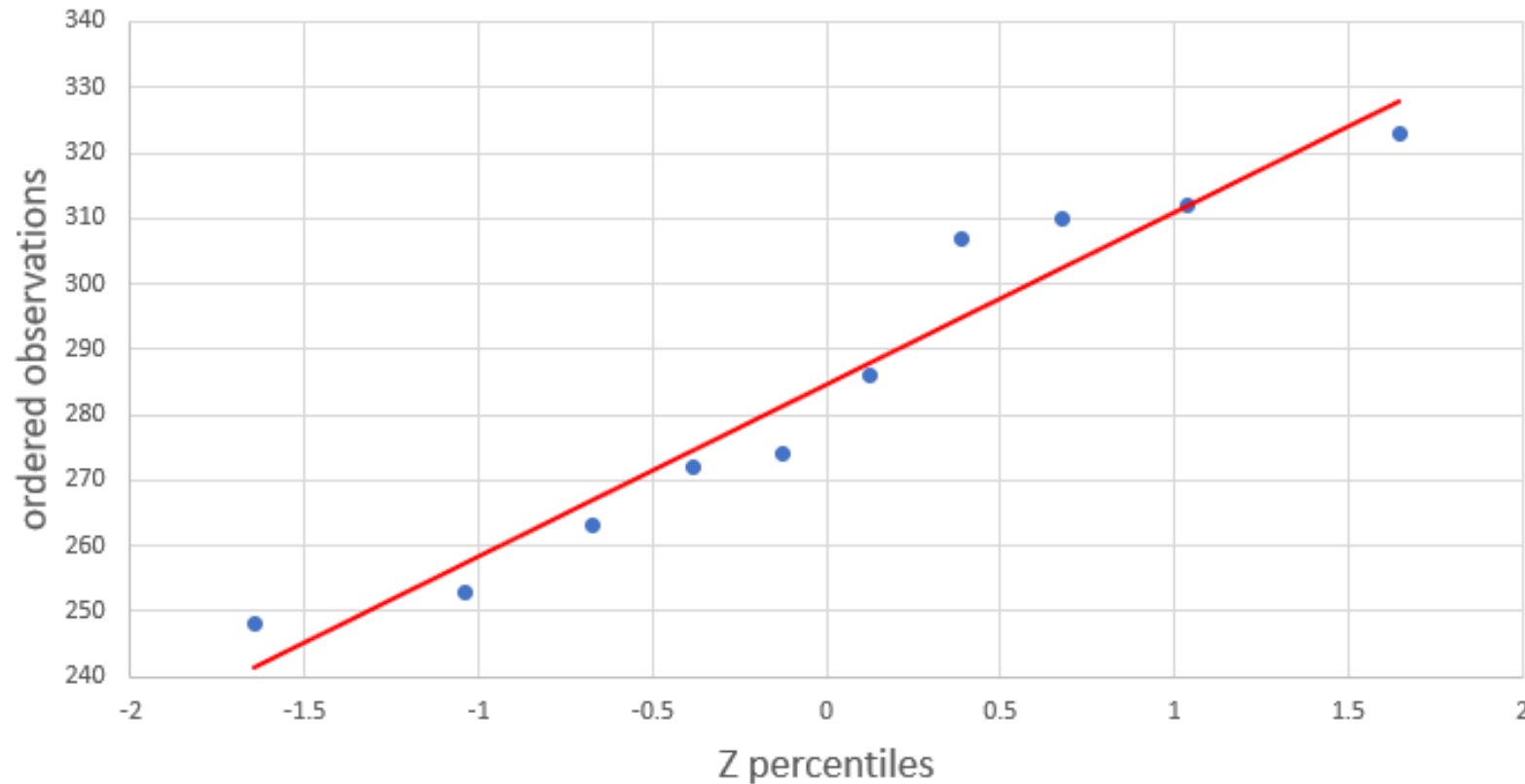


(-1.645,248)

(1.645,323)

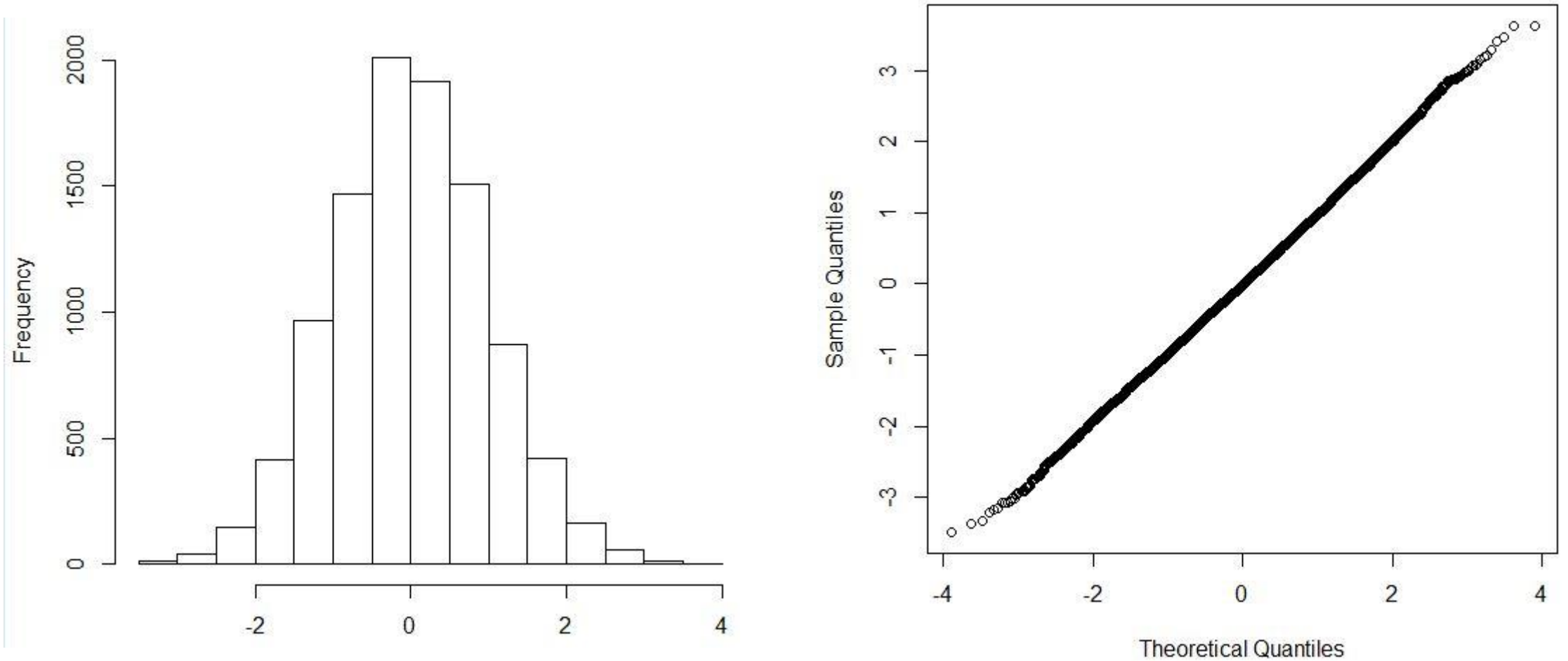
Data from a normal distribution should yield a linear QQ Plot. Unfortunately it is not always obvious whether or not a plot is “linear” so frequently we must use our best judgement.

QQ Plot for n = 10 football players weights

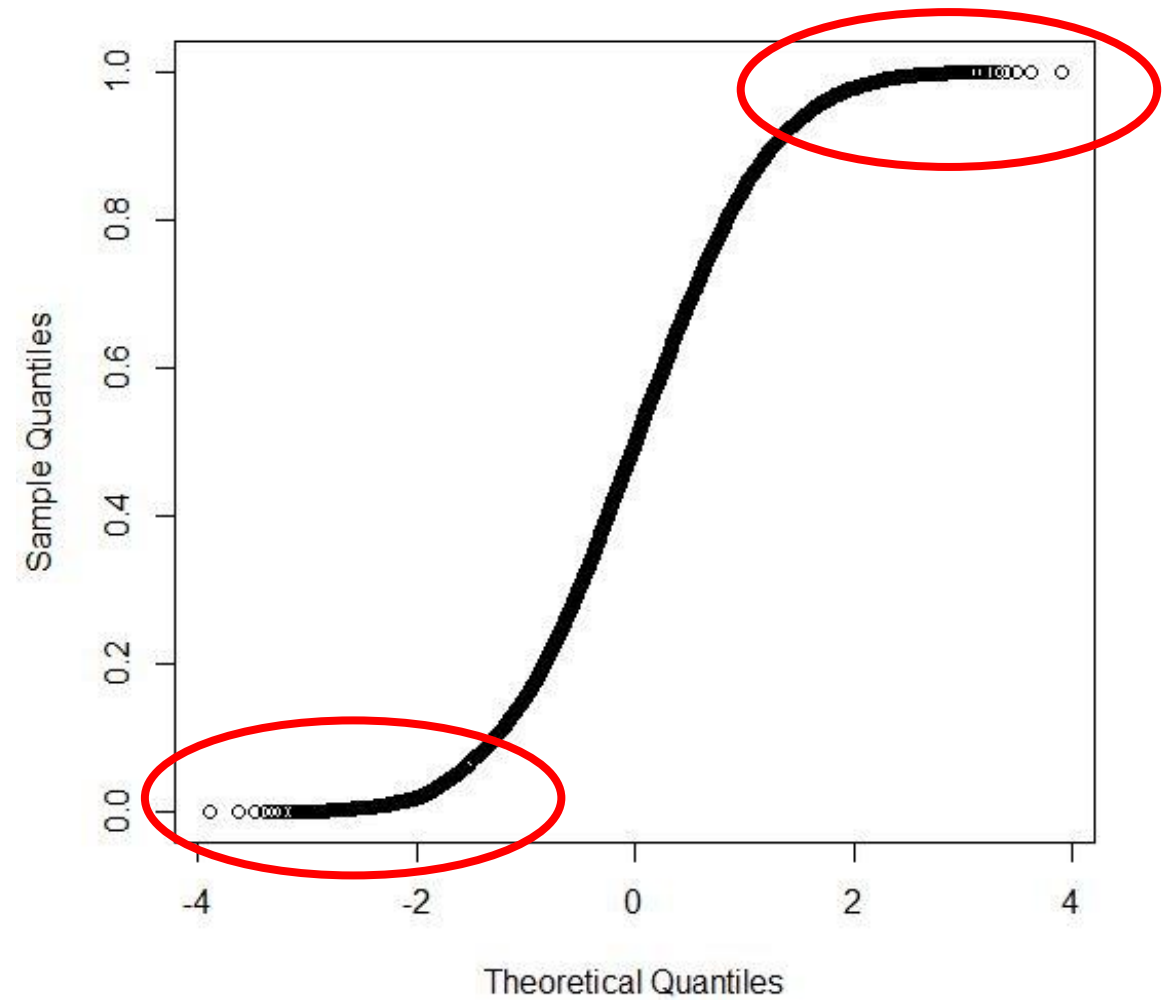
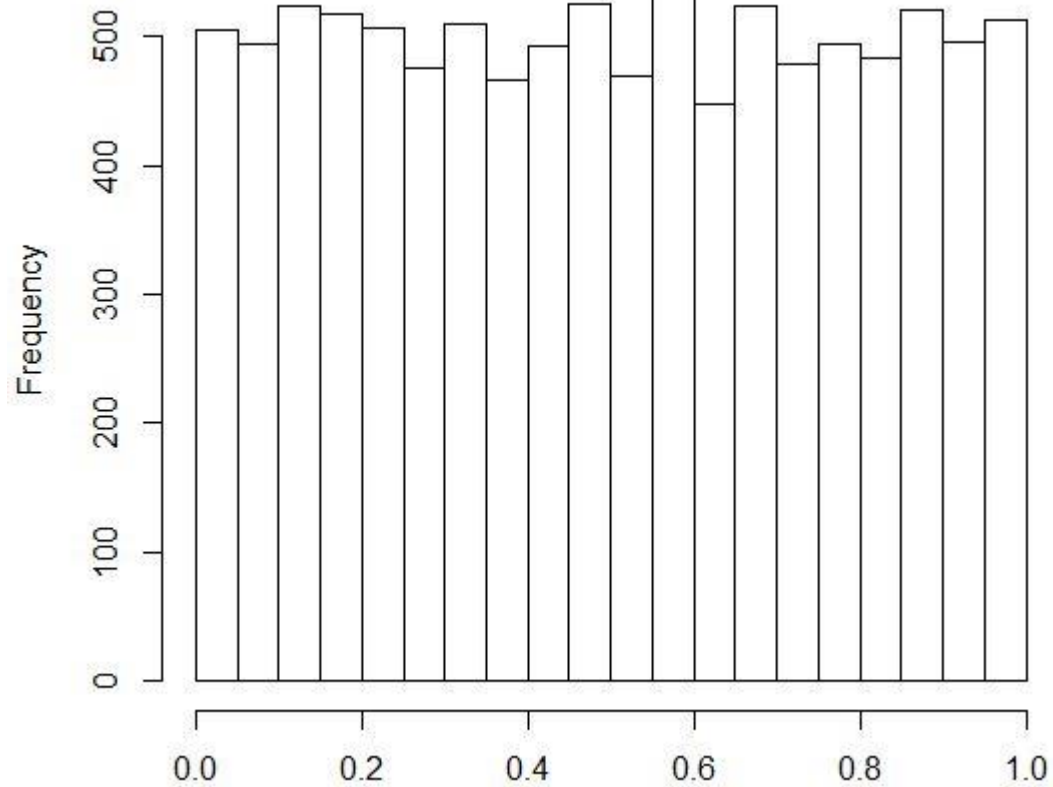


If a QQ Plot isn't linear, it may provide clues as to what kind of random variable you actually observed. Let's look at some examples.

First, here is a histogram and QQ Plot for 10,000 observations
from $Z \sim N(\mu = 0, \sigma = 1)$.



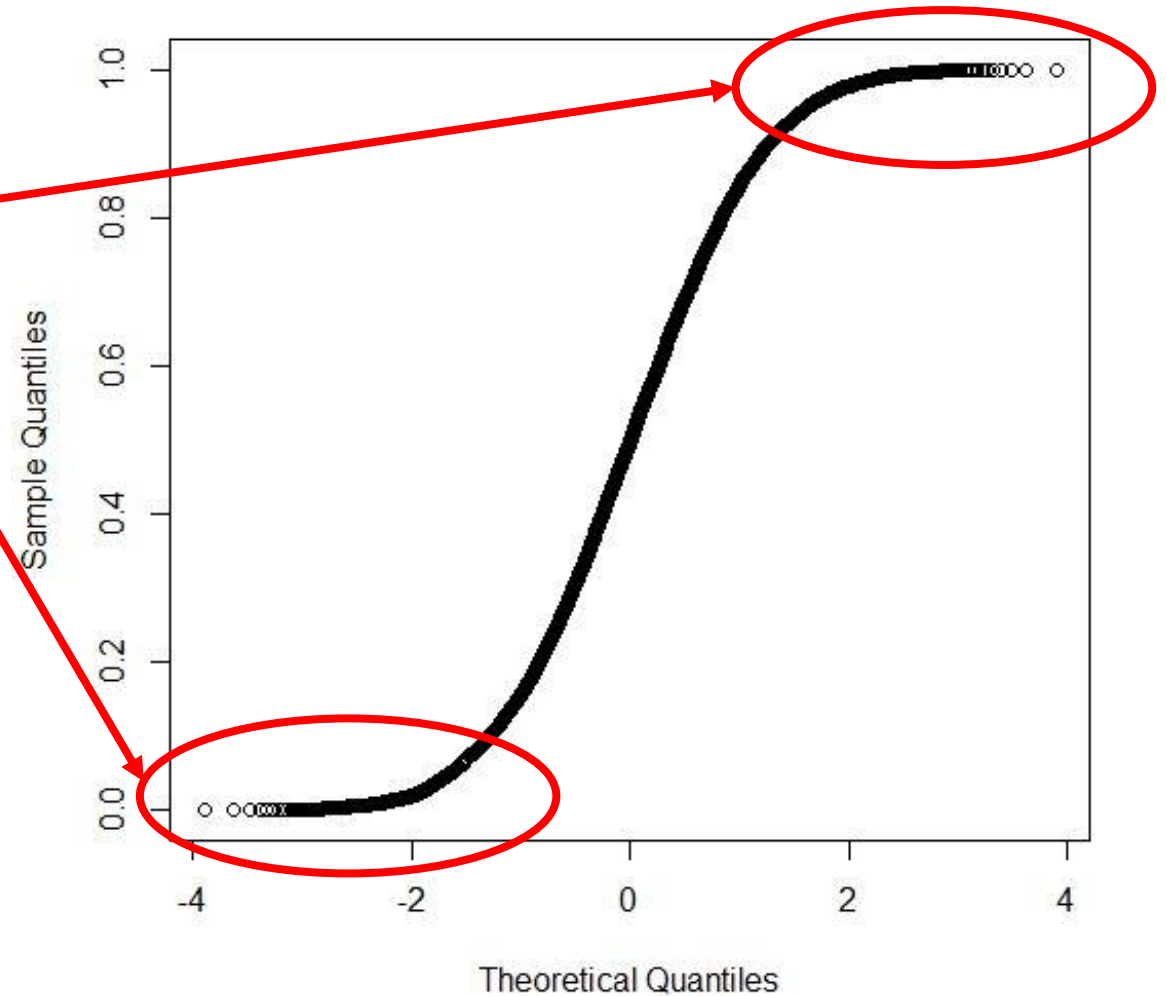
Histogram and QQ Plot for 10,000 observations from $UNIF(0,1)$



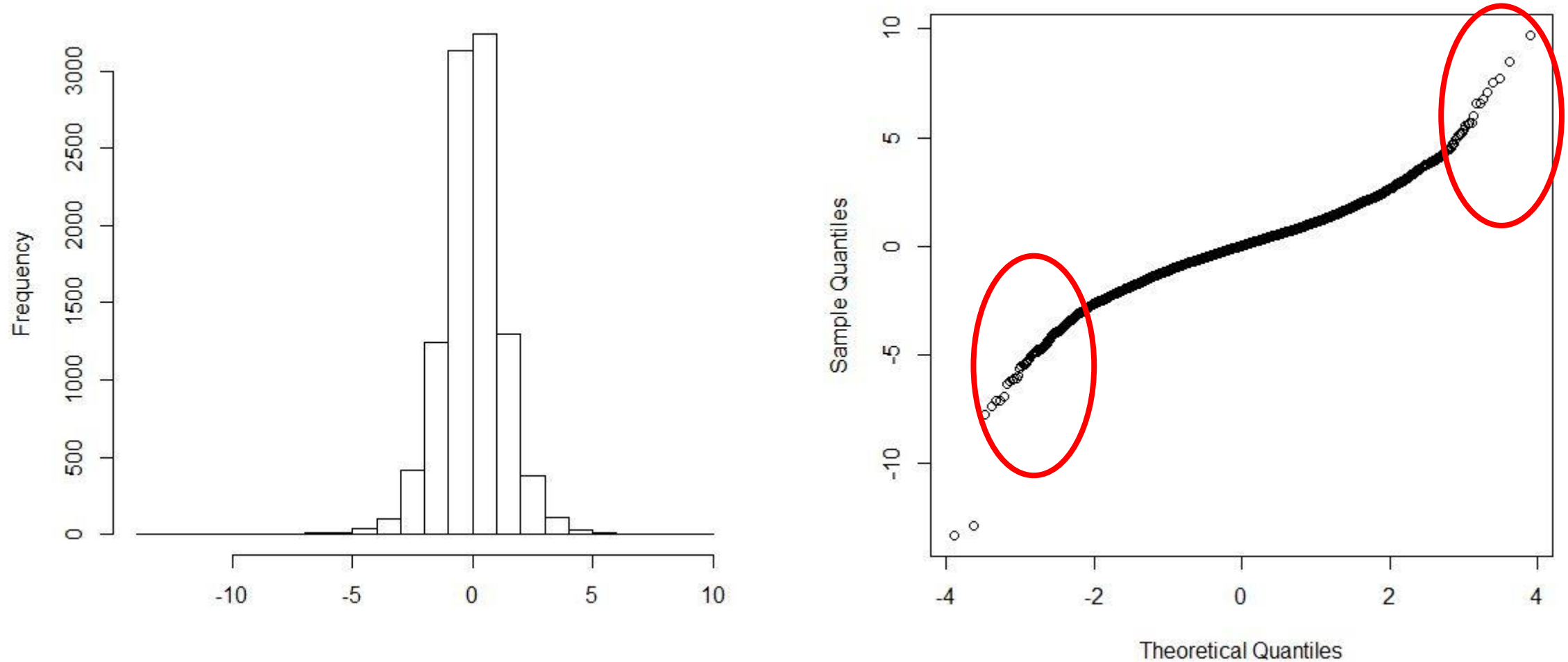
Histogram and QQ Plot for 10,000 observations from $UNIF(0,1)$

S-shaped “squiggle”

An S-shaped “squiggle” is an indication that the data came from a weak-tailed distribution.



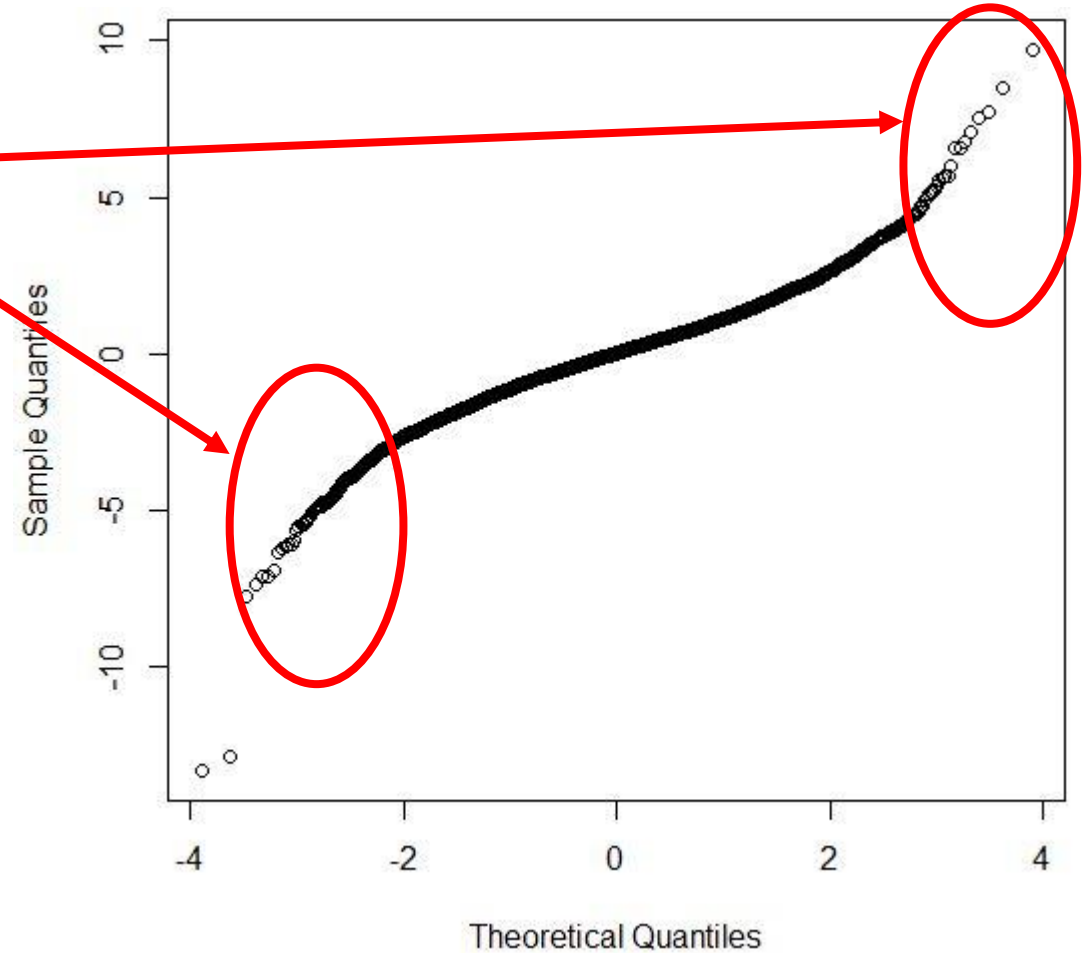
Histogram and QQ Plot for 10,000 observations from T_5



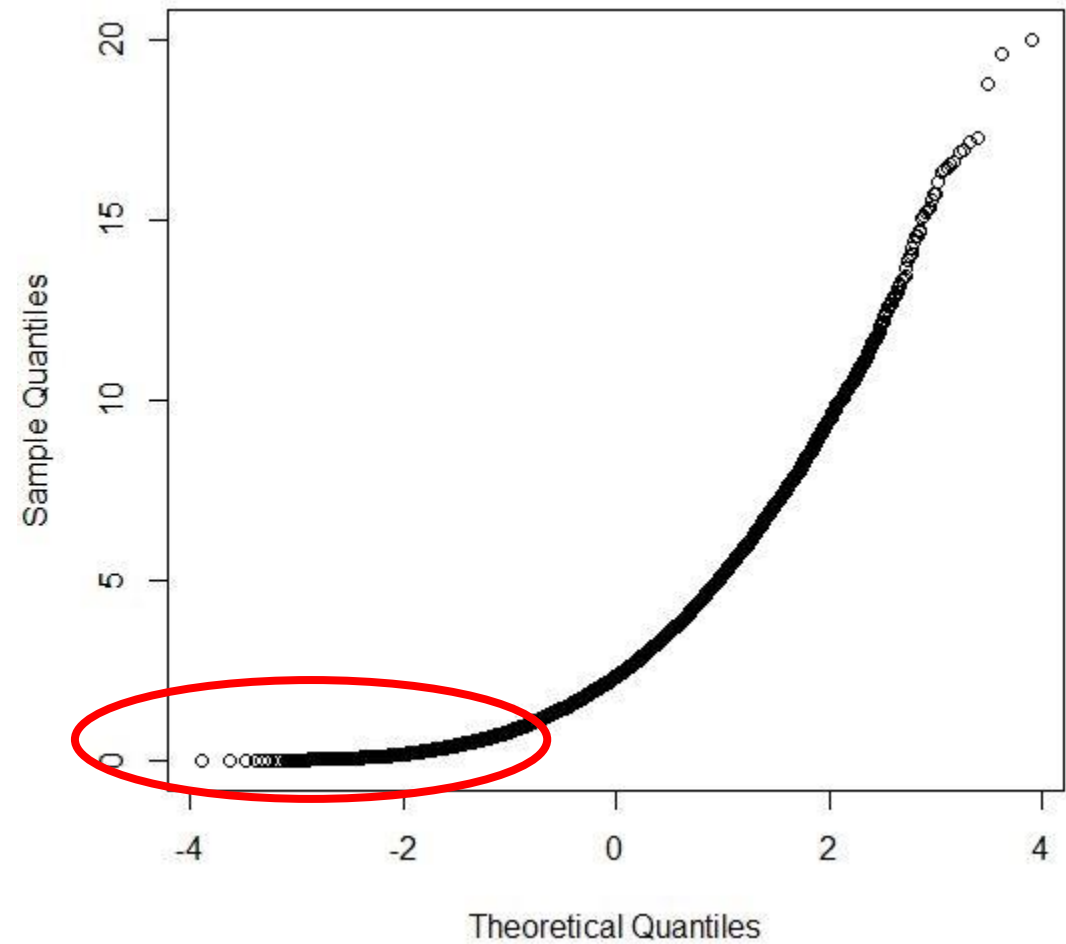
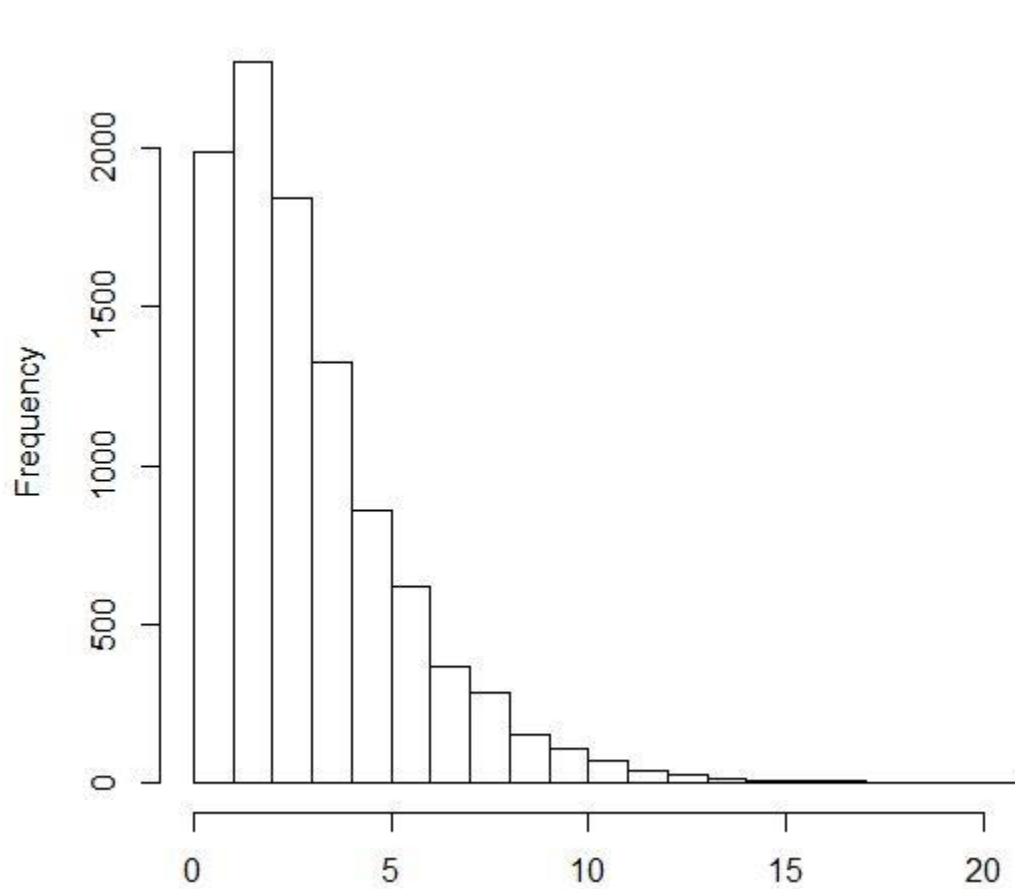
Histogram and QQ Plot for 10,000 observations from T_5

**opposite-shaped
“squiggle”**

A “squiggle” that is opposite of an S-shape is an indication that the data came from a heavy-tailed distribution.

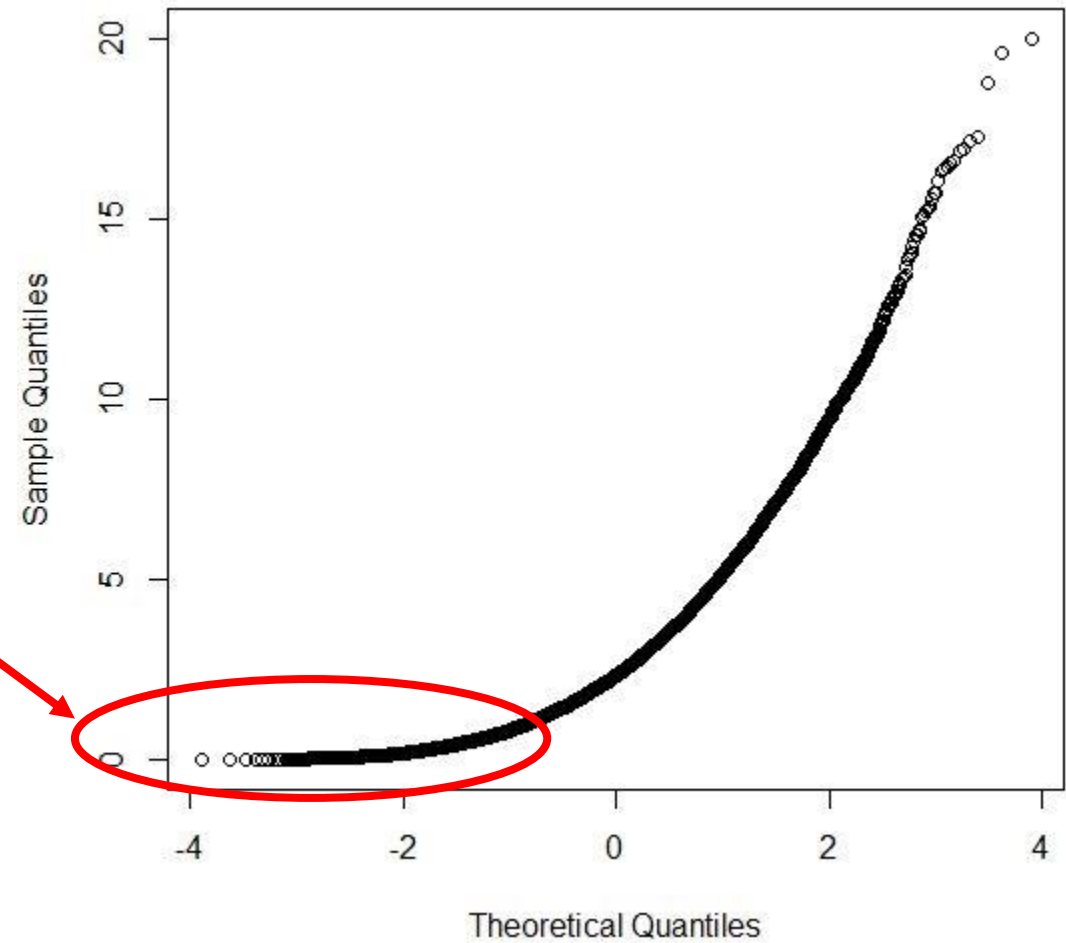


Histogram and QQ Plot for 10,000 observations from χ^2_3

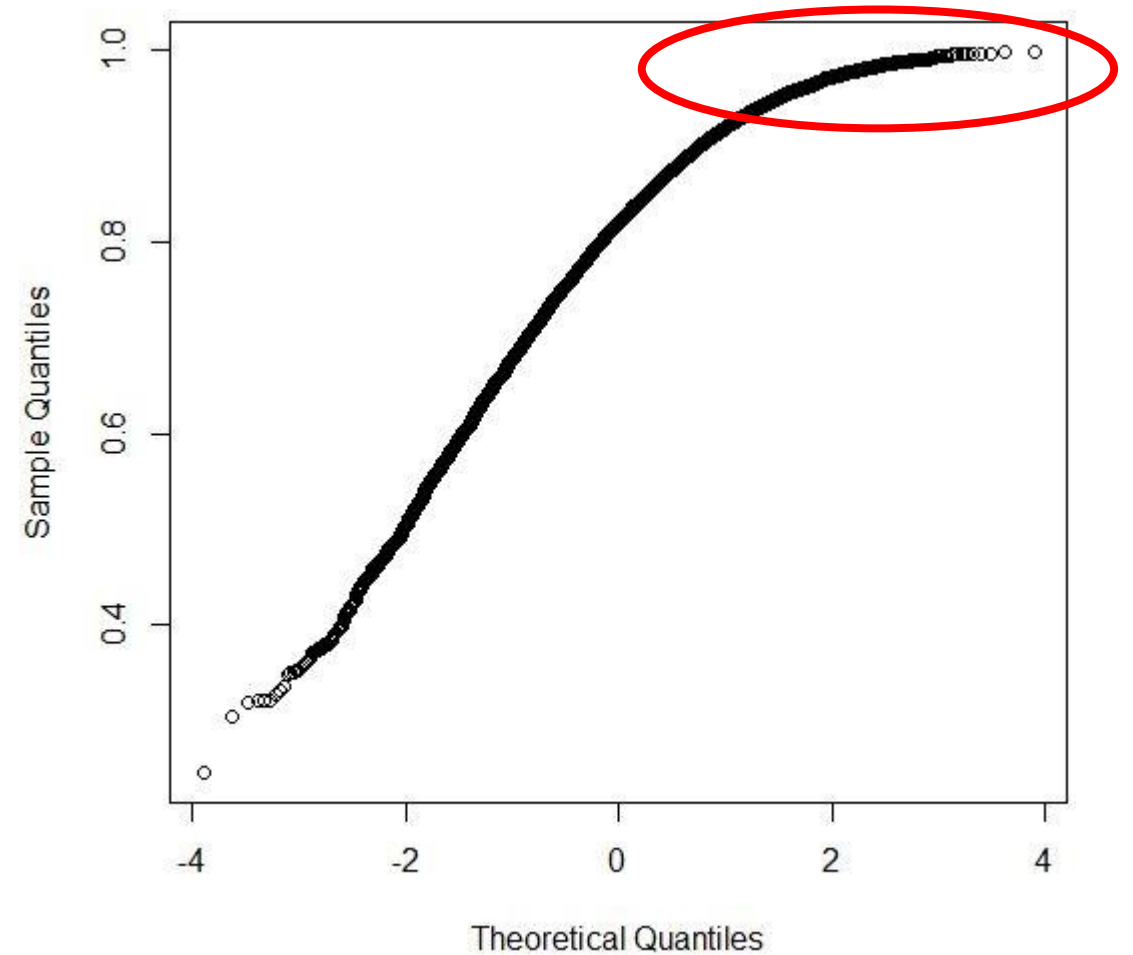
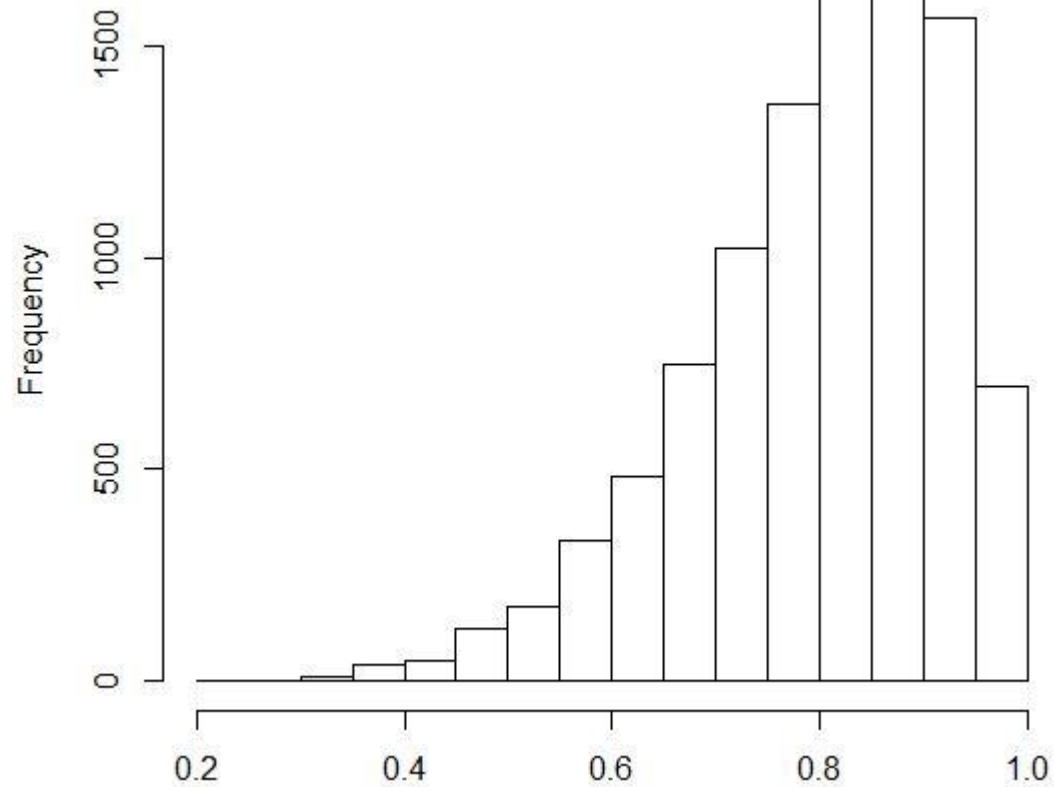


Histogram and QQ Plot for 10,000 observations from χ^2_3

**right-skewed distribution:
weak-tailed on left side**

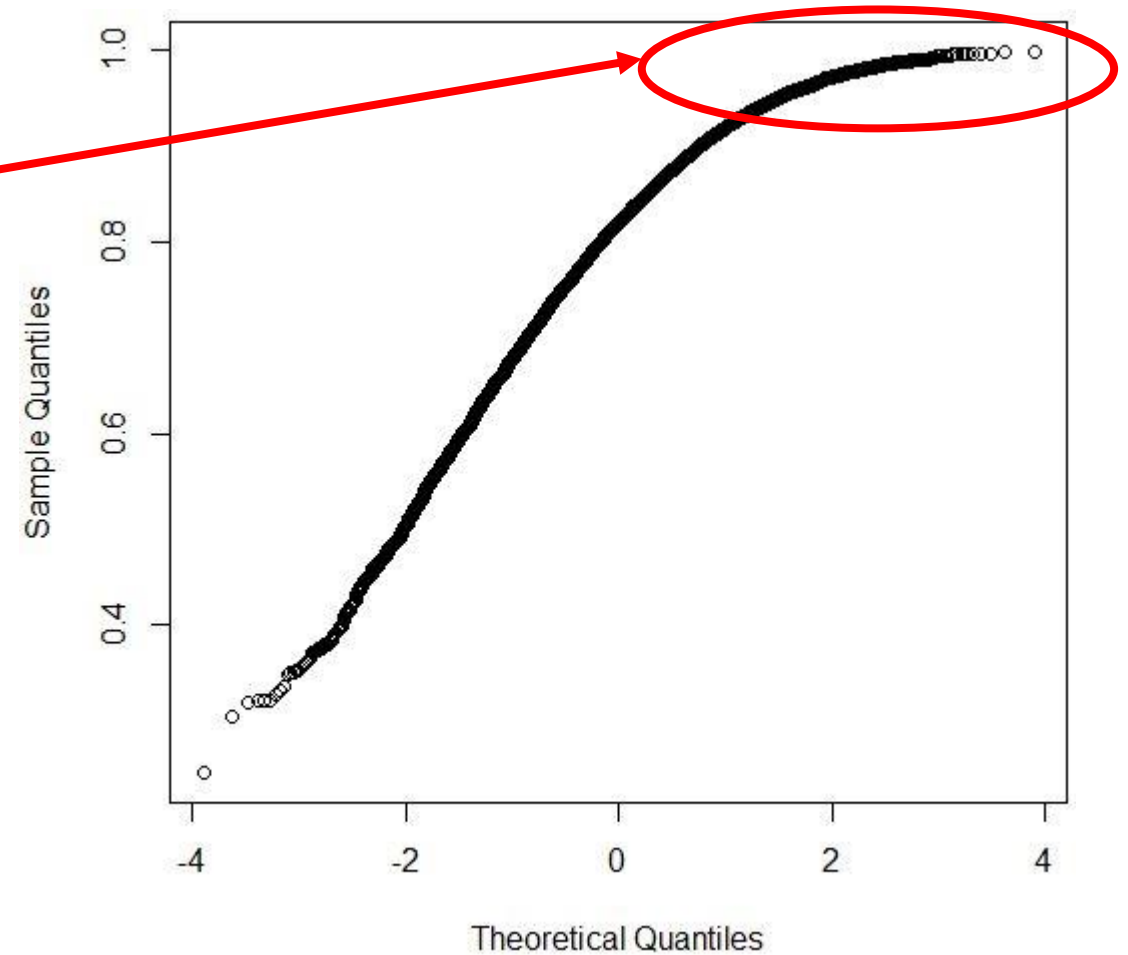


Histogram and QQ Plot for 10,000 observations from $BETA(\alpha = 8, \beta = 2)$

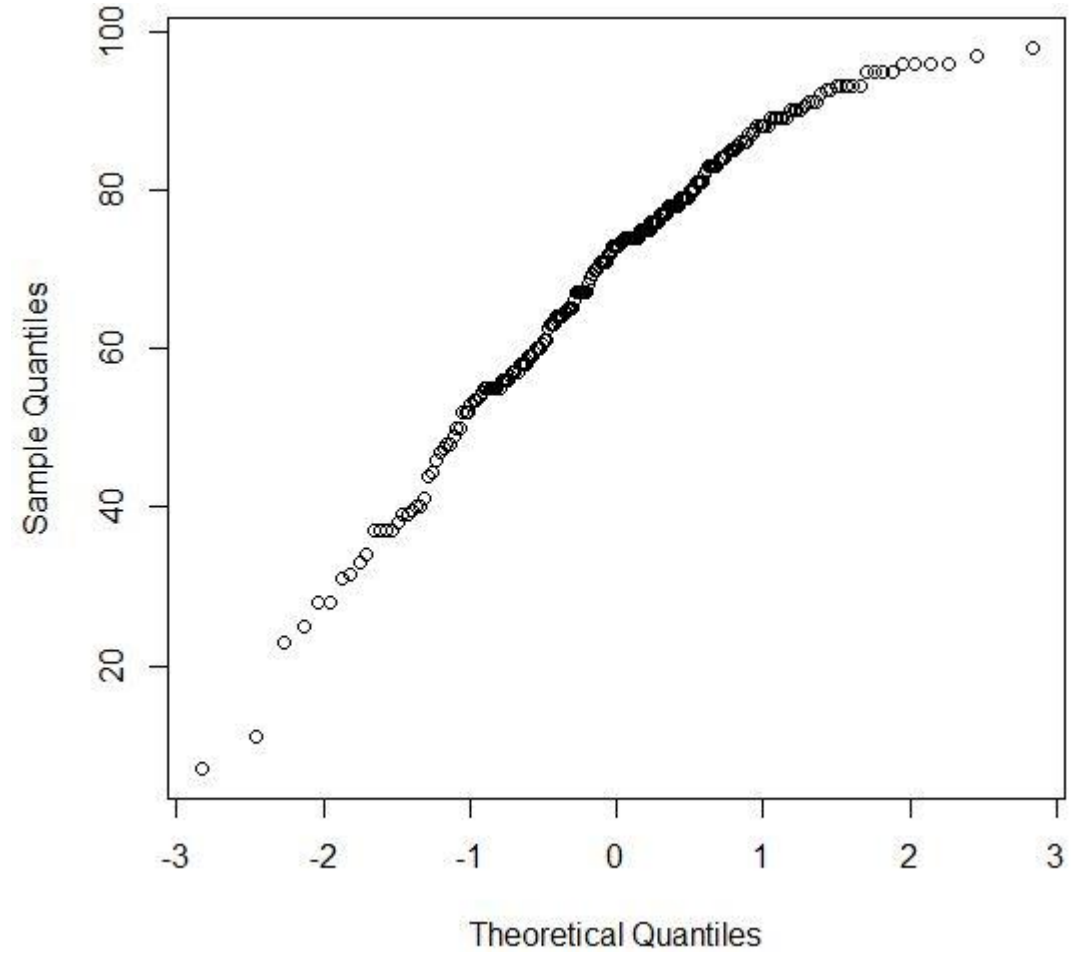
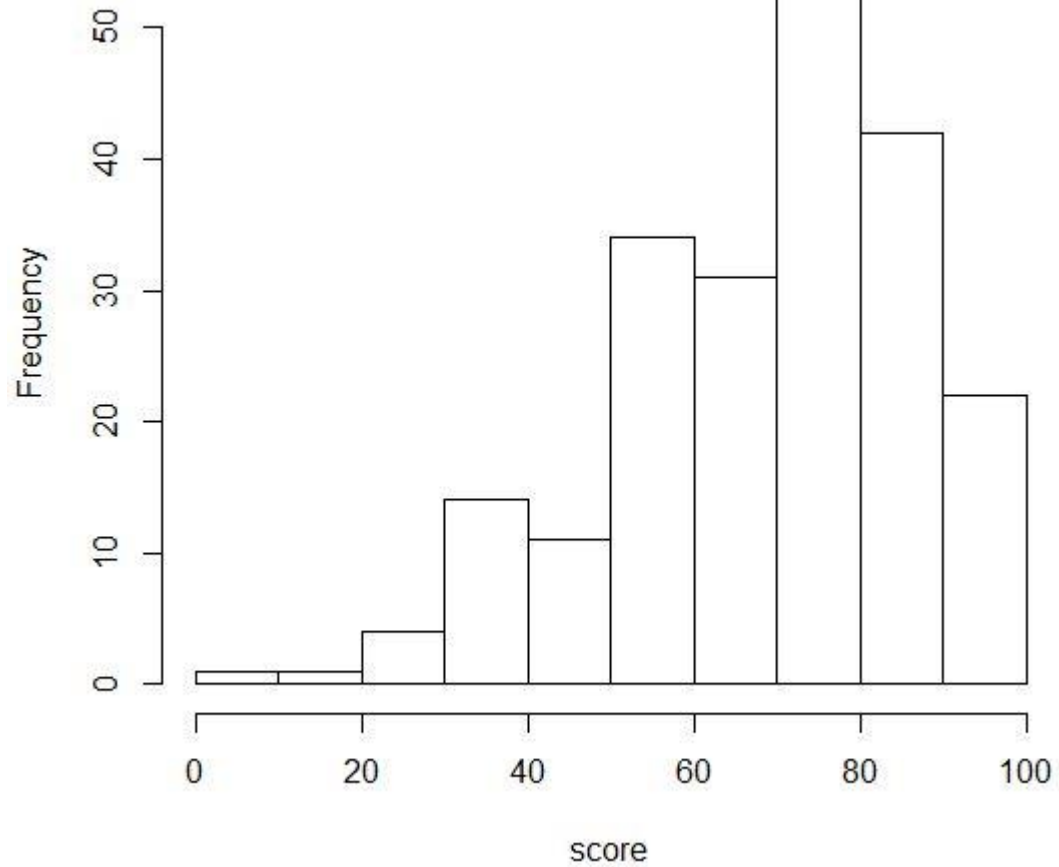


Histogram and QQ Plot for 10,000 observations from $BETA(\alpha = 8, \beta = 2)$

**left-skewed distribution:
weak-tailed on right side**



Calc 1 Final Exam scores ($n = 215$)



Warning: some courses/textbooks and software use X for the observations and Y for the Z percentiles! This results in the opposite conclusion regarding the weak-tailed and heavy-tailed “squiggles”!

Sometimes these are switched so be careful.

